Prague University of Economics and Business

Faculty of Economics

THE LAFFER CURVE IN THE CZECH REPUBLIC

Bachelor's Thesis

May 16, 2025

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Affidavit

I hereby declare on my honour that I have prepared my Bachelor's Thesis independently and using the literature cited.

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In Prague, May 16, 2025

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I would like to express my gratitude to my thesis supervisor, Ing. Pavel Potužák Ph.D., for his guidance, valuable and prompt expert feedback, and patience throughout the entire process. Furthermore, I would like to thank my family and friends for their continuous encouragement and support. A special appreciation goes to my girlfriend, whose constant belief in my abilities has been a significant source of motivation.

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BACHELOR'S THESIS TOPIC

Author of thesis: **Tomáš Haufer**

Study programme: Economics

Topic: **The Laffer Curve in the Czech Republic**

Guides to writing a thesis:

1. The main objective of this bachelor thesis is to determine the overall tax rate that would generate the highest potential tax revenues for the government of the Czech Republic, based on the Laffer curve theory. The analysis will be conducted for overall tax and welfare system and selected individual taxes.
2. So far, many papers have been published dealing with the Laffer curve theory from many different perspectives. These various approaches can be seen in Trabandt & Uhlig, (2012), Spiegel & Templeman (2004) or Fève, et al. (2018). Nevertheless, a few papers (Karas 2013, Rotschedl et al. 2012) have been written about the Laffer curve in the context of the Czech Republic, which used less data than this thesis. The core of the Czech tax system was issued in 1993, therefore it is more than 30 years old. The COVID crisis and the energetic crisis have made multiple governments to start reviewing their fiscal policies, including tax and social welfare systems. All those key points show the need to evaluate the Czech tax system. This thesis does that from the tax revenues perspective.
3. The theoretical part of the thesis will firstly present some basis of the tax theory and the Laffer curve theory. It will also introduce alternative perspectives on the Laffer curve theory and present selected empirical studies of it across various countries including the Czech Republic. This section will identify the gaps in the existing research on this topic in the Czech Republic. Secondly, it will provide brief overview of current Czech tax and social welfare system, and it will shortly summarize its development from the year 1993 when the Czech Republic was formed.
4. The empirical part will initially present available data about tax rates and revenues in the selected time period and potential control variables. Consequently, models showing the connection of tax rates to tax revenues will be built. For that purpose, polynomial Ordinary Least Squares (OLS) regression with selected control variables will be utilized. The primary sources of empirical data will be the Czech statistical office, the Financial Administration and Czech legislature. Based on our findings, we will estimate the tax rate that would maximize government's tax revenues.
5. Keywords: Laffer curve, tax system, Czech Republic, optimal tax rate

   JEL classification: H21, H55, K34

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Selected bibliography:

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Abstract

Taxation is a frequent and important topic of discussion in politics, economics and the public debate as well. This thesis analyses the overall tax system, the Personal Income Tax and the Corporate Income Tax from the Laffer Curve perspective with the objective of finding a revenue-maximising tax rate for each of the mentioned. Because of the non-stationarity of some data, two branches of models are estimated. One takes the stationarity as an assumption and finds the revenue-maximising rates at 28.6% for the overall tax system, 17.5% for the Personal Income Tax, and 27.5% for the Corporate Income Tax. The second branch finds linear relationships between the growth rates of the tax revenues and the growth rates of the tax rates, which are all stationary. Three methods are utilised for the estimation, Ordinary Least Squares, Two Stage Least Squares with Instrumental Variables, and the Generalised Method of Moments. The thesis uses quarterly data from 1996 Q1 to 2023 Q4. Comments on other determinants of tax revenues accompany the main findings. The research presented in this thesis provides the latest empirical evidence on the relationship between tax rates, macroeconomic factors, and their impact on tax revenues in the Czech Republic.

Key Words

Laffer Curve, tax system, Czech Republic, tax revenue-maximising tax rate, growth rate of the tax revenue, personal income tax, corporate income tax

JEL classifications: H21, H24, H25, K34

Abstrakt

Zdanění patří mezi častá a zásadní témata diskusí v politice, ekonomii i ve veřejné debatě. Tato práce analyzuje celkovou daňovou soustavu, daň z příjmů fyzických osob a daň z příjmů právníckých osob z pohledu Lafferovy křivky s cílem najít daňovou sazbu maximalizující příjmy plynoucí z každé ze zmíněných daní. Vzhledem k nestacionaritě některých dat se odhadují dvě větve modelů. Jedna vychází z předpokladu stacionarity a zjišťuje, že sazby maximalizující příjmy činí 28,6 % pro celkový daňový systém, 17,5 % pro daň z příjmů fyzických osob a 27,5 % pro daň z příjmů právnických osob. Druhá větev nachází lineární vztahy mezi mírami růstu výnosů a mírami růstu daňových sazeb, které jsou všechny stacionární. Pro odhad jsou využity tři metody, metoda nejmenších čtverců, dvoustupňová metoda nejmenších čtverců s instrumentálními proměnnými a zobecněná metoda momentů. Práce využívá čtvrteltní data od 1996 Q1 do 2023 Q4. K hlavním zjištěním jsou připojeny poznámky k dalším determinantům daňových příjmů. Výzkum prezentovaný v této práci přináší nejnovější empirické poznatky o vztahu mezi daňovými sazbami, makroekonomickými faktory a jejich vlivem na daňové příjmy v České republice.

Klíčová slova

Lafferova křivka, daňová soustava, Česká republika, daňová sazba maximalizující daňové příjmy, míra růstu daňových příjmů, daň z příjmů fyzických osob, daň z příjmů právníckých osob

JEL klasifikace: H21, H24, H25, K34

Contents

1 Introduction

Taxes have accompanied humankind for centuries. Sooner or later, a form of taxation emerged in every developed society. Taxes have evolved in their forms and have taxed various assets throughout the years, and their inception has led to some significant historical events. Over the years, they have also gained a crucial role in the state governance as the most important source of the government's revenue.

The tax debate is perhaps more important today than ever as the developed world's population ages. After decades of building social welfare systems, many of those structures begin to fail as they require more and more money to be funded. Since the government is the one running these systems, multiple forms of taxation are the most common and the most straightforward way to obtain the money.

Nevertheless, unceasing increases in the tax rates might not result in unceasing increases in the tax revenue. This idea was made famous by Arthur Laffer (2004) and Jude Wanniski (1978) in the 1970s, when they revived a centuries-old idea of the Laffer Curve. Laffer noted that at a certain point, a rise in the tax rate would lead to a decrease in the tax revenue. Therefore, there should be a tax rate for which the tax revenue is maximised, a peak of the hill. If a government wants to raise as much funding as possible, this should be the rate they would aim for. Finding this revenue-maximising rate for the Czech Republic is also one of the aims of this thesis.

Such a rate differs from the optimal tax rate. Taxes come with many consequences, and should therefore be imposed carefully and sensibly. Studying the various ways of taxation to find the optimal one has been an economic topic for decades. The forthcoming section introduces, after a brief overview of some basic terms, the basic tax theory and describes the desirable and optimal tax principles.

Very often, governments pursue short-term rather than long-term goals. Maximising the tax revenue might thus become the centre of the leaders' attention. The issue of maximising the tax revenue is further explored in the section on Laffer Curve Theory, where we examine the theoretical foundations and explore the origins and the proofs of the theory and some extensions.

After this introduction, we present a short historical overview of the development of the Czech tax system, which we want to analyse. We will particularly focus on income taxes, as we will also build separate models for them. Similar studies have already been produced, most notably by Karas (2013) or Rotschedl et al. (2012), but none of them could use as long time series as we could. What is more, thanks to the work of people at the Czech National Bank and the Ministry of Finance, we are able to utilise quarterly data for our analysis.

The fifth section delves into the analysis itself. Firstly, it describes the data provided by public databases and institutions. Then, the obtained data is processed, some of the key variables are created, and the major issues are addressed. The period from 1996 Q1 to 2023 Q4 is studied to preserve the possibility of analysing quarterly data.

The fifth section also states that because of stationarity, two branches of models will be developed. The first will use stationarity as an assumption, and the second will address it by using the growth rates of variables instead of their levels. Thus, the analysis will also show the relationships between growth rates of tax-related variables and relevant macroeconomic indicators.

Unlike the previous studies on this topic in the Czech context, this thesis utilises not only the Ordinary Least Squares method for estimation, but also challenges its crucial endogeneity assumption and employs Two Stage Least Squares with Instrumental Variables. These results are furthermore reestimated using the Generalised Method of Moments in order to obtain the most robust results. The second part of the fifth section is dedicated to the outputs of these models, enriched by first comments on the results. Another essential element in our analysis is the incorporation of multiple control variables, such as the Unemployment Rate or Annual Gross Domestic Product (GDP) Growth Rate.

The sixth section then connects and highlights the most important findings, presents the revenue-maximising tax rates and compares them to the latest values from the dataset. We also comment on noticeable links between utilised variables. And show the different determinants for the growth rates of individual analysed taxes. Finally, the last section summarises the whole process and concludes with the most critical findings.

2 Introduction to Tax Theory

Taxation is as old as the state itself. The government always needs some revenue to finance its expenditures. Even the Bible preaches an elementary form of taxation (“Leviticus 27”, 2011). Furthermore, the government has been expected to act on the market since the dawn of economic theories. Mercantilists were among the first who publicly promoted a significant role of the government and its interventions in order to accumulate wealth and promote industry (Stiglitz & Rosengard, 2015).

Decades later, in his magnum opus *Wealth of Nations*, Adam Smith advocates for a limited role of the government. He emphasises the idea that if individuals pursued their desires, their actions would serve the public interest (Smith, 1937).

His ideas have influenced economics for centuries and led to the support of *laissez-faire* ideas. Nevertheless, as Stiglitz and Rosengard (2015) note, not all economists were persuaded. Karl Marx became probably the most influential scholar arguing for state-controlled economies. The debate over the government’s economic role evolved over the years and ultimately shifted with the recognition of market failures and the rise of welfare states.

Stiglitz and Rosengard (2015) show that it was the government that intervened during the most severe economic crises to mitigate their impacts. They also acknowledge the fact that wrong interventions may lead to even worse outcomes. At the same time, Social Security programs have developed extensively since their origins and nowadays require more and more funds to stay sustainable.

The above-mentioned reasons are just a few notable ones that have led to an increase in government spending over the past decades. Naturally, when the expenditures increase, the government needs additional income. And that is where taxation steps in. Taxes are the most common source of the government’s revenue.

Taxes, therefore, fulfil the so-called fiscal function - they generate revenue for the public budgets. Governments then cover their expenses and reallocate the money in the economy - the allocation function of taxes. This function is closely related to the stabilisation function of taxes - governments influence the unstable economies in the right direction by using the tax revenues well (Zídková et al., 2024).

They might also be utilised to redistribute income in the society - redistributive function. Imposing a tax affects entrepreneurs and therefore creates or destroys economic incentives. The same applies to spending on ”sin” products - the regulation function (Zídková et al., 2024).

2.1 Basic Terms

Let us clarify some tax-related terms. A tax itself is defined as “a compulsory levy, imposed by government or other tax raising body, on income, expenditure, wealth or people, for which the taxpayer receives nothing specific in return.” (Lymer et al., 2016, p. 3).

The absence of a direct return for taxpayers’ payments illustrates the principle of non-equivalence. They also do not know what their money will be used for, meaning that taxes are non-specific charges. Taxes are paid in monetary form, and failure to pay them is punishable (Zídková et al., 2024).

As for the Social Security System and its contributions, they fail to fulfil the non-equivalence condition. Even though we consider them taxes. This approach is common in the economic society, as we can see in Stiglitz and Rosengard (2015) or Friedman (1999). Zídková et al. (2024) add that the contributions to the Social Security Systems are classified as taxes by the Organisation for Economic Co-operation and Development (OECD). Further debate on whether the social security contributions can be considered as taxes can be seen in Thuronyi (1996).

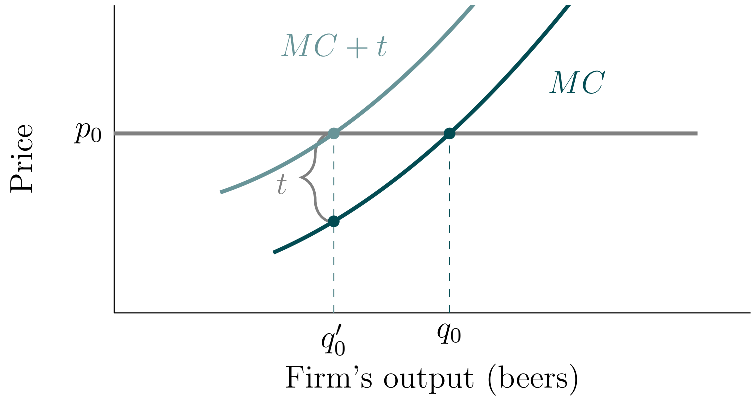
Taxes are paid either directly or indirectly. Direct taxes are levied on individuals or organisations and are paid directly to the state without a shift to another payer. They are usually calculated from income, property value, etc. Indirect taxes are imposed on taxpayers who charge the tax to their customers together with the price of their goods sold or services provided. They act as an intermediary. As a result, it is always the consumer who bears the tax burden[1] (Lymer et al., 2016; Stiglitz & Rosengard, 2015; Tax Foundation, 2025).

The final tax is calculated from a tax base which is "the total amount of income, property, assets, consumption, transactions, or other economic activity subject to taxation by a tax authority" (Tax Foundation, 2025). Tax deductions allow the subject of the tax, the taxpayers, to decrease their tax base by a certain statutory amount in specified cases. Another way to reduce the tax is by using a tax credit, a method applied to the final calculated tax bill, lowering it dollar-for-dollar. Tax rates are either flat (the whole tax base is subjected to the same tax rate) or progressive (average tax rates increase with tax base - these systems usually use tax brackets to tax different levels of tax base at different rates)[2] (Tax Foundation, 2025).

2.2 Consequences of a Tax

When the government levies a tax, it should be aware that taxation comes with various economic consequences and reactions. Different taxes might vary in their effects, and that sets the stage for this chapter, in which we will look at the economic implications of taxation. Later examples assume perfect competition.

Figure 2.1: Tax Incidence for a Single Firm



Note: Author's own elaboration based on Stiglitz and Rosengard (2015). p_0... Chosen price, q_0... Supplied quantity before tax, q'_0... Supplied quantity after tax, t... Tax, MC... Marginal Cost curve ∼ Supply curve, $MC + t$... Marginal Cost curve after tax.

Let us start with the effect of consumption taxation on a firm. Suppose that a firm produces only beers and the government levies a general Value Added Tax (VAT)

[1] The real cost of tax.

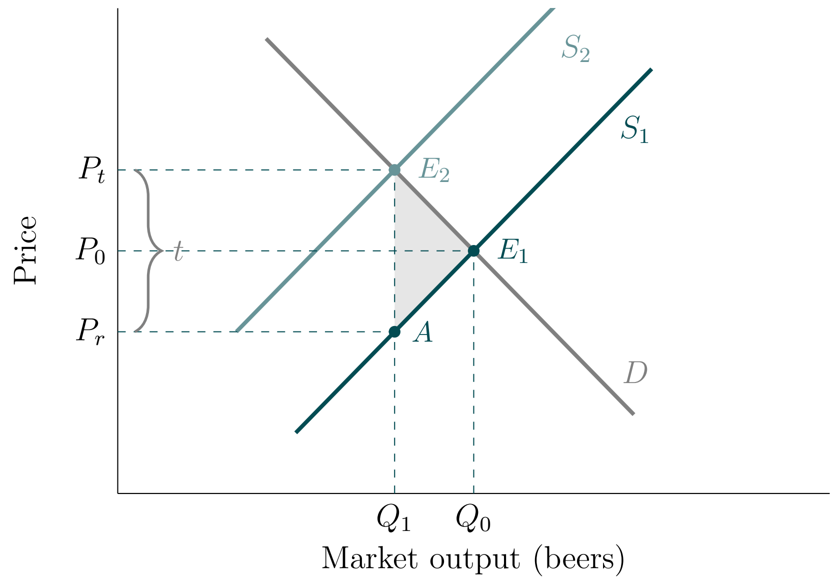
[2] For a broader glossary featuring expanded explanatory notes, see Tax Foundation (2025).

on the entire economy. Figure 2.1 plots the situation of the firm on a graph.

Before the tax was imposed, the firm supplied a quantity q_0 at a price p_0. Its supply function was characterised by the MC curve. The government then levies the VAT on all goods, including beer. Consequently, the cost of production rises and so does the marginal cost. The whole MC curve shifts to the left as denoted by the $MC + t$ curve. At the same price p_0, the firm is able to supply only quantity q_0'.

The same situation happens in all beer-producing firms in the economy concerned. Figure 2.2 presents the implications for the beer market in the country. Before the tax, the economy operated at equilibrium E_1. As the VAT is imposed, the supply decreases, leading to a new equilibrium E_2. That results in a fall in the quantity supplied from Q_0 to Q_1 and a rise in the price of the commodity from P_0 to P_t. We must not forget to include the tax itself in the new price - consumers pay the new price at P_t, but firms now only receive the after-tax price, P_r. The size of the VAT is captured by the vertical space between P_r and P_t. Rectangle $AE_2P_TP_r$ captures the tax revenue from the tax imposed on beer. In this scenario, both firms and consumers are worse off. The welfare loss is shown by the gray triangle AE_1E_2. Who bears the tax burden and how much of it depends on the elasticity of demand and supply curves.

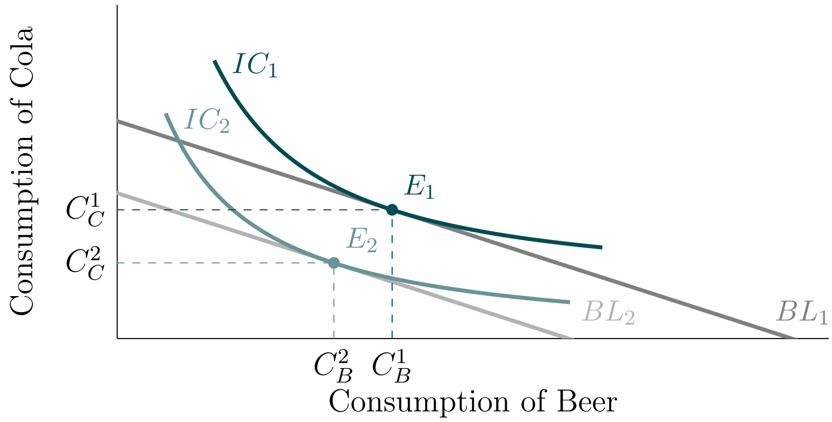
Figure 2.2: Tax Incidence for a Single Market



Note: Author's own elaboration based on Nicholson and Snyder (2012) and Stiglitz and Rosengard (2015). P_0... Initial market price, P_t... Price paid by consumer after tax, P_r... Price received by firms after tax, t... Tax, Q_0... Initial quantity supplied, Q_1... After tax quantity supplied, S_1... Initial supply curve, S_2... Supply curve after tax, D... Demand curve, E_1... Initial market equilibrium, E_2... After tax market equilibrium.

Figure 2.2 also partly shows the impact of consumption taxation on consumers. To elaborate on this topic, see Figure 2.3. The figure plots the situation of a consumer, choosing between the consumption of 2 goods - beer and cola. They are not perfect substitutes, and the government imposes a VAT at the same rate on both of them. Therefore, only the budget constraint (the budget line) of the consumer decreases from BL_1 to BL_2, and the lower indifference curve is attainable (IC_2). In case the government imposed an income tax on personal income, the results would be the same. Both mentioned taxes thus affect the real income of an individual.

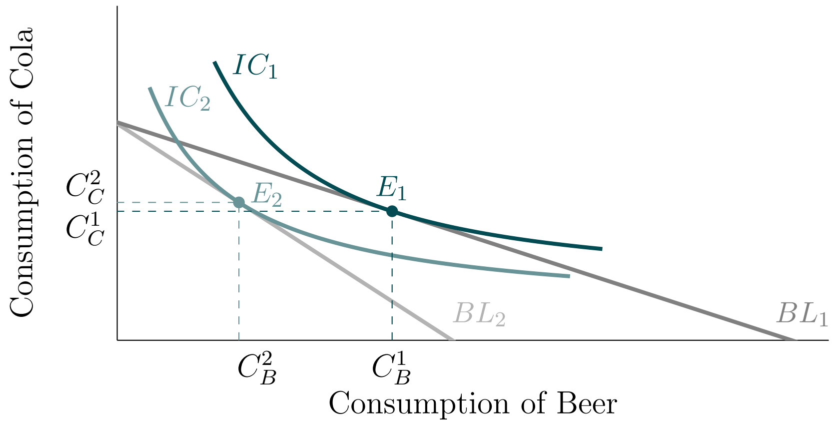
Figure 2.3: Value Added Tax Effect and Income Tax Effect on a Consumer



Note: Author's own elaboration based on Nicholson and Snyder (2012) and Stiglitz and Rosengard (2015). BL_1... Initial budget line, BL_2... After tax budget line, C_B^1... Initial consumption of beer, C_B^2... After tax consumption of beer, C_C^1... Initial consumption of cola, C_C^2...After tax consumption of cola, E_1...Initial personal equilibrium, R_2... After tax personal equilibrium, IC_1... Utility maximising indifference curve before tax, IC_2... Utility maximising indifference curve attainable after tax

A different situation would occur if the government issued a consumption tax on only a few strategically selected goods - this is the case of an excise duty. Figure 2.4 plots the situation of a consumer known from Figure 2.3. This time, the government imposes only an excise duty on beer. The consumption pattern of our consumer will therefore change. Consumption of beer decreases from C_B^1 to C_B^2, and simultaneously the consumption of cola rises from C_C^1 to C_C^2. This tax leads to a reduction in the consumer's utility, too.

Figure 2.4: Excise Duty Effect on a Consumer

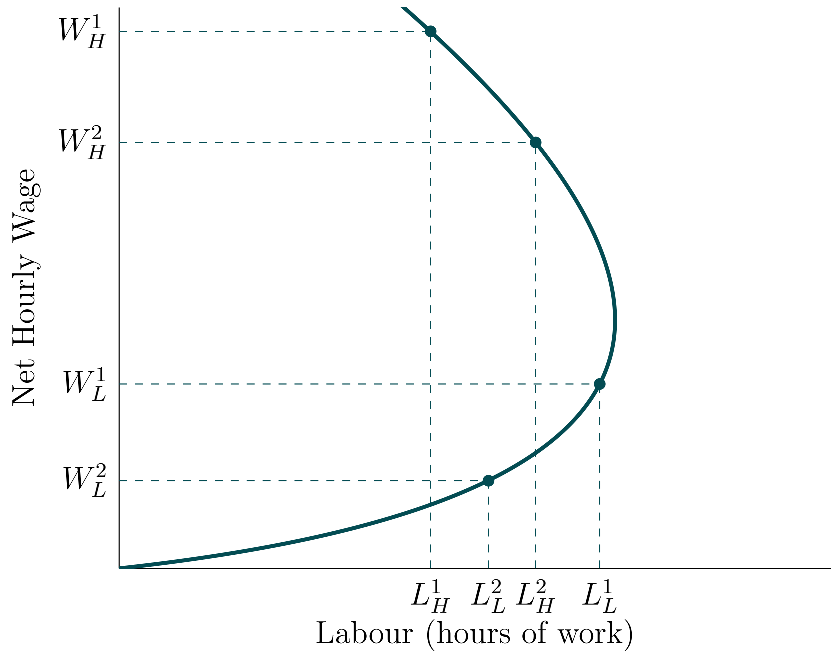


Note: Author's own elaboration based on Nicholson and Snyder (2012) and Stiglitz and Rosengard (2015). BL_1... Initial budget line, BL_2... After tax budget line, C_B^1... Initial consumption of beer, C_B^2... After tax consumption of beer, C_C^1... Initial consumption of cola, C_C^2...After tax consumption of cola, E_1...Initial personal equilibrium, R_2... After tax personal equilibrium, IC_1... Utility maximising indifference curve before tax, IC_2... Utility maximising indifference curve attainable after tax

The last situation we present shows an individual making a choice in the labour market and facing the imposition of an income tax. As we see in Figure 2.5, because of the specific shaped individual labour supply, the arising situations may differ. If an individual earns a high hourly wage W_H^1 and this income is subjected to a tax, his net wage decreases to W_H^2. This leads to an increase in the supplied labour hours from L_H^1

to L_H^2. On the contrary, if a person receives a low wage W_L^1, which shrinks further to W_L^2 due to taxation, he will work fewer labour hours (L_L^2) than before the introduction of the tax (L_L^1).

Figure 2.5: Income Tax Effect on Individual Labor Supply



Note: Author's own elaboration based on Borjas (2013) and Stiglitz and Rosengard (2015). L_H^1... Initial labour hours supplied by an individual with high wage, L_H^2... Labour hours supplied by an individual with high wage after tax, L_L^1... Initial labour hours supplied by an individual with low wage, L_L^2... Labour hours supplied by an individual with low wage after tax, W_H^1... Initial net hourly wage received by an individual with high wage, W_H^2... Net hourly wage received by an individual with high wage after tax, W_L^1... Initial net hourly wage received by an individual with low wage, W_L^2... Net hourly wage received by an individual with low wage after tax.

Thus, imposing a tax can cause both an increase and a decrease in the number of hours of work an individual provides. In the aggregate market, this effect is neglected because the labour supply curve rises by default. A reduction in the net hourly wage would only lead to a reduction in the number of hours of labour provided. Derivation of the individual and market supply curve can be found in Borjas (2013).

As we will see in Section 4, consumption and income taxation are the most common forms of taxation in the Czech Republic, and for that reason, their implications deserve special attention in this thesis.

2.3 Desirable Features of a Tax System

Multiple economic consequences presented in Chapter 2.2 raise the question of whether, and under what circumstances, it is socially beneficial to implement a tax system. Economists have been discussing it for decades and have presented several findings we will review in this chapter.

Smith (1937) premises 4 elementary "maxims" regarding taxes to be effective. The first of them can be summarised as equality.

> The subjects of every state ought to contribute towards the support of the government, as nearly as possible, in proportion to their respective

> abilities; that is, in proportion to the revenue which they respectively enjoy under the protection of the state (Smith, 1937, p. 777).

Secondly, tax and its properties should be certain. Both the contributor and the receiver must be aware of the time and the manner of payment, and the quantity to be paid. Thirdly, the payment should be convenient. (Smith, 1937)

> Every tax ought to be levied at the time, or in the manner, in which it is most likely to be convenient for the contributor to pay it. A tax upon the rent of land or of houses, payable at the same term at which such rents are usually paid, is levied at the time when it is most likely to be convenient for the contributor to pay, or when he is most likely to have wherewithal to pay (Smith, 1937, p. 778).

Fourthly, effectiveness must be maintained. That is, the administrative costs related to the tax collection must not surpass the gains from taxation. Furthermore, taxation might discourage certain choices - ideal taxation should distort the decisions as little as possible. If tax evasion occurs, penalties should arise proportionally to the "temptation of evasion". Effectiveness also means that tax controls must not be used to oppress people deliberately (Lymer et al., 2016; Smith, 1937).

Lymer et al. (2016) in their textbook remind that these "Canons of taxation" have not changed much since the first mention by Adam Smith. In their own textbook on public economics, Stiglitz and Rosengard (2015) build on those foundations and establish *The Five Desirable Characteristics of Any Tax System*: efficiency, administrative simplicity, flexibility, political responsibility, and fairness.

Efficiency aligns with Smith's views - the system should not distort the decision-making of economic agents. By administrative simplicity, authors mean low costs of running the system and motivation for voluntary compliance. Flexibility consists in the adaptability to changing circumstances, political responsibility lies in the system's transparency. Finally, fairness stands for the system treating similar individuals similarly. For a more elaborate explanation, see Stiglitz and Rosengard (2015).

2.3.1 Optimal Tax Structures

Many economists explored the taxation theory to find the most optimal system and its necessary features. Presented works from Smith and Stiglitz & Rosengard explained the desirabilities in a non-mathematical way. The subsequent paragraphs present some of the most influential papers on optimal tax systems whose authors base their conclusions on mathematical insights.

Ramsey (1927, p. 54) demonstrates that "the taxes should be such as to diminish the production of all commodities in the same proportion." Furthermore, he shows that for commodities exhibiting high inelasticity of either supply or demand, the distribution of taxation across them is irrelevant, as such taxes do not distort producer or consumer behaviour. Optimal taxation should therefore be structured according to the elasticities of demand and supply (Ramsey, 1927).

Diamond and Mirrlees (1971) establish that the optimal tax system does not affect production choices. This principle is violated when the government imposes taxes on intermediate goods, thereby disrupting production efficiency. Thus, legislators should focus taxation efforts on the final goods and outcomes of production processes.

Mirrlees (1971) pursued the tax theory in the questions of the income taxes as well and confirms that the optimal income tax schedule is sensitive to the distribution of skills among the population. To his surprise, his utilitarian analysis of income taxation did not provide an argument for high progressive tax rates. His numerical results suggest that, under reasonable assumptions, a nearly linear tax schedule may be close to optimal. This view is also supported by Stiglitz (1987).

P. A. Diamond, former co-author of J. A. Mirrlees, explored the area of income taxation as well. In his paper, he adds that "the optimal marginal tax rate at some income level depends on the elasticity of labour supply at that income level." (Diamond, 1998, p. 92) He also challenges the use of the lognormal distribution of skills in Mirrlees (1971), arguing that the nearly linear optimal tax schedule found there is a consequence of this specific distributional assumption. He also shows that the optimal marginal tax rate can be U-shaped, high both at low and high incomes (Diamond, 1998).

Atkinson and Stiglitz (1976) examined the optimal structure of direct and indirect taxation. They demonstrate that if the government had no distributional objectives, only direct taxation would be sufficient. Moreover, they highlight the need to study tax systems as a whole since the part-by-part analyses may be misleading.

Feldstein (1978) advocates for eliminating the capital income tax and replacing it with consumption or labour income taxes. He finds that capital income taxes distort savings and consumption timing. This view is supported by Lucas (1990), who presents quantitative general equilibrium analysis and concludes that removing capital taxation increases welfare.

Finally, Mankiw et al. (2009) analyse whether the presented theoretical insights from literature were incorporated into actual tax policies. The authors note that some progress has been made, but significant gaps between theory and practice still remain. As they observe:

> On the one hand, some trends in tax policy look like at least partial victories for optimal tax theory. Perhaps the most important is the worldwide trend toward reduced taxation of capital income, at least in statutory tax rates. In addition, the worldwide trend toward tax systems with flatter tax rates might be seen as a reflection of lessons from theoretical work (Mankiw et al., 2009, p. 171).

3 Laffer Curve Theory

So far, we have been concerned with the economic consequences of taxes and the design of a potentially optimal tax system. But the government could just want to use its current tax system to find the tax rate at which it would raise the highest revenues to cover its rising deficits and debts. The rising tax rate might not lead to a further increase in tax yields.

This idea is not new to humankind. Renowned Muslim philosopher Ibn Khaldun proposed a similar thought in the 14th century:

> It should be known that at the beginning of the dynasty, taxation yields a large revenue from small assessments. At the end of the dynasty, taxation yields a small revenue from large assessments (Khaldun, 2015, p. 355).

A few centuries later, David Hume pondered similarly in his essay, *Of Taxes*:

> Exorbitant taxes, like extreme necessity, destroy industry by producing despair; and even before they reach this pitch they raise the wages of the labourer and manufacturer, and heighten the price of all commodities. An attentive, disinterested legislature will observe the point when the emolument ceases and the prejudice begins; but as the contrary character is much more common, it is to be feared that taxes all over Europe are multiplying to such a degree as will entirely crush all art and industry; though perhaps their first increase, together with circumstances, might have contributed to the growth of these advantages (Hume, 1752).

Even Adam Smith reminded society of destructive effects of excessive taxation:

> Secondly, it may obstruct the industry of the people, and discourage them from applying to certain branches of business which might give maintenance and employment to great multitudes. While it obliges the people to pay, it may thus diminish, or perhaps destroy, some of the funds which might enable them more easily to do so (Smith, 1937, pp. 778-779).

However, this idea has never received as much attention as it has since 1978, when J. Wanniski published his original paper about the so-called Laffer Curve (Wanniski, 1978).

3.1 Origins of the Modern Theory

The story traces back to the 1974 dinner of Arthur Laffer, Jude Wanniski, Donald Rumsfeld and Dick Cheney where while discussing proposed tax increases, A. Laffer sketched a curve on his napkin, illustrating the trade-off between tax rates and tax revenues. Wanniski then called the trade-off "The Laffer Curve" (Laffer, 2004).

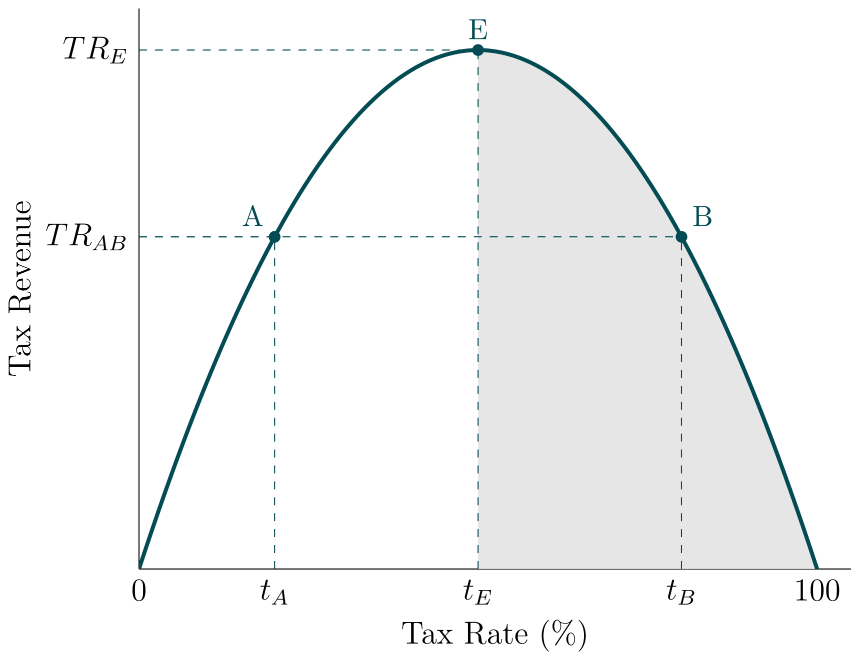
According to Laffer, "there are always 2 tax rates that yield the same revenues." (Wanniski, 1978) At the same time, when the tax rate is 0%, there would be no tax revenues. This is naturally intuitive as consumers and firms are allowed to keep all of the products of their labour and production. Furthermore, if the tax rate reached 100%, there would not be any tax revenues as well. Any incentives to work and to produce would disappear since all products of labour would be confiscated by the government.[3] All legal production would cease, and people would tend to find some illegal means for living or start bartering (Laffer, 2004; Wanniski, 1978).

In between the 2 extremes, the Laffer Curve lies. As we raise the tax rate from 0%, people begin to pay taxes and therefore the tax revenue increases. This continues until a certain tax rate, where the revenues reach their maxima. On the illustrative plot in Figure 3.1, this point on the curve is denoted as "E". In case the government keeps raising taxes further, earnings from taxation diminish as people leave their jobs, turn to a barter economy and firms close or eventually find illegal ways of operation (Laffer, 2004; Wanniski, 1978).

As mentioned above, there are always the same exact revenues for 2 certain tax rates. To illustrate, on the graph in Figure 3.1, those are points "A" and "B". In such a case, only point "A" is desirable as it generates the same revenue but at a lower tax rate. We could perform a similar analysis for each of the points on the Laffer Curve, and we would eventually find out that points from 0 to "E" are preferable since they use lower tax rates. The grey area from "E" to 100 is the so-called "prohibitive range". No rational government[4] would set the tax rate in that area because a tax increase would result in a fall in tax revenues, and a tax cut would lead to a rise in the revenues when considering the rates in the prohibitive range (Laffer, 2004).

The shape of the curve is generally a concave polynomial function of tax revenues with respect to the tax rate. Its maximum is not necessarily 50% as it could seem from Figure 3.1. The revenue-maximising tax rate is a subject of analysis for every tax system, and determining it for the Czech Republic is the ultimate goal of this thesis.

Figure 3.1: The Laffer Curve



Note: Author's own elaboration based on Laffer (2004) and Wanniski (1978). t_A... Tax rate at point A; t_B... Tax rate at point B; t_E... Tax rate at point E; TR_{AB}... Tax revenue at point A and B; TR_E... Tax revenue at point E.

[3]Alternatively, the government could force people to work and use a rationing system.

[4]Assuming a government whose objective is to maximise tax revenue.

The reasoning described above works quite well for economies of all different scales. Wanniski himself elaborates further on this thought:

> This is true whether the political leader heads a nation or a family. The father who disciplines his son at point B[5], imposing harsh penalties for violating both major and minor rules, only invites sullen rebellion, stealth, and lying (tax evasion, on the national level). The permissive father who disciplines casually at point A[6] invites open, reckless rebellion: His son's independence and relatively unfettered growth comes at the expense of the rest of the family. The wise parent seeks point E, which will probably vary from one child to another, from son to daughter (Wanniski, 1978, p. 5).

Wanniski (1978) also realises that the revenue-maximising tax rate is not fixed in time. Economies are dynamic, and when, for example, a war is declared, the structure of the economies concerned transforms rapidly. People aware of military risks will be willing to pay higher taxes in exchange for their security. The revenue-maximising tax rate will therefore rise. Similar changes, though not as rapid and intense, are occurring all the time.

3.2 Evidence from the Real World

The Laffer Curve has fascinated economists for more than 4 decades, during which time many studies have been published on its implications for the real world. Scientists used various datasets from multiple countries to provide empirical evidence for the existence of the Laffer Curve.

Laffer (2004) in his original contribution describes multiple tax cuts that should prove the validity of the theory. He illustrates the idea on the Harding-Coolidge Tax Cut from the 1920s. During this period, income tax progressivity got lowered, which led to a rapid boost in the GDP growth while the tax revenues increased a bit as well. A similar story goes for the Kennedy Tax Cut in the 1960s. While the tax rates generally decreased, the GDP growth after the cut increased, and the total tax revenue boomed in this case even more. The Reagan Tax Cut in the 1980s provides an almost identical story.

Wanniski (1978) provides multiple historical cases when a tax cut resulted in an increase in tax revenues. He emphasises the role of lowering the tax rates on economic growth in Germany after World War II, or in the British Empire in 1816, leading to unprecedented growth.

Those observations are merely a comparison of the economies before and after the cuts. For better evidence of the theory, we ought to look for more complex analyses over longer time periods using more sophisticated methods. Thankfully, many studies from various authors have been published since the theory's inception.

Trabandt and Uhlig (2012) in their paper build on their research from 2011 and examine the differences in the Laffer Curves across 14 European Union (EU) countries

[5] In the original paper this point is referred to as "A" and it lies within the "prohibitive range". The same exact point is labelled as "B" in Figure 3.1 and therefore we have modified this quotation to maintain consistency in our notation.

[6] In the original paper this point is referred to as "B" and it does not lie within the "prohibitive range". The same exact point is labelled as "A" in Figure 3.1 and therefore we have modified this quotation to maintain consistency in our notation.

and the United States (US). Their work determines the Laffer Curve for each described country and extends the analysis to computing the possible additional income from labour and capital taxes that would occur if the country used tax revenue-maximising tax rates. According to the paper, European countries are, on average, closer to the revenue-maximising tax rates than the US. In case of labour taxes, they could, on average, improve their earnings by 4.3% of their GDP[7]. The possible improvement for the US reaches 9.0% of GDP. Considering the capital taxes, EU countries could earn, on average, 1.2% of their GDP more, for the US it would be 2.6% of their GDP.

The work of Clausing (2007) shows the existence of a non-linear relationship between tax rates and tax revenues for the Corporate Income Tax (CIT) among the OECD countries over the time period 1979-2002. She finds the revenue-maximising rate at 33%. Ideas in this paper are further developed by Brill and Hassett (2007), who extend the period and find evidence for the maximising rate at 34% in the 1980s and 26% from 2000 to 2005.

In 2019, a study about the Laffer Curve in Chinese conditions was released and found the revenue-maximising tax rate at 40% (Lin & Jia, 2019). Another major Asian country, Japan, was investigated by Nutahara (2015), who pointed out that the labour tax rate is smaller than the revenue-maximising rate. At the same time, the capital tax rate was very close to the peak rate.

Oliveira and Costa (2015) estimated the Laffer Curve for the overall EU separately in times of expansion and recession. The paper shows that the maximising rate in expansion (22%) exceeds the rate in recession (21.5%) and points out that most of the EU countries operated outside the "prohibitive range". Authors also emphasise that both the VAT tax rates and the revenue-maximising rates have been decreasing over time.

This phenomenon was previously studied by Czech authors as well. Černohorský (2011) analyses the system with data from 1995 to 2009 and determines the revenue maximising tax rate for the Personal Income Tax (PIT) at 48.94%, for the Corporate Income Tax, the rate ought to be 12.92%. Rotschedl et al. (2012) finds the peaking rate for the Corporate Income Tax at 20.3% using data from 1998 to 2008. Finally, Karas (2013) models the Laffer Curve for the Personal Income Tax and contributions to the Social Security System of the Czech Republic, utilising data from 1993 to 2010. He finds the revenue maximising tax rate at 33.14% of annual gross income.[8]

3.3 Extensions of the Theory

In addition to looking for evidence of the existence of the Laffer Curve itself, many scholars have turned to more general analysis. They have subjected the theory to certain conditions to obtain a clearer understanding of its possible shape, behaviour, etc. In this part, we are going to introduce several major extensions of the original basic theory.

Spiegel and Templeman (2004) note that the most Laffer Curve studies do not distinguish between individual and aggregate Laffer Curves. Consequently, they show that under certain assumptions, the Laffer Curve is likely to have multiple peaks,

[7]GDP baseline for numbers in the paper is calculated as the country-specific yearly average of 1995-2010.

[8]Papers presented by Rotschedl et al. (2012) and Karas (2013) provide relevant t-statistics and p-values for their findings. Černohorský (2011) fails at providing those key statistics.

especially when the inequality in wage distribution is high.

Under the assumptions of incomplete markets and liquidity-constrained agents, the Laffer Curve may become horizontally S-shaped as proved by Fève et al. (2018). This happens when the Laffer Curve is conditional on public debt - that is, when the government is adjusting taxes to balance the budget, while varying the debt and keeping transfers fixed.

Sanyal et al. (2000) have incorporated corruption and tax evasion into the theory. Their paper points out that the tax revenue can decrease not necessarily because of entering the "prohibitive range", but also because auditing intensity is not optimally chosen (the government audits too little or even too much) or because the corruption rate is significantly increased by a rise in the tax rate.

The assumptions of household homogeneity and flat tax rates, commonly adopted in Laffer Curve studies, are relaxed by Holter et al. (2019). They work with house heterogeneity and a progressive tax regime and explore their interactions with tax revenues and the shape of the Laffer Curve itself. They conclude that the US peaking rate could be increased if the progressive tax regime was replaced by a flat labour income tax. They also highlight that joint taxation of married couples can lead to different labour participation and different Laffer Curves under progressive and flat tax regimes.

The Laffer Curve also behaves differently under imperfect competition. As demonstrated by Miravete et al. (2018), when firms have market power and act as price-makers, they strategically respond to tax changes. If taxes are lowered, firms tend to increase their wholesale prices, capturing most of the potential gains in tax revenue as increased profits. This strategic pricing behaviour reduces the revenue gain from lowering taxes compared to what would be expected. Such behaviour "flattens" the Laffer Curve, making tax revenues less sensitive to changes in tax rates.

Similar findings about the market power and firm behaviour are also presented by Hollenbeck and Uetake (2021) who specifically highlight how significant retail market power in the marijuana industry leads to substantial tax incidence on consumers and keeps Washington state's marijuana tax on the upward-sloping region of the Laffer Curve despite its high rate.

4 Tax System in the Czech Republic

Before moving on to the empirical part and its methodology, there is one more topic to be discussed. So far, we have talked about taxation in theory but have not mentioned any real-world examples of a system. Since this thesis focuses on the Laffer Curve in the Czech Republic, we will briefly review the history of the tax system in the Czech Republic.

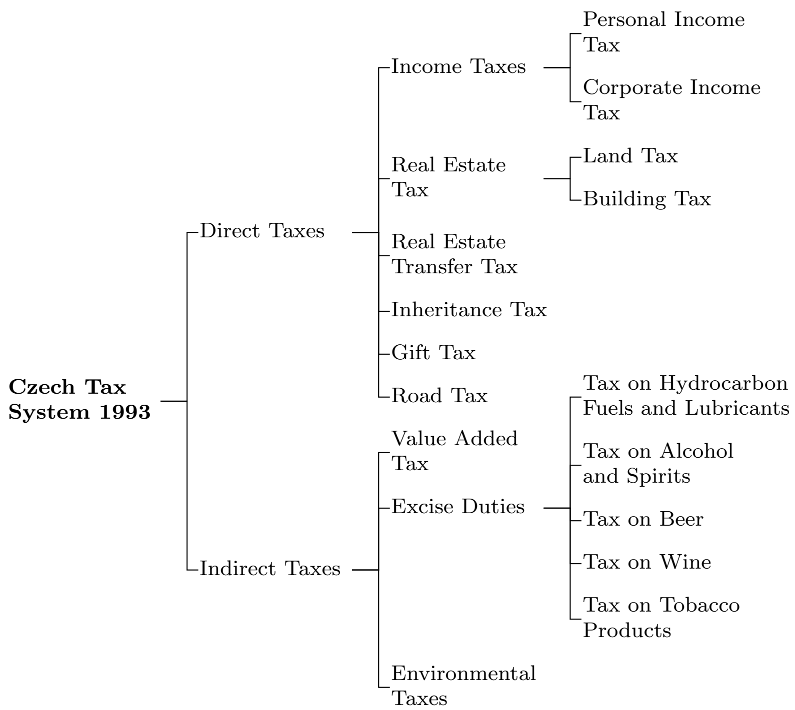
The tax system in the Czech Republic essentially consists of two major pillars. The first one is represented by all the taxes themselves and is supervised by the Ministry of Finance through the Financial Administration. This system was transferred from the former Czechoslovak Republic (Czech and Slovak Federative Republic, 1990, 1992a; Czech Republic, 2011).

For the purpose of this thesis, we can consider the Social Security System as the second pillar of the Czech tax system.

4.1 Overview of Taxes

After the Velvet Revolution in 1989, Czechoslovakia transformed into a democratic republic with a market-based economic system. A new tax regime was introduced in 1992, effective January 1993. That year, Czechoslovakia also split into 2 new countries - Czechia and Slovakia.

Figure 4.1: Czech Tax System in 1993

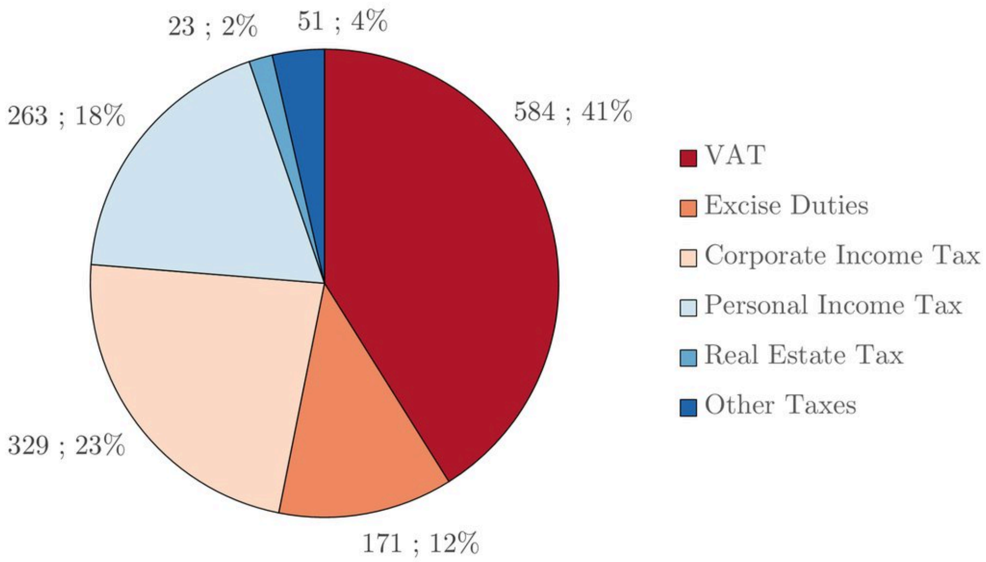


Note: Author's own elaboration based on Czech tax laws from 1992.

The system originally used in socialist Czechoslovakia required transformation as it had been suited for a centrally planned economy. The *Tax System Act* introduced 8 important taxes, 5 direct and 3 indirect, which will be described in the following paragraphs. A visualised overview of the system can be seen in Figure 4.1 (Czech and Slovak Federative Republic, 1992a).

The Czech tax system heavily relies on indirect taxation - 53% of tax revenue come from Value Added Tax and Excise Duties. Income Taxes account for 41% of the collected amount. This leaves only 6% for the rest of the taxes. The structure of collected taxes in 2024 is observable in Figure 4.2 (Czech National Bank, 2025).

Figure 4.2: The Structure of Tax Revenues (2024)(in bil. CZK & %)



Note: Author's own elaboration based on Czech National Bank (2025)

Our later analysis focuses primarily on the overall tax system and the income taxes. Therefore, our description of these taxes needs to be more detailed in their sections. This applies in particular to the tax rates.

4.1.1 Income Taxes

As shown in Figure 4.1, the tax system is divided into direct and indirect taxes. In the case of direct taxes, the taxpayer pays the tax directly to the tax authority. Indirect taxes are paid by the intermediary (usually a legal entity), who adds the statutory amount to the price of the goods sold or services provided (Atkinson, 1977).

One part of direct taxes is income taxes, both on personal and corporate income. Both were introduced by the *Income Tax Act* with effect from January 1993. The core of those taxes has remained unchanged to this day[9], as this Act is still valid. (Czech and Slovak Federative Republic, 1992d).

The Personal Income Tax applies to all incomes of its taxpayers - Czech residents and non-residents earning income from Czechia. Inheritance and gifts were formerly exempt from this tax because of their own tax until 2014. As with all taxes, there are certain exceptions, tax credits, and deductions (Czech and Slovak Federative Republic, 1992d).

The tax base is the amount by which income exceeds necessary expenditures. From the beginning of 2008 until the end of 2020, this tax base was increased by the social security contributions paid by the employer. The tax base was therefore the

[9]As of March 2025.

so-called super-gross salary. In practice, this meant an increase in the effective tax rate (Czech Republic, 2007, 2020b).

The tax rate for the Personal Income Tax started as progressive in 1993, as observable in table 4.1. The progressivity was continuously reduced in 1994, 1995, 1996, 1997, 1998, 1999, 2000, and 2005. From 2008, the tax rate was set as a flat 15%. With the effect from 2013, a new concept called "solidarity tax increase" was introduced. This increase affected the part of yearly income exceeding 48 times the average monthly wage, which got taxed by an additional 7%. The concept has been replaced by a progressive structure since 2021. The basic rate remained at 15% and amounts above the original threshold were affected by a 23% rate. Since 2024, the threshold has been lowered to 36 times the average monthly wage (Czech and Slovak Federative Republic, 1992d; Czech Republic, 1994, 1995a, 1996, 1997, 1998, 1999, 2000, 2005, 2012, 2020b).

Table 4.1: Personal Income Tax Rates in 1993

From Tax Base (CZK)	To Tax Base (CZK)	Tax Rate
0	60,000	15%
60,000	120,000	20%
120,000	180,000	25%
180,000	540,000	32%
540,000	1,080,000	40%
1,080,000	and above	47%

Note: Author's own elaboration based on Czech and Slovak Federative Republic (1992d).

From 2005 to 2008, the law allowed married couples with at least 1 child to tax their income jointly. They used this option to lower their overall tax burden. This practice is quite common in other countries (Kalíšková, 2014). Its effect on the labour market in the Czech Republic was studied in Kalíšková (2014).

Table 4.2: Corporate Income Tax Rates from 1993 to 2024

Time Period	Tax Rate	Time Period	Tax Rate	Time Period	Tax Rate
1993	45%	1998-1999	35%	2006-2007	24%
1994	42%	2000-2003	31%	2008	21%
1995	41%	2004	28%	2009	20%
1996-1997	39%	2005	26%	2010-2023	19%
				2024	21%

Note: Author's own elaboration based on Czech and Slovak Federative Republic (1992d) and Czech Republic (1993a, 1994, 1995a, 1997, 1999, 2003c, 2007, 2023).

The Corporate Income Tax shares its basic principles with the Personal Income Tax, with the key difference being the taxpayer. All legal entities established in the Czech Republic are subject to the tax. The same applies to foreign companies that generate revenue from sources located in the Czech Republic. The tax base is the taxpayers' profit (Czech and Slovak Federative Republic, 1992d). Even though the tax rate has been flat since 1993, it has changed dramatically since the introduction of the tax, as table 4.2 shows.

4.1.2 Other Direct Taxes

The Real Estate Tax is one of the most commonly used taxes worldwide. The Czech Republic incorporated this tax into its system in 1993 through the *Real Estate Tax Act*. Total tax is constructed from 2 taxes - the Land Tax and the Building Tax. It is also the only tax in the system that generates revenues directly for the municipality through the location of property (Czech and Slovak Federative Republic, 1992b).

The basic principles of the tax remain unchanged to this day as well. The taxpayer is typically the owner of the property or its user. The tax base for the Land Tax equals the total area of the plot and the total built-up area for the Building Tax (Czech and Slovak Federative Republic, 1992b).

The tax rate is a certain sum per square metre determined by the type of land/building and multiplied by 2 coefficients. The first one is set according to the population of the related municipality. The second one was introduced in 2008 and is chosen directly by the municipality itself (Czech and Slovak Federative Republic, 1992b; Czech Republic, 2007).

Another tax belonging to the system is the Road Tax. It has been paid by the owners of vehicles used for business since 1993. The tax rate for passenger cars used to be set according to the engine displacement, but this part of the tax was abolished by the amendment from 2022. For lorries, the total weight and number of axles matter (Czech Republic, 1993b, 2022b).

The last direct 3 taxes to be discussed were issued by the same law from 1993. Another common feature is that none of them are valid nowadays. The first two of them, the Inheritance Tax and the Gift Tax, were abolished in 2014. Since then, inheritance has become a tax-free income, and gifts have been incorporated into the Personal and Corporate Income Tax (Czech and Slovak Federative Republic, 1992c; Czech Republic, 2013a, 2013b).

The Inheritance Tax was paid by the heirs from the total value of their inheritance. The tax rates were progressive and also distinguished the relation of the deceased to the heirs: close relatives were subjected to lower tax rates. The same tax rates used to be applied to gifts under the Gift Tax, which was paid by the recipients. These taxes were abolished in 2014, and their subjects have been incorporated into the income tax system under which they gained tax exemption (Czech and Slovak Federative Republic, 1992c; Czech Republic, 2013a).

The Real Estate Transfer Tax was levied on real estate transfers. The taxpayer was generally the seller of the property, while the buyer acted as a guarantor. Initially, tax was calculated using the same tax rates as the Inheritance Tax and the Gift Tax, with the tax base equal to the property's value. From 2014, this Tax has been replaced by the Property Acquisition Tax. They shared basic principles but shifted the tax liability primarily to the acquirer. The tax rate was flat at 4%. This tax was abolished in 2020. Since then, real estate transfers have become tax-free (Czech and Slovak Federative Republic, 1992c; Czech Republic, 2013b, 2020a).

4.1.3 Indirect Taxes

The second part of taxation consists of indirect taxation being represented by the Value Added Tax, the Excise Duties, and the Environmental Taxes, which are essentially just special cases of the Excise Duties.

As shown in Figure 4.2, the Value Added Tax is the most profitable tax in the system. As previously mentioned, the tax is collected from intermediaries[10] but consumers ultimately pay it daily. Effective from 1993, the tax has increased the prices of most goods and services by a statutory percentage. Originally, there were 2 tax rates - 23% for most goods and some specified services and 5% for the majority of services and other specified items. The tax rates have changed multiple times since then (Czech and Slovak Federative Republic, 1992h).

One of the major changes occurred in 2004 when a new law regulating the VAT was issued. That year, the Czech Republic officially joined the EU and, therefore, had to align its legislation with European regulations. Tax rates changed to 19%[11] and 5%[12]. In 2014, a second reduced tax rate was introduced. Since then, the Value Added Tax has had three tax rates: a basic rate of 21%, a first reduced rate of 15%, and a second reduced rate of 10%. As of 2024, the second reduced tax rate has been abolished, and the first reduced rate has decreased to 12% (Czech Republic, 2004, 2014, 2023).

The Excise Duties differ from other types of taxes as they are strategically imposed on certain goods or services where the government wants to steer consumption or generate income (Zídková et al., 2024). Before the Czech Republic entered the EU, the *Act on the Excise Duties* from 1992 regulated their structure. Since 2004, a new law has been introduced. At the beginning, there were originally 5 excise duties - Tax on Hydrocarbon Fuels and Lubricants, Tax on Alcohol and Spirits, Tax on Beer, Tax on Wine, and Tax on Tobacco Products. Throughout the years, Tax on Heated Tobacco Products, Tax on Other Tobacco Products, Tax on Tobacco-Related Products, and Tax on Raw Tobacco were also introduced (Czech and Slovak Federative Republic, 1992e; Czech Republic, 2003a, 2015a, 2019, 2023).

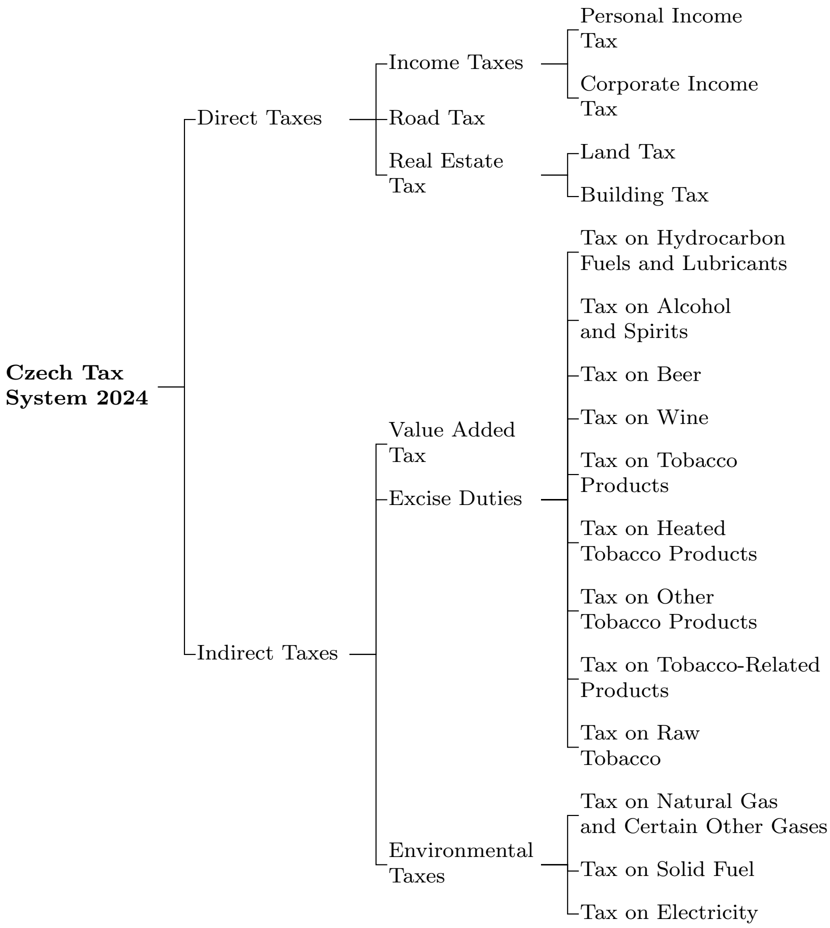
The tax rates vary depending on the specific Excise Tax. Generally, an absolute statutory amount is always added to the basic price of the goods based on their volume, quantity, etc. These taxes are paid by those who put the goods into circulation (Czech Republic, 2003a).

In 1993, the legal groundwork for Environmental Taxes was laid. Nevertheless, they were only actually introduced into the tax system in 2008. All 3 of them, Tax on Natural Gas and Certain Other Gases, Tax on Solid Fuels and Tax on Electricity, share their principles with Excise Duties, as they add an absolute amount to a certain quantity of a good. The taxpayer is the distributor of the taxable good to the public (Czech Republic, 2007).

For a visualised overview of the Czech Tax System effective at the end of 2024, see Figure 4.3.

[10] On a monthly or quarterly basis.
[11] The Basic Rate
[12] The Reduced Rate

Figure 4.3: Czech Tax System in 2024



Note: Author's own elaboration based on Czech tax laws from 1992 to 2024.

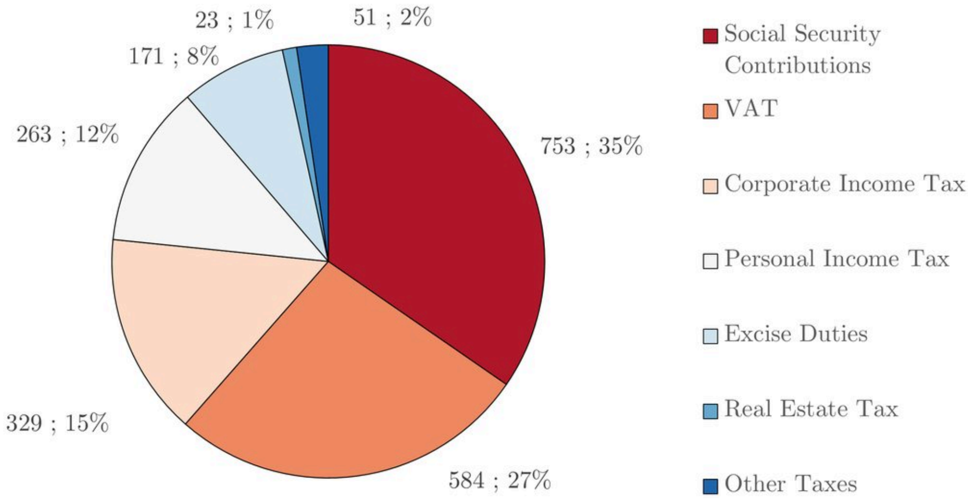
4.2 Overview of Contributions to the Social Security System

In 1993, the Social Security System underwent major changes as well. Since then, it has become an essential part of the state budget, accounting for 35% of state budget income in 2024 (Ministry of Finance of the Czech Republic, 2025). For the structure of tax revenues and social security contributions, see 4.4. As we have already mentioned, this system can be considered the second pillar of the Czech Tax System and therefore deserves a short historical review.

The origins of the current system trace back to 1992 when the *Act on Social Security Contributions and Contributions to State Employment Policy* was enacted. The organisation and the way of implementing the system had already been set the year before. Greatest responsibility was given to the *Czech Social Security Administration*, which is supervised by the *Ministry of Labour and Social Affairs* (Czech and Slovak Federative Republic, 1991, 1992f).

The System consists of 3 types of contributions: (1) Sickness Insurance Contribution, (2) Pension Contribution, and (3) State Employment Policy Contribution. Their burden is split between employees and employers. The self-employed individuals contribute at the full scale (Czech and Slovak Federative Republic, 1992f).

Figure 4.4: The Structure of Tax Revenues and Social Security Contributions (2024)(in bil. CZK & %)



Note: Author's own elaboration based on Czech National Bank (2025)

The assessment base[13] was determined as the gross salary for working individuals. Self-employed people could decide how large assessment base they want to use. It had to be at least half of their average monthly net profits and no more than 45,000 CZK per month (Czech and Slovak Federative Republic, 1992f).

The rates were originally 27 % for employers, decomposed as 3.6% as the Sickness Insurance Contribution, 20.4% as the Pension Contribution, and 3% as the State Employment Policy Contribution. Employees paid 9% broken down as 1.2%, 6.8% and 1% for the same respective contributions. The self-employed contributed the percentage of employers and employees combined (Czech and Slovak Federative Republic, 1992f).

Throughout the years, the system was affected by multiple changes. In 1995, the Pension Insurance System was overhauled. These changes mainly concerned the expenditure side of the budget, as the eligibility and pension calculations were updated. In 2004, the legislation was amended to bring it into line with EU law. The contribution regime for the self-employed was reformed, too. The law now differentiated between *primary* and *secondary* employment, each with its own assessment base (Czech Republic, 1995b, 2003b).

Starting from 2008, the Maximum Assessment Bases have been introduced, limiting the maximum yearly assessment base to 48 times the average monthly wage. The rates have evolved through the years as well. The employees' State Employment Policy Contribution was abolished in 2009, together with the Sickness Insurance Contribution. Nevertheless, the second one returned in 2024 (Czech Republic, 2007, 2008, 2023).

The employers' contribution rates had decreased to 24.8% by 2025. This rate is made up of 2.1% for Sickness Insurance Contribution, 21.5% for Pension Contribution, and 1.2% for State Employment Policy Contribution. The employees contribute 7.1% out of which 0.6% is the Sickness Insurance Contribution and 6.5% consist of Pension Contribution. The self-employed pay 29.2% of their assessment base, 28% being the

[13] Analogy to the tax base

Pension Contribution, and 1.2% State Employment Policy Contribution. If they participate in the sickness insurance program as well, they add another 2.3% from their assessment base (Czech Republic, 2015b, 2023).

Since the healthcare system is publicly provided, there must also be a health insurance program. It has been established since 1993 as well. However, funds from the Public Health Insurance Contribution go directly to insurers, and therefore, they are not a part of the state budget. The assessment base is the same as for the other contributions, the rate is 13.5%. One third is paid by the employee, two thirds by the employer. The self-employed may opt into the program and pay the whole amount. This contribution will not be considered in our later analysis, as its revenues are not a part of the government budget (Czech and Slovak Federative Republic, 1992g).

5 Estimating Laffer Curves for the Czech Republic

Until now, we have been theoretically discussing various topics relevant to our study. The forthcoming section is devoted to the analysis of the Tax System in the Czech Republic from the perspective of the Laffer Curve.

Firstly, we describe our final compiled dataset and then show how we processed it. Secondly, we delve into the adjusted data and look for obvious issues. Thirdly, we develop multiple models using the Ordinary Least Squares (OLS) methodology, which we also present. In the next part, we challenge some of the assumptions from the OLS analysis, and the Instrumental Variable Regression (IVR) is used to adjust for issues in our previous models. Finally, we improve our models by re-estimating them using the Generalised Method of Moments (GMM), which creates the most robust findings.

For the models, a short comment on the most important aspects will be provided. A deeper discussion about the results is situated in Section 6.

5.1 Data Description

Our dataset uses quarterly data from 1996 Q1 to 2023 Q4. Because of the nature[14] of the data, the analysis will be performed for 2 different periods, firstly from 1996 Q1 to 2023 Q4 and secondly from 2001 Q1 to 2023 Q4.

Tables 5.1, 5.2, and 5.3 present basic descriptive statistics of the processed variables in our dataset. All statistics are computed from quarterly, seasonally adjusted values. Variables expressed in Czech Koruna (CZK) were adjusted for inflation, with 2015 being the baseline. The data manipulation process is described in the next section, Data Processing.

Table 5.1: Descriptive Statistics for the Initial Data from 1996 Q1 to 2023 Q4

Statistic	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
TTR	284,560	59,265	180,877	242,721	329,637	384,848
TR	185,929	43,988	105,051	154,527	217,735	255,114
SSC	98,631	16,291	72,400	86,383	109,488	130,470
PITR	24,788	7,052.800	11,132	21,714	28,734	39,810
CITR	25,649	7,337.600	11,229	20,883	29,412	60,513
GDP	1,035,792	201,476	734,075	826,726	1,191,488	1,338,953
TG	0.360	0.101	0.132	0.296	0.426	0.696
u	0.056	0.022	0.018	0.035	0.073	0.093
PITrate	0.129	0.033	0.083	0.101	0.167	0.177
CITrate	0.243	0.068	0.190	0.190	0.310	0.390
Trate	0.274	0.020	0.233	0.260	0.285	0.317
g	0.022	0.032	−0.108	0.002	0.042	0.092

Note: Author's own elaboration based on collected data. $TTR = (TR + SSC)$... Total Tax Revenue, TR... Tax Revenue, SSC... Social Security Contributions, $PITR$... Personal Income Tax Revenue, $CITR$... Corporate Income Tax Revenue, GDP... Gross Domestic Product, TG... Trust in the Government, u... Unemployment Rate, $PITrate$... Average Effective Personal Income Tax Rate, $CITrate$... Corporate Income Tax Rate, $Trate$... Average Overall Tax Rate, g... Annual GDP Growth Rate, $TTR - GDP$ in mil. CZK.

[14] Presence of a structural break.

Table 5.2: Descriptive Statistics for the Initial Data from 2001 Q1 to 2023 Q4

Statistic	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
TTR	304,392	45,165.000	188,559	278,012	337,500	384,848
TR	200,875	32,878.000	109,679	182,878	225,288	255,114
SSC	103,517	13,694.000	78,881	94,833	114,517	130,470
PITR	27,362	4,794.500	17,912	24,158	29,683	39,810
CITR	27,235	6,949.100	12,258	23,746	30,208	60,513
GDP	1,098,417	165,088.000	795,974	1,022,589	1,264,013	1,338,953
TG	0.357	0.104	0.132	0.295	0.426	0.696
u	0.054	0.022	0.018	0.030	0.073	0.085
PITrate	0.121	0.031	0.083	0.097	0.133	0.177
CITrate	0.218	0.044	0.190	0.190	0.240	0.310
Trate	0.278	0.019	0.237	0.263	0.290	0.317
g	0.024	0.033	−0.108	0.008	0.047	0.092

Note: Author's own elaboration based on collected data. $TTR = (TR + SSC)$... Total Tax Revenue, TR... Tax Revenue, SSC... Social Security Contributions, $PITR$... Personal Income Tax Revenue, $CITR$... Corporate Income Tax Revenue, GDP... Gross Domestic Product, TG... Trust in the Government, u... Unemployment Rate, $PITrate$... Average Effective Personal Income Tax Rate, $CITrate$... Corporate Income Tax Rate, $Trate$... Average Overall Tax Rate, g... Annual GDP Growth Rate, $TTR - GDP$ in mil. CZK.

Table 5.3: Descriptive Statistics for the Initial Data from 1996 Q1 to 2000 Q4

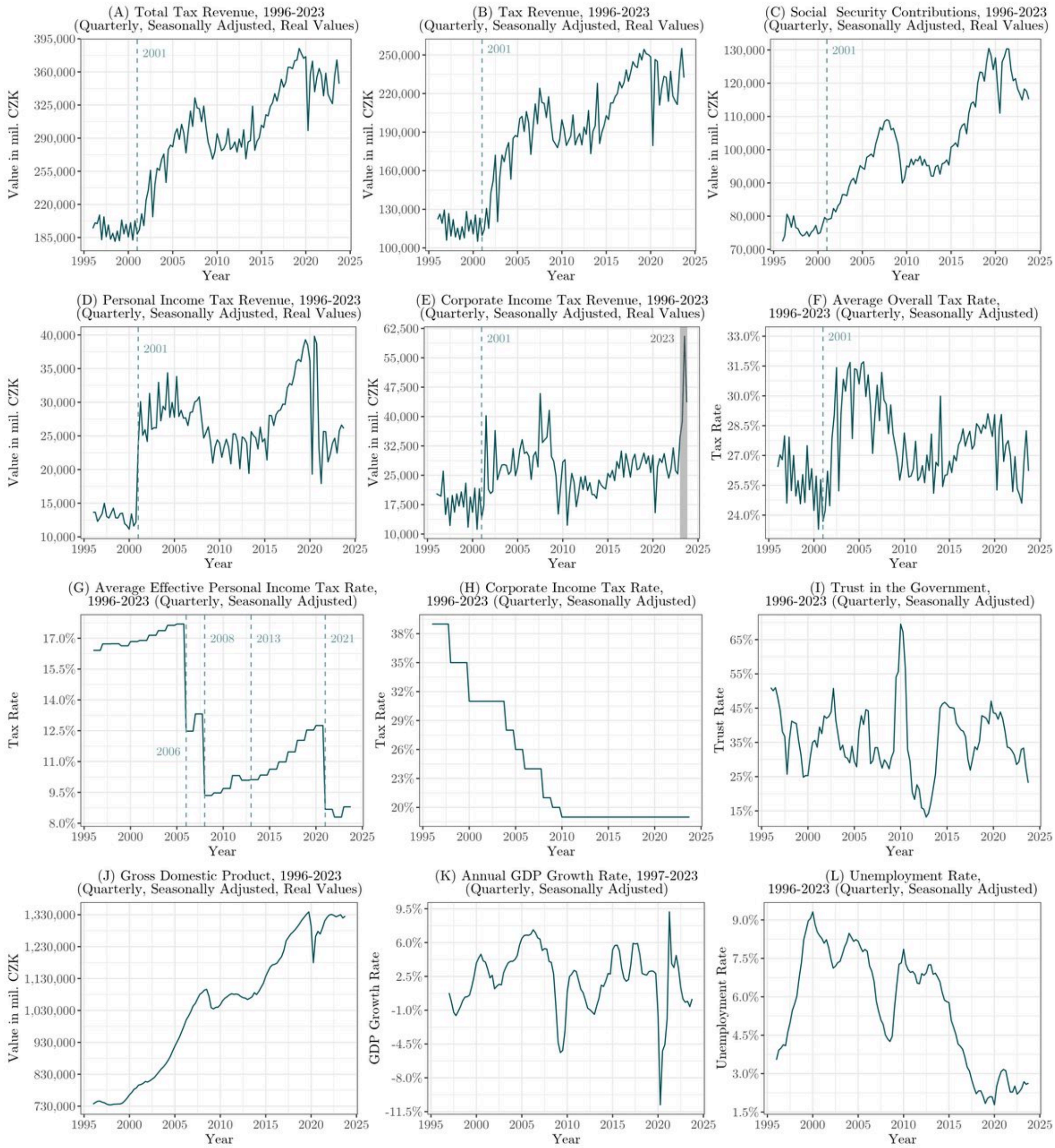
Statistic	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
TTR	193,331	9,184.800	180,877	184,930	200,516	208,733
TR	117,174	8,307.900	105,051	109,039	123,825	129,527
SSC	76,157	2,263.700	72,400	74,607	77,244	80,582
PITR	12,949	947.290	11,132	12,269	13,507	14,989
CITR	18,357	3,887.100	11,229	15,576	20,422	26,062
GDP	747,717	16,339.000	734,075	736,613	749,250	786,658
TG	0.373	0.086	0.248	0.312	0.420	0.510
u	0.065	0.021	0.035	0.045	0.085	0.093
PITrate	0.167	0.001	0.164	0.166	0.167	0.168
CITrate	0.358	0.031	0.310	0.350	0.390	0.390
Trate	0.259	0.013	0.233	0.249	0.268	0.280
g	0.012	0.021	−0.016	−0.002	0.029	0.048

Note: Author's own elaboration based on collected data. $TTR = (TR + SSC)$... Total Tax Revenue, TR... Tax Revenue, SSC... Social Security Contributions, $PITR$... Personal Income Tax Revenue, $CITR$... Corporate Income Tax Revenue, GDP... Gross Domestic Product, TG... Trust in the Government, u... Unemployment Rate, $PITrate$... Average Effective Personal Income Tax Rate, $CITrate$... Corporate Income Tax Rate, $Trate$... Average Overall Tax Rate, g... Annual GDP Growth Rate, $TTR - GDP$ in mil. CZK.

Part of the original Tax Revenue (TR), Social Security Contributions (SSC), Personal Income Tax Revenue (PITR), and Corporate Income Tax Revenue (CITR) values were found in Czech National Bank (2025). Values for these variables from 1995 to 2000 were not publicly available. Still, the Ministry of Finance provided us with the data in private communication.[15] The GDP data was obtained from the Czech Statistical Office (2025a, 2025b) together with the data on the Unemployment Rate (u). The data on the Trust in the Government (TG) were provided by the Public Opinion Research Center (2022) and later Public Opinion Research Center press releases.

The time development of our processed key variables is shown in Figure 5.1. The same plots, but scaled up, can be found in the Appendix Section A.2. For comments on the data structure and progression, see Section 5.3. The intuition for including those selected variables follows in Table 5.4.

Figure 5.1: Development of our Key variables



Note: Author's own elaboration based on data provided by the institutions mentioned in Section 5.1 and on variables calculated by the procedures described in Section 5.2.

[15]To see the English translation of the received tables, see the Data Provided by the Czech Ministry of Finance in Appendix.

Table 5.4: Intuition for Including Selected Variables

Variable	Intuition
TTR	Dependent variable for some of our models.
PITR	Dependent variable for some of our models.
CITR	Dependent variable for some of our models.
TG	Higher trust in the government could lead to increased confidence that the money will be well spent, and therefore less tax evasion.
u	Higher Unemployment Rate means fewer tax-payers for PIT, and therefore a smaller possible tax revenue.
PITrate	Higher tax rate would standardly lead to higher revenues, for the opposite effect, squared terms are used.
CITrate	Higher tax rate would standardly lead to higher revenues, for the opposite effect, squared terms are used.
Trate	Higher tax rate would standardly lead to higher revenues, for the opposite effect, squared terms are used.
g	Annual GDP Growth Rate leads to possible larger tax bases for some taxes and therefore larger tax revenues.

Note: $TTR = (TR + SSC)$... Total Tax Revenue, $PITR$... Personal Income Tax Revenue, $CITR$... Corporate Income Tax Revenue, TG... Trust in the Government, u... Unemployment Rate, $PITrate$... Average Effective Personal Income Tax Rate, $CITrate$... Corporate Income Tax Rate, $Trate$... Average Overall Tax Rate, g... Annual GDP Growth Rate, $TTR - CITR$ in mil. CZK.

5.2 Data Processing

The obtained raw data required preprocessing as they were in multiple formats, with various frequencies and in non-comparable terms. Let us start with the adjustments made to tax-related variables, TR, SSC, PITR, CITR, which were supplied at a monthly frequency. The forthcoming approach holds for each mentioned variable; therefore, let $\mathbf{x}$ denote the vector of variables affected by this adjustment.

$$\mathbf{x} = (x_1, x_2, x_3, x_4), \quad \text{where} \begin{cases} x_1 = \text{TR} \\ x_2 = \text{SSC} \\ x_3 = \text{PITR} \\ x_4 = \text{CITR} \end{cases} \tag{5.1}$$

At first, it is necessary to disaggregate the cumulative monthly data into individual monthly data.[16] Let $\mathbf{C}_{m,y}$ denote a matrix of values for a variable from $\mathbf{x}$, where $\mathbf{m}$ is the vector of numbers representing a month in the year and $\mathbf{y}$ is the vector of years.

$$\mathbf{m} = (1, 2, \ldots, 12) \tag{5.2}$$

$$\mathbf{y} = (1995, 1996, \ldots, 2024) \tag{5.3}$$

[16]Values from 1995 Q1 to 2024 Q4 were utilised to process TR, SSC, PITR, and CITR data.

Therefore the matrix $\mathbf{C}_{my}$ can be visualised as

$$\mathbf{C}_{m,y} = \begin{bmatrix} c_{1,1995} & c_{1,1996} & \cdots & c_{1,2024} \\ c_{2,1995} & c_{2,1996} & \cdots & c_{2,2024} \\ \vdots & \vdots & \ddots & \vdots \\ c_{12,1995} & c_{12,1996} & \cdots & c_{12,2024} \end{bmatrix} . \tag{5.4}$$

To obtain the matrix of individual monthly values, $\mathbf{I}_{m,y}$, we must compute its elements. Since the original values are cumulative, we subtract the cumulative sum from the previous month from the current month's cumulative value to find the true value for a specific month,

$$i_{m,y} = \begin{cases} c_{m,y} & \forall m = 1 \\ c_{m,y} - c_{m-1,y} & \forall m \in [2, 12] . \end{cases} \tag{5.5}$$

Adding them together forms a new matrix I,

$$\mathbf{I}_{m,y} = \begin{bmatrix} i_{1,1995} & i_{1,1996} & \cdots & i_{1,2024} \\ i_{2,1995} & i_{2,1996} & \cdots & i_{2,2024} \\ \vdots & \vdots & \ddots & \vdots \\ i_{12,1995} & i_{12,1996} & \cdots & i_{12,2024} \end{bmatrix} . \tag{5.6}$$

This procedure is applied to all x_i in $\mathbf{x}$. Once we have monthly data for all of our variables, we can transform them from nominal to real values. To do so, we use the monthly Consumer Price Indexes (CPIs) from 1995 to 2024 provided by the Czech Statistical Office (2025c, 2025d).[17] Let $\mathbf{R}_{m,y}$ denote a new matrix with the real monthly values, such as

$$\mathbf{R}_{m,y} = \begin{bmatrix} r_{1,1995} & r_{1,1996} & \cdots & r_{1,2024} \\ r_{2,1995} & r_{2,1996} & \cdots & r_{2,2024} \\ \vdots & \vdots & \ddots & \vdots \\ r_{12,1995} & r_{12,1996} & \cdots & r_{12,2024} \end{bmatrix} . \tag{5.7}$$

The real values of each element in the matrix for all x_i in $\mathbf{x}$ are therefore calculated as

$$r_{m,y} = \frac{i_{m,y}}{CPI_{m,y}} \times 100 . \tag{5.8}$$

We intend to use this data together with quarterly data on GDP and u. A new matrix $\mathbf{Q}_{q,y}$ with q representing the vector of quarters in a year, and y representing the vector of years, contains quarterly values created from $\mathbf{R}_{m,y}$

$$\mathbf{Q}_{q,y} = \begin{bmatrix} q_{1,1995} & q_{1,1996} & \cdots & q_{1,2024} \\ q_{2,1995} & q_{2,1996} & \cdots & q_{2,2024} \\ q_{3,1995} & q_{3,1996} & \cdots & q_{3,2024} \\ q_{4,1995} & q_{4,1996} & \cdots & q_{4,2024} \end{bmatrix} , \tag{5.9}$$

where:

$$q_{1,y} = \sum_{m=1}^{3} r_{m,y} , \tag{5.10}$$

[17] The CPIs use the average of 2015 as a baseline; we take over this approach.

$$q_{2,y} = \sum_{m=4}^{6} r_{m,y} \ , \tag{5.11}$$

$$q_{3,y} = \sum_{m=7}^{9} r_{m,y} \ , \tag{5.12}$$

$$q_{4,y} = \sum_{m=10}^{12} r_{m,y} \ . \tag{5.13}$$

Special adjustments must be made to the Trust in the Government. Since reported values are neither on a monthly nor a quarterly basis,[18] we use the latest observed value for each of the following months with unobserved values. Then we calculate the mean of the values in a quarter to obtain the final values for each quarter.

Let TG_t denote the vector of values of the Trust in the Government (TG), where $t \in (1, 2, \ldots, n)$ represents months between 1996 Q1 and 2023 Q4 and $O \subset (1, 2, \ldots, n)$ represents the subset of months for which we have observed TG values, then

$$TG_t = TG_{t'} \ , \tag{5.14}$$

with t' being the most recent point before t with an observed value,

$$t' = \max(j \in O \mid j \leq t) \ , \tag{5.15}$$

where j represents any month index in the set O of months with observed values. For quarterly data TG_q, these equations hold,

$$TG_{q=1} = \frac{1}{3} \sum_{i=1}^{3} m_i \ , \tag{5.16}$$

$$TG_{q=2} = \frac{1}{3} \sum_{i=4}^{6} m_i \ , \tag{5.17}$$

$$TG_{q=3} = \frac{1}{3} \sum_{i=7}^{9} m_i \ , \tag{5.18}$$

$$TG_{q=4} = \frac{1}{3} \sum_{i=10}^{12} m_i \ , \tag{5.19}$$

where m_i denotes the value in a month i.

At this point, we have TR, SSC, PITR, CITR in real values and on a quarterly basis. Our dataset also contains quarterly data in percentage terms on u and TG. GDP is so far the only seasonally adjusted variable.

[18]They are reported once in a couple of months.

5.2.1 Seasonal Adjustment

Seasonal patterns are redundant for our later analysis; therefore, we intend to eliminate them. We follow the approach described by Hyndman and Athanasopoulos (2021).

> The additive decomposition is the most appropriate if the magnitude of the seasonal fluctuations, or the variation around the trend-cycle, does not vary with the level of the time series. When the variation in the seasonal pattern, or the variation around the trend-cycle, appears to be proportional to the level of the time series, then a multiplicative decomposition is more appropriate. Multiplicative decompositions are common with economic time series (Hyndman & Athanasopoulos, 2021).

Based on the behaviour of our time series, multiplicative decomposition is utilised for TR, SSC, PITR, CITR. On the contrary, we use additive decomposition for TG and u. The data on GDP have already been seasonally adjusted by the Czech Statistical Office.

Assuming a multiplicative decomposition, we can write,

$$y_t = S_t \times T_t \times R_t \ , \tag{5.20}$$

where y_t is our time series with the data, S_t is the seasonal component, T_t is the trend component and R_t is the remainder component, all at time t. When adjusting for seasonality, we divide the whole equation by S_t

$$\frac{y_t}{S_t} = \frac{S_t \times T_t \times R_t}{S_t} = T_t \times R_t \ . \tag{5.21}$$

Alternatively, the additive decomposition can be written as

$$y_t = S_t + T_t + R_t \ . \tag{5.22}$$

To remove the seasonal component, we subtract it from the equation,

$$y_t - S_t = S_t + T_t + R_t - S_t = T_t + R_t \ . \tag{5.23}$$

These approaches are sequentially applied to all relevant variables mentioned. From now on, when referring to TR, SSC, PITR, CITR, GDP, and u, we mean these adjusted time series unless otherwise specified.

5.2.2 Calculation of the Rest of Our Variables

In Data Description, we have already described the calculated variables which were not retrieved from public databases. This section provides the procedure for calculating the Total Tax Revenue (TTR), the Average Overall Tax Rate (Trate), the PITrate, the Corporate Income Tax Rate (CITrate), and the Annual GDP Growth Rate (g).

TTR represents the tax revenues from all taxes and contributions to the social security system. Therefore, they can be expressed as

$$TTR_t = TR_t + SSC_t \ , \tag{5.24}$$

where t denotes the period at which the sum is calculated.

The corresponding tax rate, Trate, is calculated as follows and represents the average tax burden at time t. A similar approach for obtaining the tax rate can be seen in Clausing (2007) and Karas (2013),

$$Trate_t = \frac{TTR_t}{GDP_t} \ . \tag{5.25}$$

Annual GDP Growth Rate (g) represents the year-on-year change in the quarterly real GDP and therefore is provided by the following expression

$$g_t = \frac{GDP_t - GDP_{t-4}}{GDP_{t-4}} \ . \tag{5.26}$$

The Corporate Income Tax Rates (CITrates) are provided by the effective laws for each observed year, as the basic tax rate for each year has been retrieved from the law text.

The Average Effective Personal Income Tax Rates (PITrates) cannot be directly taken from the law, as over the years, the tax bases have changed, progressivity has been altered, and a basic tax credit came into effect. Because of those reasons, we calculate the PITrates from the data on wages and the number of employees across the sectors in the Czech Republic. The data was retrieved from the Czech Statistical Office (2024a, 2024b, 2024c, 2024d).[19]

First, we calculate the effective tax rate for each group in a given year by using the average wage as the tax base. For years, when the super-gross wage as a tax base has been effective, we have used the super-gross wage. We obtain it by adding the employers' contributions to social security to the basic wage. Then we subject it to the tax rate[20] and subtract the applicable tax credit. This final tax is divided by the original average wage in order to obtain a percentage rate. This procedure repeats for each group and each year.[21]

Mathematically, let $\mathbf{W_{s,y}}$ be a matrix of wages, where s denotes a sector of the economy and y denotes the relevant year. The total amount of sectors is noted as σ. $\mathbf{L_{s,y}}$ is the matrix containing the number of workers in sector s and year y,

$$\mathbf{W}_{s,y} = \begin{bmatrix} w_{1,1996} & w_{1,1997} & \cdots & w_{1,2023} \\ w_{2,1996} & w_{2,1997} & \cdots & w_{2,2023} \\ \vdots & \vdots & \ddots & \vdots \\ w_{\sigma,1996} & w_{\sigma,1997} & \cdots & w_{\sigma,2023} \end{bmatrix} , \tag{5.27}$$

$$\mathbf{L}_{s,y} = \begin{bmatrix} l_{1,1996} & l_{1,1997} & \cdots & l_{1,2023} \\ l_{2,1996} & l_{2,1997} & \cdots & l_{2,2023} \\ \vdots & \vdots & \ddots & \vdots \\ l_{\sigma,1996} & l_{\sigma,1997} & \cdots & l_{\sigma,2023} \end{bmatrix} , \tag{5.28}$$

$$ETR_{s,y} = \frac{\sum_{j=1}^{B_y} \tau_{j,y} \times \min(\max(TB_{s,y} - b_{j-1,y}, 0), b_{j,y} - b_{j-1,y}) - TC_y}{w_{s,y}} , \tag{5.29}$$

[19]The data overlap from 2005 to 2009, for this overlapping period, data for 2005-2023 is used.
[20]Either linear or progressive, if effective.
[21]The table that was utilised for tax rates, tax credits and brackets can be found in A.3.

where

$$TB_{s,y} = \begin{cases} w_{s,y} & \text{if standard wage is the tax base} \\ w_{s,y} \times (1 + c_y) & \text{if super-gross wage is the tax base} \end{cases} . \tag{5.30}$$

For the explanation of variables in equations (5.29) and (5.30), see Table 5.5.

Table 5.5: Explanation of Variables in Equation (5.29)

Variable	Explanation
$ETR_{s,y}$	Effective Tax Rate for sector s in year y
$TB_{s,y}$	Tax Base for sector s in year y
$TC_{s,y}$	Personal Tax Credit for sector s in year y
$\tau_{j,y}$	Tax Ratefor bracket j in year y
$b_{j,y}$	Upper Cap of bracket j in year y
$b_{j-1,y}$	Lower Cap of bracket j in year y
B_y	Number of brackets in year y
c_y	Employers' part of contributions to social security

Tax Rate $\tau_{j,y}$ can be found in the relevant text of the law for year y. To get the Average Effective Personal Income Tax Rates, we compute the weighted average of ETR_y by using the number of workers in the relevant sector.

$$PITrate_y = \frac{\sum_{s=1}^{\sigma} n_{s,y} \times ETR_{s,y}}{\sum_{s=1}^{\sigma} n_{s,y}} \tag{5.31}$$

The PITrates and CITrates described in this part are calculated or retrieved at a yearly frequency. We therefore apply the annual $PITrate_y$ value to each quarter q within year y to complete our dataset.

The analysis might use alternative forms of our variables (e.g., logarithms, lagged values or differences). In such cases, the form of the variable will be clearly stated. Now that we have described the data and its origin, we can move to its analysis.

5.3 Issues with the Data

Tracking the evolution of the Time Series (TS) of our variables raises several questions. The first one is related to a surprising increase in TTR, TR, PITR, and CITR after 2001 Q1. This increase cannot be explained by any change in the methodology or a change in the tax rate. We believe this deviation happened because of the introduction of the *Act on Budgetary Determination of Taxes*.

This Act permanently stated how the collected taxes will be distributed over public budgets (Czech Republic, 2000). Therefore, we believe the data provided by the Czech Ministry of Finance accounted only for the state budget before 2001, and they have accounted for all public budgets since 2001. Therefore, we introduce a binary variable *partial*, such as

$$partial_t = \begin{cases} 1 & \forall t < 2001\,Q1 \\ 0 & \forall t \geq 2001\,Q1 \end{cases} , \tag{5.32}$$

where t denotes the time period in which the variable is observed.

The Chow test for a structural break at this point further provides evidence for the existence of a structural break. The difference in β_i between the periods before and after the break, which is tested by this test, was found statistically significant for developed models at the 5% significance level. For further information about the test principle, see Stock and Watson (2020) and Wooldridge (2013).

There is also an atypical increase for PITR in all quarters of 2023. We attribute this increase to the introduction of the windfall tax (Czech Republic, 2022a). Therefore we create a dummy variable to control for this spike,

$$windfall_t = \begin{cases} 1 & \forall t \in [2023\,Q1, 2023\,Q4] \\ 0 & \forall t \notin [2023\,Q1, 2023\,Q4] \end{cases}. \tag{5.33}$$

A quite unusual evolution is observable for Average Effective Personal Income Tax Rate. We marked some important policy changes that affected the calculation and the final size of the tax rate. The first one is traced back to 2006, where a first form of the personal tax credit was introduced. Another drop occurred in 2008, when the flat tax rate came into force. From 2013 to 2021, a solidarity increase was applied, after which the tax officially reverted to progressivity. Since 2021, the tax base has been determined as gross wage and not super-gross wage, as it was from 2008 to 2020. This explains the last drop in data (Czech Republic, 2005, 2007, 2012, 2020b).

5.3.1 Stationarity

Lastly, we have to address stationarity. TS is stationary when its probability distribution is constant over time (Stock & Watson, 2020). The possible stationarity of some variables raises suspicions at first glance at the graphs.

For determining whether our data are stationary, we use 3 tests suited for testing stationarity. Two of them, Augmented Dickey-Fuller Test (ADF) and Phillips-Perron Test (PP) use non-stationarity as their null hypothesis, the third one, Kwiatkowski–Phillips–Schmidt–Shin Test (KPSS), uses stationarity as its null hypothesis. Test statistics whose absolute value is greater than the absolute value of the critical values are in favour of rejecting the null hypothesis.

Tables 5.6 and 5.7 provide test statistics and critical values for each of the mentioned variables. They also show the decision about stationarity made based on comparing the statistic to the critical value. Because of the structural break in 2001 Q1, we performed the test for 2 versions of our original TS for the variables affected by the issue with the data. Non-stationarity might be often observed from the graph as well - see Figure 5.1. For more information about stationarity and its testing, see Dickey and Fuller (1979), Kwiatkowski et al. (1992), Phillips and Perron (1988), Stock and Watson (2020), and Wooldridge (2013).

Table 5.6: Stationarity Test Results for the period 1996-2023

Comp	**Var**	**ADF**			**KPSS**			**PP**		
		TST	C	S/N	TST	C	S/N	TST	C	S/N
I,T	TTR	-2.51	-3.43	N	0.182	0.146	N	-3.95	-3.45	S
I,T	TR	-3.04	-3.43	N	0.225	0.146	N	-5.15	-3.45	S
I,T	SSC	-2.18	-3.43	N	0.134	0.146	S	-2.26	-3.45	N
I,T	PITR	-3.02	-3.43	N	0.199	0.146	N	-3.68	-3.45	S
I,T	CITR	-3.33	-3.43	N	0.172	0.146	N	-6.47	-3.45	S
I,T	Trate	-2.87	-3.43	N	0.201	0.146	N	-5.45	-3.45	S
I,T	PITrate	-2.03	-3.43	N	0.247	0.146	N	-2.07	-3.45	N
I,T	CITrate	-0.99	-3.43	N	0.557	0.146	N	-0.81	-3.45	N
I	TG	-3.93	-3.43	S	0.068	0.146	S	-3.66	-3.45	S
I,T	GDP	-2.36	-3.43	N	0.135	0.146	S	-2.39	-3.45	N
I	g	-3.73	-3.43	S	0.135	0.146	S	-3.58	-3.45	S
I,T	u	-2.98	-3.43	N	0.284	0.146	N	-3.01	-3.45	N
I	Trate^2	-2.89	-2.88	S	0.202	0.463	S	5.30	-2.89	S
I,T	PITrate^2	-1.97	-3.43	N	0.263	0.146	N	-2.01	-3.45	N
I,T	CITrate^2	-1.46	-3.43	N	0.552	0.146	N	-1.25	-3.45	N
I,T	ln(TTR)	2.08	-3.43	N	0.260	0.146	N	-3.53	-3.45	S
I,T	ln(PITR)	-2.76	-3.43	N	0.254	0.146	N	-3.14	-3.45	N
I,T	ln(CITR)	-3.77	-3.43	S	0.178	0.146	N	-7.35	-3.45	S

Note: All tested on 5% significance level. I... Constant included for the purpose of the test, T... Trend included for the purpose of the test, TST... Test Statistic, C... Critical Value, S/N... Stationary/Non-stationary Decision.

Table 5.7: Stationarity Test Results for the period 2001-2023

Comp	**Var**	**ADF**			**KPSS**			**PP**		
		TST	C	S/N	TST	C	S/N	TST	C	S/N
I,T	TTR	-3.18	-3.45	N	0.149	0.146	N	-4.01	-3.46	S
I,T	TR	-3.98	-3.45	S	0.150	0.146	N	-5.30	-3.46	S
I,T	SSC	-2.03	-3.45	N	0.205	0.146	N	-1.99	-3.46	N
I,T	PITR	-3.52	-3.45	S	0.186	0.146	N	-4.66	-3.46	S
I,T	CITR	-3.60	-3.45	S	0.224	0.146	N	-5.68	-3.46	S
I,T	Trate	-4.27	-3.45	S	0.145	0.146	S	-5.45	-3.46	S
I	Trate^2	-3.38	-2.89	S	0.58	0.463	N	4.82	-2.89	S
I,T	ln(TTR)	-3.50	-3.45	S	0.16	0.146	N	-4.08	-3.46	S
I,T	ln(PITR)	-3.71	-3.45	S	0.19	0.146	N	-4.99	-3.46	S
I,T	ln(CITR)	-4.51	-3.45	S	0.22	0.146	N	-6.15	-3.46	S

Note: All tested on 5% significance level. I... Constant included for the purpose of the test, T... Trend included for the purpose of the test, TST... Test Statistic, C... Critical Value, S/N... Stationary/Non-stationary Decision.

After the review of our test results, we can say that most of our original TS are non-stationary. Stationary are definitely g, TG and possibly also Trate^2. For multiple variables, an interesting situation occurs as some tests in a certain period indicate stationarity, while some others indicate otherwise. This question is crucial for the logarithmic form of TTR, PITR, CITR, as they seem stationary in the shorter TS without the structural break.

The theoretical explanation could be that both tax rates and tax revenues tend to return to some "equilibrium level" as either an optimal tax rate or a revenue-maximising tax rate exists and the governments, in addressing contemporary demands, issue such policies, which return both the tax rates and the tax revenues to the equilibrium level, ceteris paribus. At the same time, there is some potential output in the economy, which limits the possible growth of the tax revenues as well. For the case of the Unemployment Rate, it should naturally return to the natural Unemployment Rate unless it changes over time as well.

This would lead us to assume trend-stationarity for our mentioned variables, including the Unemployment Rate. This allows us to build models in levels but under a very strong assumption of stationarity that is just partly supported by the data. Thus, we will develop another set of models that will solve for stationarity even though the interpretation might change.

One way to solve non-stationarity is to differentiate the variables. This is the simplest approach, though it results in a loss of information about the levels, and the interpretation of the model crucially changes. Since we want to estimate the revenue-maximising tax rate for the Czech tax system, this approach would stop us in our efforts. Nevertheless, it is suitable for addressing the issue with the Unemployment Rate. Therefore, we define a new variable, D_{u_t}, which represents the annual percentage change[22] in the Unemployment Rate,

$$D_{u_t} = \frac{u_t - u_{t-4}}{u_{t-4}}. \tag{5.34}$$

Another way to control for stationarity is testing for cointegration and, if possible, developing an Error Correction Model (ECM).

> Suppose that X_t and Y_t are integrated of order one. If, for some coefficient θ, $Y_t - \theta X_t$ is integrated of order zero, then X_t and Y_t are said to be cointegrated. The coefficient θ is called the cointegrating coefficient. If X_t and Y_t are cointegrated, then they have the same, or a common, stochastic trend. Computing the difference $Y_T - \theta X_t$ eliminates this common stochastic trend (Stock & Watson, 2020, p. 664).

Table 5.8: Results of the Engle–Granger Augmented Dickey–Fuller test

$Y =$ $\ln(Y) =$	**TTR** $\alpha + \theta_1 Trate +$ $+\theta_2 partial + z_t$			**PITR** $\alpha + \theta_1 PITrate +$ $+\theta_2 PITrate^2 +$ $+\theta_3 partial + z_t$			**CITR** $\alpha + \theta_1 CITrate +$ $+\theta_2 CITrate^2 +$ $+\theta_3 partial + z_t$		
	TST	C	COI	TST	C	COI	TST	C	COI
1996-2023	-1.387	-3.80	NO	-5.209	-4.16	YES	-4.286	-4.16	YES
2001-2023	-2.413	-3.41	NO	-4.674	-3.80	YES	-4.863	-3.80	YES

Note: All tested on 5% significance level. Binary variable *partial* utilised only for testing in the longer period. TST... test statistic, C... Critical Value, COI... Information if the variables are cointegrated.

[22] Normally, the method uses simple first differences. We use the annual growth rate for interpretation reasons as they make more sense in context of tax-policy changes.

To test for cointegration, the Engle–Granger Augmented Dickey–Fuller test, described by Engle and Granger (1987), is often employed. It regresses potentially cointegrated variables and then subjects their residuals to the ADF test. If the test result proves to be statistically significant, the subjected TS are supposed to be cointegrated. Table 5.8 presents the result of testing several of our models.

To obtain relevant test statistics, we included only the non-stationary variables with the control variable, *partial*, to adjust for the observed structural break. This test showed us that we can develop an ECM only for the analysis of the PITR and the CITR. Another issue arises as some of the crucial regressors for the CITR in the tested equation prove statistically insignificant without adding other control variables. This could raise questions about the validity of potential ECM models. Broadening the test for cointegration by other variables is impossible, as the test formula should contain only the non-stationary variables.

These findings prevent us from constructing an ECM model for the TTR because of no cointegration and from constructing an ECM model for the CITR because of the insignificance. We also aim to use the same methods for all 3 analysed constructs.

Therefore, we apply the differencing method described earlier to stationarize the variables and remove the possibility of spurious regression from the model. This will stop us from finding the revenue-maximising tax rate for the overall system, as the interpretation will change. Nevertheless, it will allow us to observe the dynamics of the system and find the change in the tax rate that would accelerate the growth in the tax revenue the most.[23]

Our new variables are then calculated as

$$D_{x_{i,t}} = \frac{x_{i,t} - x_{i,t-4}}{x_{i,t-4}}, \tag{5.35}$$

for

$$\mathbf{x} = (x_1, x_2, x_3, x_4, x_5, x_6), \quad where \begin{cases} x_1 = TTR \\ x_2 = PITR \\ x_3 = CITR \\ x_4 = Trate \\ x_5 = PITrate \\ x_6 = CITrate \\ x_7 = TG \end{cases}. \tag{5.36}$$

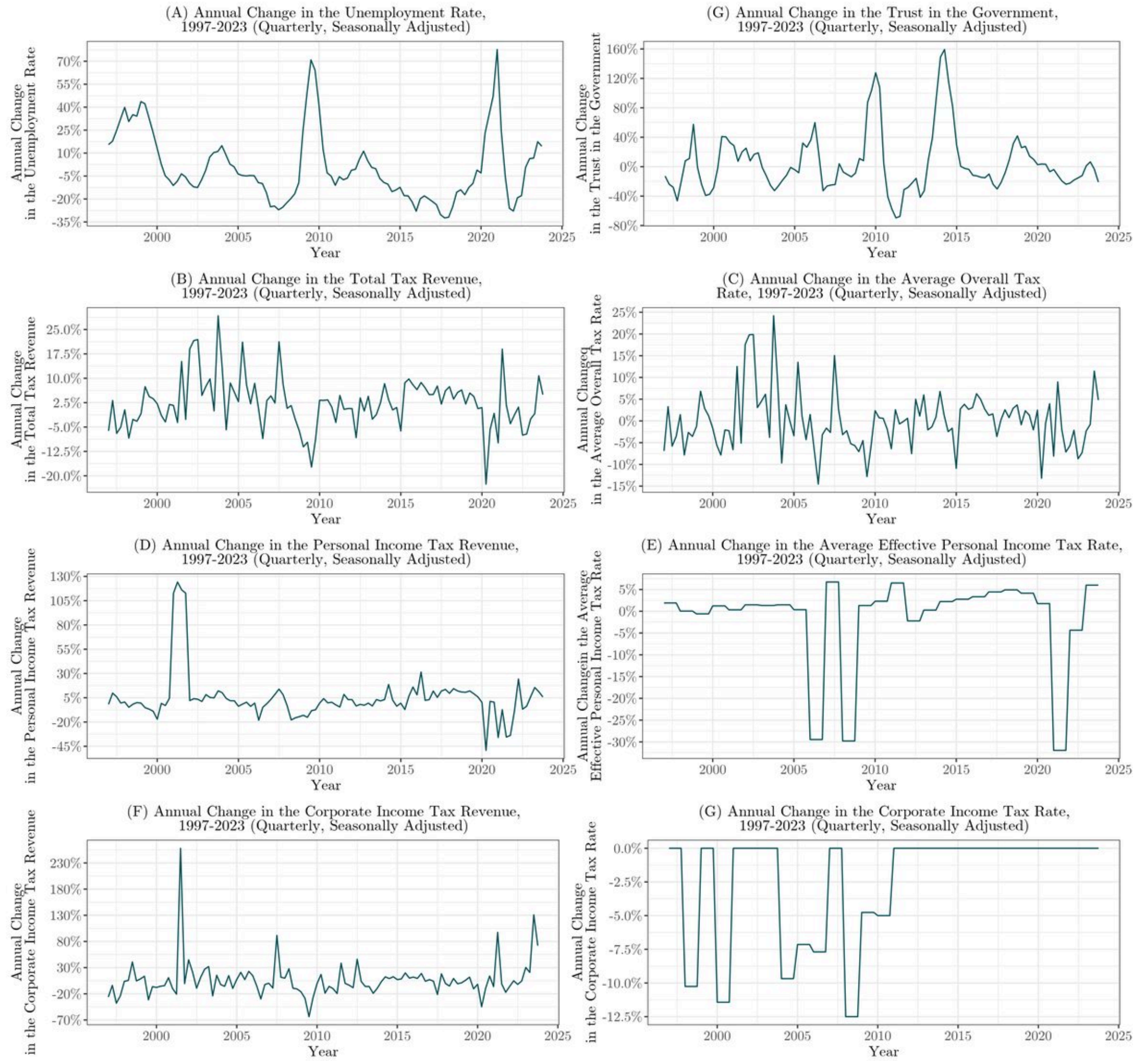
The standard approach for the differencing method would be to calculate the first differences. We used annual percentage growth rates, as their economic interpretation is more convenient.

The time evolution of variables, D_{u_t}, D_{TG_t} D_{Trate_t}, D_{TTR_t}, $D_{PITrate_t}$, D_{PITR_t}, $D_{CITrate_t}$ and D_{CITR_t}, is observable in Figure 5.2.[24] As a squared term of $D_{x_{i,t}}$, we will use $(D_{x_{i,t}})^2$. Formal tests of stationarity for the new variables are in Tables 5.9 and 5.10.

[23] We also create a growth rate for the Trust in the Government as we believe the growth rate of tax revenue would depend more on the change of trust than on the level of trust.

[24] For scaled up plots, see Appendix Section A.2.

Figure 5.2: Development of our New Growth Rates



Note: Author's own elaboration based on data provided by the institutions mentioned in Section 5.1 and on variables calculated by the procedures described in Section 5.2 and 5.3.1.

Table 5.9: Stationarity Test Results for the Growth Rates in the period 2001-2023

Comp	**Var**	**ADF**			**KPSS**			**PP**		
		TST	C	S/N	TST	C	S/N	TST	C	S/N
I	D_{u_t}	-4.16	-2.88	S	0.100	0.146	S	0.09	-2.89	S
I	D_{TTR_t}	-4.92	-2.88	S	0.097	0.146	S	-7.37	-2.89	S
I	D_{Trate_t}	-5.44	-2.88	S	0.071	0.146	S	-8.25	-2.89	S
I	D_{PITR_t}	-4.23	-2.88	S	0.061	0.146	S	-4.44	-2.89	S
I	$D_{PITrate_t}$	-4.46	-2.88	S	0.089	0.146	S	-4.23	-2.89	S
I	D_{CITR_t}	-5.61	-2.88	S	0.099	0.146	S	-9.41	-2.89	S
I	$D_{CITrate_t}$	-3.96	-2.88	S	0.115	0.146	S	-3.79	-2.89	S

Note: All tested on 5% significance level. I... Constant included for the purpose of the test, T... Trend included for the purpose of the test, TST... Test Statistic, C... Critical Value, S/N... Stationary/Non-stationary Decision.

Table 5.10: Stationarity Test Results for the Growth Rates in the period 2001-2023

Comp	**Var**	**ADF**			**KPSS**			**PP**		
		TST	C	S/N	TST	C	S/N	TST	C	S/N
I	D_{TTR_t}	-4.67	-2.89	S	0.14	0.146	S	-6.8191	-2.89	S
I	D_{Trate_t}	-5.15	-2.89	S	0.14	0.146	S	-7.5909	-2.89	S
I	D_{PITR_t}	-5.81	-2.89	S	0.18	0.146	N	-4.7979	-2.89	S
I	D_{CITR_t}	-5.30	-2.89	S	0.17	0.146	N	-8.8369	-2.89	S

Note: All tested on 5% significance level. I... Constant included for the purpose of the test, T... Trend included for the purpose of the test, TST... Test Statistic, C... Critical Value, S/N... Stationary/Non-stationary Decision.

There is still a noticeable spike in revenues after 2001, especially for D_{PITR_t} and D_{CITR_t}, which was previously controlled for by our dummy variable $partial$. This dummy will still be utilised later to control for the structural break in original level variables. All calculated growth rates are in decimal format.

5.4 Ordinary Least Squares (OLS)

To begin our analysis, we perform regression using the Ordinary Least Squares (OLS) method, which has already been frequently used in the Laffer Curve studies such as Brill and Hassett (2007), Clausing (2007), Karas (2013), and Wang et al. (2021).

OLS stands on the essential assumption of exogeneity, $\mathbb{E}(u \mid \mathbf{X}) = 0$, where u denotes the error term and $\mathbf{X}$ stands for a $(t \times k)$ matrix of explanatory variables. Furthermore, t represents a time period (a quarter in a year[25]) and k represents the number of explanatory variables.

The general multiple regression model is

$$y_t = \beta_0 + \beta_1 x_{1,t} + \beta_2 x_{2,t} + \cdots + \beta_k x_{k,t} + \epsilon_t, \tag{5.37}$$

where y_t are the observations of the dependent variable in t periods, $x_{i,t}$ are explanatory variables at time t, coefficients β_i measure the effect of each explanatory variable after taking into account the effects of all the other regressors in the model, and ϵ_t is the error term (Hyndman & Athanasopoulos, 2021).

In practice, the true values of β_i are not observable, and therefore we estimate them. This is where the OLS method steps in. It estimates them so that the sum of squared residuals is minimised,

$$\min_{\beta_0, \beta_1 \beta_2, \ldots, \beta_k} \sum_{t=1}^{T} \epsilon_t^2 = \sum_{t=1}^{T} (y_t - \beta_1 x_{1,t} - \beta_2 x_{2,t} - \cdots - \beta_k x_{k,t}) \ , \tag{5.38}$$

we then obtain the estimated function

$$\hat{y}_t = \hat{\beta}_0 + \hat{\beta}_1 x_{1,t} + \hat{\beta}_2 x_{2,t} + \cdots + \hat{\beta}_k x_{k,t} \ . \tag{5.39}$$

[25] For simplicity, we transform our $(m \times y)$ matrices into $(t \times 1)$ vectors, so that the time period would seamlessly follow.

A unit change in x_i, t leads to a $\beta_i \times 1$ change in y_t, ceteris paribus (all else held constant). This can be shown by taking the partial derivative of the regression equation with respect to $x_{i,t}$

$$\frac{\partial y_t}{\partial x_i} = \beta_i \tag{5.40}$$

We have already mentioned the crucial assumption of endogeneity. For the OLS method to give us unbiased results, other assumptions must also hold, see Table 5.11.

Table 5.11: Assumptions for the OLS regression for TS

No.	Assumption	Short Explanation
TS.1	Linearity in Parameters	
TS.2	No Perfect Collinearity	no independent variable is neither a constant nor a perfect linear combination of the others
TS.3	Zero Conditional Mean	$\mathbb{E}(u \mid \mathbf{X}) = 0$
TS.4	Homoskedasticity	$var(u_t \mid \mathbf{X}) = var(u_t) = \sigma^2$
TS.5	No Serial Correlation	$corr(u_t, u_\tau \mid \mathbf{X}) = 0$, for all $t \neq \tau$
TS.6	Normality of the Data	$\epsilon_t \sim \mathcal{N}(0, \sigma^2)$

Note: Author's own elaboration based on Wooldridge (2013).

If one or more assumptions are not held, specific measures must be applied in order to keep the model unbiased. For more about the assumptions and the principle of Ordinary Least Squares regressions, see Wooldridge (2013), Stock and Watson (2020) or Hyndman and Athanasopoulos (2021).

5.4.1 Stationarity Assuming OLS Models

Following the discussion from the previous section, we create multiple OLS models aiming to find the evidence for the Laffer curve. The models are estimated with data from 2 different periods, firstly from 1997 Q1 to 2023 Q4, and secondly from 2001 Q1 to 2023 Q4, to isolate the effect of the structural break. In the longer case, we utilise the *partial* variable as explained in the Issues with the Data section.

We start with simpler models and consequently add other variables controlling for external effects. The shortest and best-fitted estimated model for both periods will be subsequently estimated by the Instrumental Variable Regression and the Generalised Method of Moments to address potential endogeneity. More on this can be found in sections 5.5 and 5.6.

As stated before, the analysis is performed for the overall tax system, the Personal Income Tax and the CIT. Therefore, the formal specification of our model to be estimated is as follows,

$$\ln(y_t) = \beta_0 + \beta_1 x_t + \beta_2 x_t^2 + \delta_1 c_{1,t} + \delta_2 c_{2,t} + \cdots + \delta_k c_{k,t} + u, \tag{5.41}$$

where

$$x_1 = \begin{cases} Trate & \forall y_t = TTR \\ PITrate & \forall y_t = PITR \\ CITrate & \forall y_t = CITR \end{cases}, \tag{5.42}$$

and where

$$\mathbf{c} = (c_{1,t}, c_{2,t}, \ldots, c_{k,t}), \quad where \begin{cases} c_1 = partial \\ c_2 = windfall \\ c_3 = CITrate \\ c_4 = PITrate \\ c_5 = TG \\ c_6 = g \\ c_7 = u \end{cases} . \tag{5.43}$$

To find the revenue-maximising tax rate for each of the taxes, we need to find the maximum of the function. Therefore, we differentiate it with respect to x_t,

$$\frac{\partial \ln(y)}{\partial x_t} = \beta_1 + 2\beta_2 x_t. \tag{5.44}$$

To obtain the extreme value, we need to set the equation equal to zero and calculate the maximising value of x_t,

$$x_t = \frac{\beta_1}{-2\beta_2}. \tag{5.45}$$

This extreme is a maximum when the function is concave, that is, the second derivative of $\ln(y)$ with respect to x_t is negative,

$$\frac{\partial^2 \ln(y)}{\partial x_t^2} = 2\beta_2 \Rightarrow \beta_2 < 0. \tag{5.46}$$

To sum up, when the β_2 is negative, the function has a local maximum which denotes the revenue-maximising tax rate we want to obtain (Chiang, 1984).

To obtain robust results, each regression is tested for heteroskedasticity by the Breusch-Pagan Test (BP test). If the findings are statistically significant, we reject the null hypothesis of homoskedasticity and utilise White's Heteroskedasticity Consistent Standard Errors (HC).[26] We also test for the presence of autocorrelation with the Durbin-Watson Test (DW test). If we reject the null hypothesis of no autocorrelation, we use Newey-West's Heteroskedasticity and Autocorrelation Consistent Standard Errors (HAC). The results of all the tests mentioned are reported in the tables together with the computed revenue-maximising tax rate and other relevant statistics. Standard Errors can be found in parentheses. For more on these topics, see Breusch and Pagan (1979), Durbin and Watson (1950), Newey and West (1987), Stock and Watson (2020), White (1980), and Wooldridge (2013).

Looking at tables 5.12 and 5.13, OLS regressions in both periods prove a statistically significant Laffer Curve effect for the overall tax system with the revenue-maximising tax rate ranging from 28.6% to 30.3%.[27] For the longer data, the windfall tax effect shows statistical significance. The Unemployment Rate is statistically significant in all cases, too. All the regressions prove a statistically insignificant relationship between the level of the TG and the TTR. To our surprise,

[26]We automatically report them for relevant regressions instead of non-robust in the output.

[27]The revenue-maximising tax rate for each estimation, in decimal format, is observable in each regression output in the row *Rev-max Tax Rate*.

the Annual GDP Growth Rate is not a statistically significant contributor to the TTR when assessing based on the OLS regression results.

Table 5.12: OLS Regression Output for the Total Tax System from 1997 Q1 to 2023 Q4

	Dependent variable:				
	ln(TTR)				
	(1)	(2)	(3)	(4)	(5)
Trate	82.952***	84.009***	88.222***	30.212***	36.339***
	p = 0.001	p = 0.001	p = 0.000	p = 0.001	p = 0.002
	(21.688)	(21.916)	(21.814)	(8.068)	(11.088)
$Trate^2$	−144.947***	−146.842***	−154.532***	−49.809***	−60.807***
	p = 0.001	p = 0.001	p = 0.000	p = 0.001	p = 0.003
	(38.606)	(39.101)	(38.629)	(14.984)	(20.425)
Partial	−0.367***	−0.365***	−0.350***	−0.327***	−0.284***
	p = 0.000	p = 0.000	p = 0.000	p = 0.000	p = 0.001
	(0.065)	(0.065)	(0.063)	(0.072)	(0.076)
Windfall	0.189***	0.182***	0.190***	0.041	0.049**
	p = 0.001	p = 0.000	p = 0.000	p = 0.109	p = 0.045
	(0.049)	(0.046)	(0.048)	(0.026)	(0.024)
TG		−0.098	−0.072	−0.200	−0.133
		p = 0.523	p = 0.642	p = 0.330	p = 0.405
		(0.154)	(0.155)	(0.206)	(0.160)
g			0.285		0.435
			p = 0.540		p = 0.163
			(0.465)		(0.312)
u				−4.743***	−5.033***
				p = 0.000	p = 0.000
				(0.973)	(1.033)
Constant	0.802	0.691	0.100	8.408***	7.540***
	p = 0.791	p = 0.821	p = 0.974	p = 0.000	p = 0.000
	(3.025)	(3.045)	(3.062)	(1.111)	(1.523)
Rev-max Tax Rate	0.286	0.286	0.285	0.303	0.299
BP test (p-value)	0.057	0.064	0.174	0.000	0.000
DW test (p-value)	0.000	0.000	0.000	0.000	0.000
Observations	112	112	108	112	108
R^2	0.722	0.724	0.704	0.883	0.892
Adjusted R^2	0.712	0.711	0.686	0.877	0.884
Residual SE	0.120	0.120	0.122	0.078	0.074
F Statistic	69.539***	55.642***	39.947***	132.669***	117.515***

Note: Performed on regressors in decimal format. *p<0.1; **p<0.05; ***p<0.01

Table 5.13: OLS Regression Output for the Total Tax System from 2001 Q1 to 2023 Q4

	Dependent variable:				
	ln(TTR)				
	(1)	(2)	(3)	(4)	(5)
Trate	108.465***	109.097***	109.516***	38.409***	38.815***
	p = 0.001	p = 0.001	p = 0.001	p = 0.009	p = 0.009
	(28.550)	(29.361)	(29.080)	(14.702)	(14.712)
Trate^2	−189.616***	−190.767***	−191.632***	−63.514**	−64.350**
	p = 0.001	p = 0.001	p = 0.001	p = 0.019	p = 0.017
	(50.086)	(51.657)	(51.023)	(26.983)	(26.835)
Windfall	0.197***	0.191***	0.193***	0.031*	0.033*
	p = 0.000	p = 0.000	p = 0.000	p = 0.083	p = 0.063
	(0.044)	(0.042)	(0.045)	(0.018)	(0.018)
TG		−0.088	−0.088	−0.088	−0.089
		p = 0.596	p = 0.594	p = 0.526	p = 0.534
		(0.165)	(0.165)	(0.139)	(0.143)
g			0.106		0.102
			p = 0.832		p = 0.483
			(0.498)		(0.145)
u				−5.608***	−5.608***
				p = 0.000	p = 0.000
				(1.026)	(1.043)
Constant	−2.822	−2.877	−2.929	7.201***	7.150***
	p = 0.486	p = 0.489	p = 0.478	p = 0.001	p = 0.001
	(4.049)	(4.150)	(4.120)	(2.004)	(2.015)
Rev-max Tax Rate	0.286	0.286	0.286	0.302	0.302
BP test (p-value)	0.601	0.641	0.742	0.000	0.000
DW test (p-value)	0.000	0.000	0.000	0.000	0.000
Observations	92	92	92	92	92
R^2	0.376	0.379	0.380	0.832	0.832
Adjusted R^2	0.355	0.351	0.343	0.822	0.821
Residual SE	0.126	0.126	0.127	0.066	0.066
F Statistic	17.664***	13.282***	10.522***	85.217***	70.385***

Note: Performed on regressors in decimal format. *p<0.1; **p<0.05; ***p<0.01

Based on the statistical significance of the variables, the residual SE and the adjusted R^2, we choose the model (5) to be re-estimated alongside the basic model (1) in our later analysis. The results of the OLS regressions for the PIT are observable in tables 5.14 and 5.15, for CIT, see tables 5.16 and 5.17.

Table 5.14: OLS Regression Output for the PIT from 1997 Q1 to 2023 Q4

	Dependent variable:				
	ln(PITR)				
	(1)	(2)	(3)	(4)	(5)
PITrate	30.315***	27.090***	26.564***	22.767***	25.049***
	p = 0.000	p = 0.001	p = 0.001	p = 0.000	p = 0.000
	(7.601)	(7.002)	(7.111)	(4.850)	(4.918)
PITrate^2	−105.552***	−87.911***	−85.395***	−67.242***	−80.247***
	p = 0.001	p = 0.001	p = 0.001	p = 0.000	p = 0.000
	(28.324)	(25.258)	(25.777)	(15.840)	(17.365)
CITrate		−1.201	−1.346*	−0.962	
		p = 0.111	p = 0.098	p = 0.146	
		(0.753)	(0.813)	(0.662)	
Partial	−0.833***	−0.735***	−0.730***	−0.757***	−0.825***
	p = 0.000	p = 0.000	p = 0.000	p = 0.000	p = 0.000
	(0.034)	(0.075)	(0.077)	(0.045)	(0.040)
TG		0.079	0.060	−0.009	−0.022
		p = 0.660	p = 0.754	p = 0.926	p = 0.813
		(0.179)	(0.190)	(0.100)	(0.094)
g			0.161	0.280	0.331
			p = 0.738	p = 0.800	p = 0.692
			(0.480)	(1.102)	(0.834)
u				−4.038***	−4.143***
				p = 0.000	p = 0.000
				(0.606)	(0.563)
Constant	8.179***	8.529***	8.588***	8.919***	8.644***
	p = 0.000	p = 0.000	p = 0.000	p = 0.000	p = 0.000
	(0.473)	(0.437)	(0.453)	(0.275)	(0.303)
Rev-max Tax Rate	0.144	0.154	0.156	0.169	0.156
BP test (p-value)	0.018	0.084	0.000	0.005	0.003
DW test (p-value)	0.000	0.000	0.000	0.538	0.435
Observations	112	112	108	108	108
R^2	0.851	0.857	0.839	0.894	0.891
Adjusted R^2	0.847	0.850	0.829	0.887	0.884
Residual Std. Error	0.127	0.125	0.128	0.104	0.105
F Statistic	205.922***	127.178***	87.421***	120.689***	136.927***

Note: Performed on regressors in decimal format. *p<0.1; **p<0.05; ***p<0.01

Table 5.15: OLS Regression Output for the PIT from 2001 Q1 to 2023 Q4

	Dependent variable:				
	ln(PITR)				
	(1)	(2)	(3)	(4)	(5)
PITrate	30.245***	23.331***	22.830***	21.861***	24.820***
	p = 0.000	p = 0.000	p = 0.000	p = 0.000	p = 0.000
	(7.377)	(4.707)	(5.029)	(4.504)	(4.847)
PITrate^2	−105.280***	−66.609***	−65.079***	−62.410***	−79.156***
	p = 0.001	p = 0.000	p = 0.001	p = 0.000	p = 0.000
	(27.480)	(16.480)	(17.890)	(15.716)	(16.926)
CITrate		−2.690***	−2.743***	−1.337**	
		p = 0.001	p = 0.001	p = 0.028	
		(0.728)	(0.739)	(0.607)	
TG		0.013	0.017	−0.016	−0.027
		p = 0.936	p = 0.916	p = 0.869	p = 0.789
		(0.165)	(0.164)	(0.099)	(0.102)
g			0.487	0.382	0.331
			p = 0.462	p = 0.679	p = 0.717
			(0.662)	(0.921)	(0.912)
u				−3.866***	−4.296***
				p = 0.000	p = 0.000
				(0.671)	(0.693)
Constant	8.183***	9.000***	9.035***	9.026***	8.665***
	p = 0.000	p = 0.000	p = 0.000	p = 0.000	p = 0.000
	(0.460)	(0.282)	(0.310)	(0.285)	(0.296)
Rev-max Tax Rate	0.144	0.175	0.175	0.175	0.157
BP test (p-value)	0.067	0.225	0.001	0.008	0.003
DW test (p-value)	0.000	0.005	0.004	0.547	0.405
Observations	92	92	92	92	92
R^2	0.378	0.453	0.461	0.615	0.599
Adjusted R^2	0.364	0.428	0.430	0.588	0.576
Residual Std. Error	0.135	0.128	0.128	0.109	0.111
F Statistic	27.062***	18.012***	14.724***	22.645***	25.683***

Note: Performed on regressors in decimal format. *p<0.1; **p<0.05; ***p<0.01

Table 5.16: OLS Regression Output for the CIT from 1997 Q1 to 2023 Q4

	Dependent variable:				
	ln(CITR)				
	(1)	(2)	(3)	(4)	(5)
CITrate	11.627**	11.578*	6.039	19.820***	17.488***
	p = 0.024	p = 0.057	p = 0.336	p = 0.003	p = 0.003
	(5.036)	(6.008)	(6.238)	(6.496)	(5.658)
CITrate^2	−22.576**	−22.463**	−11.791	−35.035***	−31.743***
	p = 0.027	p = 0.045	p = 0.309	p = 0.004	p = 0.005
	(10.003)	(11.037)	(11.529)	(11.777)	(10.869)
PITrate		0.000	−0.169	−1.120	
		p = 1.000	p = 0.919	p = 0.464	
		(1.689)	(1.646)	(1.522)	
Partial	−0.299***	−0.302**	−0.316***	−0.278***	−0.268**
	p = 0.010	p = 0.011	p = 0.007	p = 0.009	p = 0.011
	(0.113)	(0.116)	(0.113)	(0.104)	(0.103)
Windfall	0.545***	0.542***	0.575***	0.441***	0.460***
	p = 0.000	p = 0.000	p = 0.000	p = 0.001	p = 0.000
	(0.115)	(0.120)	(0.117)	(0.111)	(0.108)
TG		−0.048	−0.037	−0.117	−0.128
		p = 0.830	p = 0.865	p = 0.565	p = 0.526
		(0.225)	(0.219)	(0.201)	(0.200)
g			1.928**	1.461**	1.451**
			p = 0.013	p = 0.040	p = 0.041
			(0.756)	(0.700)	(0.699)
u				−5.628***	−5.497***
				p = 0.000	p = 0.000
				(1.263)	(1.248)
Constant	8.739***	8.761***	9.410***	8.016***	8.223***
	p = 0.000	p = 0.000	p = 0.000	p = 0.000	p = 0.000
	(0.610)	(0.684)	(0.713)	(0.724)	(0.666)
Rev-max Tax Rate	0.258	0.258	0.256	0.283	0.275
BP test (p-value)	0.817	0.601	0.508	0.072	0.071
DW test (p-value)	0.080	0.054	0.065	0.686	0.683
Observations	106	106	106	106	106
R^2	0.415	0.415	0.452	0.545	0.542
Adjusted R^2	0.392	0.380	0.413	0.507	0.510
Residual Std. Error	0.222	0.224	0.218	0.200	0.200
F Statistic	17.921***	11.724***	11.536***	14.518***	16.592***

Note: Performed on regressors in decimal format. *p<0.1; **p<0.05; ***p<0.01

Table 5.17: OLS Regression Output for the CIT from 2001 Q1 to 2023 Q4

	Dependent variable:				
	ln(CITR)				
	(1)	(2)	(3)	(4)	(5)
CITrate	29.889***	29.358***	20.441**	32.122***	31.137
	p = 0.001	p = 0.002	p = 0.047	p = 0.001	p = 0.318
	(8.524)	(9.466)	(10.274)	(8.801)	(31.137)
CITrate^2	−59.458***	−58.890***	−41.173**	−60.196***	−59.283
	p = 0.001	p = 0.002	p = 0.039	p = 0.001	p = 0.318
	(17.429)	(18.266)	(19.879)	(17.545)	(−59.283)
PITrate		0.402	0.179	−0.803	
		p = 0.823	p = 0.914	p = 0.500	
		(1.788)	(1.645)	(1.190)	
Windfall	0.559***	0.564***	0.592***	0.461***	0.475
	p = 0.000	p = 0.000	p = 0.000	p = 0.000	p = 0.318
	(0.093)	(0.107)	(0.108)	(0.118)	(0.475)
TG		−0.021	−0.031	−0.082	−0.089
		p = 0.939	p = 0.897	p = 0.756	p = 0.318
		(0.266)	(0.242)	(0.262)	(−0.089)
g			1.910***	1.417*	1.407
			p = 0.007	p = 0.096	p = 0.318
			(0.697)	(0.851)	(1.407)
u				−5.484***	−5.383
				p = 0.000	p = 0.318
				(1.316)	(−5.383)
Constant	6.586***	6.632***	7.685***	6.525***	6.595
	p = 0.000	p = 0.000	p = 0.000	p = 0.000	p = 0.318
	(1.000)	(1.078)	(1.162)	(0.999)	(6.595)
Rev-max Tax Rate	0.251	0.249	0.248	0.267	0.263
BP test (p-value)	0.926	0.278	0.232	0.004	0.004
DW test (p-value)	0.005	0.003	0.003	0.320	0.339
Observations	88	88	88	88	88
R^2	0.327	0.328	0.389	0.540	0.537
Adjusted R^2	0.303	0.287	0.344	0.499	0.503
Residual Std. Error	0.194	0.197	0.189	0.165	0.164
F Statistic	13.634***	8.009***	8.603***	13.397***	15.682***
Note:					*p<0.1; **p<0.05; ***p<0.01

The results for the PIT show the statistical significance of the Laffer effect for all of the estimated models. As it was for the total tax system, even the Personal Income Tax Revenue depends on the Unemployment Rate and does not seem to be statistically significantly determined by Trust in the Government and Annual GDP Growth Rate. For some instances, the cross-effect of CIT seems to be statistically significant. Higher Corporate Income Tax Rate would then mean that when the firms have to pay more from their income, they shift the burden towards their employees, lowering their wages, which leads to a decrease in the PITR, ceteris paribus.

Based on the statistical significance of the variables, the residual standard error and the adjusted R^2, and as there is just a little evidence for the cross-effects after the introduction of the Unemployment Rate, model (4) will be utilised to re-estimate the relationship with more complex methods.

The value of the revenue-maximising tax rate for the Personal Income Tax ranges from 14.4% to 17.5%. For the simplest model, the rate is consistent for both time periods, 15.4%. The second selected model predicts 16.9 or 17.5 %. We did not include the windfall tax in this regression as it is aimed at the CIT.

Considering the Corporate Income Tax, the Laffer effect is statistically significant as well. The cross-effect is not statistically significant in this case, so it is for TG. The windfall tax proves to be statistically significant in most cases, and for multiple regressions, the Annual GDP Growth Rate is statistically significant as well.

The revenue-maximising tax rate for the CIT oscillates from 24.8% to 28.3%. We pick the model (4) alongside the simplest model (1) for our later re-estimation, as most coefficients are statistically significant for both TS and the model proves to be one of the best fitting, even considering the residual standard error and the adjusted R^2.

5.4.2 OLS Models Solving for Stationarity

To solve for stationarity, we created the growth rates of our crucial variables, which will be utilised in this part of the analysis. As we did with the variables in levels, we will introduce multiple models and select the best-fitting ones to continue with in the analysis.

The same tests will be conducted to check the robustness of our outputs, and relevant standard errors will be used. The results of the OLS analysis of the growth rates of our variables are observable in tables 5.18, 5.19, 5.20, 5.21, 5.22 and 5.23.

At first glance, it is certain that there is no Laffer-type effect for the growth rates when analysing the overall tax system. Predicted models show statistical significance only for the linear term. Besides this regressor, the Change in the Unemployment Rate seems to be consistently statistically significant.

Alarming are the values of the R^2 and consequent adjusted values. They are generally very high, with the estimation (4) being impossibly close to 1. This is most probably the result of the way the Trate was calculated.[28]

For later analysis, models (1) and (5) will be utilised based on regressor significance and the model statistics (F-statistic, R^2 and the Residual Standard Error).

[28]As explained in Section 5.2.2, it is the ratio TTR over GDP.

Table 5.18: OLS Regression Output for the Growth Rates in the Total Tax System from 1997 Q1 to 2023 Q4

	Dependent variable:				
	D_{TTR}				
	(1)	(2)	(3)	(4)	(5)
D_{Trate}	1.125***	1.122***	1.072***	1.019***	1.065***
	p = 0.000	p = 0.000	p = 0.000	p = 0.000	p = 0.000
	(0.078)	(0.078)	(0.063)	(0.008)	(0.049)
D^2_{Trate}	−0.484	−0.484	−0.089	0.056	
	p = 0.236	p = 0.246	p = 0.810	p = 0.329	
	(0.408)	(0.417)	(0.369)	(0.058)	
Partial	−0.013	−0.014	0.012	−0.001**	0.011*
	p = 0.259	p = 0.264	p = 0.152	p = 0.041	p = 0.080
	(0.011)	(0.012)	(0.008)	(0.001)	(0.007)
Windfall	−0.032***	−0.032***	−0.016**	−0.002**	−0.016***
	p = 0.000	p = 0.000	p = 0.020	p = 0.038	p = 0.007
	(0.007)	(0.007)	(0.007)	(0.001)	(0.006)
D_{TG}		−0.006	0.002	−0.000	
		p = 0.345	p = 0.800	p = 0.393	
		(0.006)	(0.006)	(0.000)	
g				0.974***	
				p = 0.000	
				(0.024)	
D_u			−0.089***	0.002	−0.089***
			p = 0.000	p = 0.355	p = 0.000
			(0.010)	(0.002)	(0.006)
Constant	0.028***	0.028***	0.022***	0.001	0.022***
	p = 0.000	p = 0.000	p = 0.001	p = 0.103	p = 0.001
	(0.007)	(0.007)	(0.007)	(0.001)	(0.006)
D_{TTR}-max D_{Trate}	1.162	1.159	6.016	-9.053	–
BP test (p-value)	0.006	0.019	0.014	0.000	0.017
DW test (p-value)	0.000	0.000	0.000	0.179	0.000
Observations	108	108	108	108	108
R^2	0.862	0.863	0.912	0.999	0.912
Adjusted R^2	0.857	0.857	0.906	0.999	0.908
Residual Std. Error	0.030	0.030	0.024	0.002	0.024
F Statistic	161.519***	128.965***	173.741***	18,334.660***	265.294***

Note: Performed on regressors in decimal format. *p<0.1; **p<0.05; ***p<0.01

Table 5.19: OLS Regression Output for the Growth Rates in the Total Tax System from 2001 Q1 to 2023 Q4

	Dependent variable:				
	D_{TTR}				
	(1)	(2)	(3)	(4)	(5)
D_{Trate}	1.143***	1.141***	1.071***	1.020***	1.064***
	p = 0.000	p = 0.000	p = 0.000	p = 0.000	p = 0.000
	(0.089)	(0.093)	(0.073)	(0.009)	(0.056)
D^2_{Trate}	−0.569	−0.579	−0.092	0.051	
	p = 0.252	p = 0.223	p = 0.826	p = 0.417	
	(0.496)	(0.475)	(0.415)	(0.063)	
Windfall	−0.032***	−0.033***	−0.017**	−0.002**	−0.016**
	p = 0.000	p = 0.000	p = 0.024	p = 0.041	p = 0.013
	(0.005)	(0.006)	(0.007)	(0.001)	(0.007)
D_{TG}		−0.009	0.001	−0.000	
		p = 0.264	p = 0.925	p = 0.384	
		(0.008)	(0.008)	(0.000)	
g				0.975***	
				p = 0.000	
				(0.025)	
D_u			−0.087***	0.002	−0.087***
			p = 0.000	p = 0.333	p = 0.000
			(0.015)	(0.003)	(0.015)
Constant	0.028***	0.029***	0.023***	0.001	0.022***
	p = 0.000	p = 0.000	p = 0.001	p = 0.115	p = 0.000
	(0.005)	(0.007)	(0.006)	(0.001)	(0.005)
D_{TTR}-max D_{Trate}	1.004	0.985	5.845	-10.027	–
BP test (p-value)	0.009	0.030	0.026	0.001	0.026
DW test (p-value)	0.000	0.000	0.000	0.196	0.000
Observations	92	92	92	92	92
R^2	0.863	0.865	0.908	0.999	0.908
Adjusted R^2	0.858	0.859	0.903	0.999	0.905
Residual Std. Error	0.031	0.031	0.026	0.002	0.026
F Statistic	184.563***	139.307***	170.087***	17,197.330***	289.737***

Note: Performed on regressors in decimal format. *p<0.1; **p<0.05; ***p<0.01

Table 5.20: OLS Regression Output for the Growth Rates in the PIT from 1997 Q1 to 2023 Q4

	Dependent variable:				
	D_{PITR}				
	(1)	(2)	(3)	(4)	(5)
$D_{PITrate}$	−0.384	−0.499	−0.831	−0.689	0.688***
	p = 0.766	p = 0.727	p = 0.609	p = 0.636	p = 0.000
	(1.286)	(1.424)	(1.621)	(1.455)	(0.150)
$D^2_{PITrate}$	−4.145	−4.140	−5.246	−5.559	
	p = 0.421	p = 0.428	p = 0.364	p = 0.335	
	(5.145)	(5.220)	(5.771)	(5.760)	
$D_{CITrate}$		0.672	0.729		
		p = 0.369	p = 0.345		
		(0.747)	(0.771)		
Partial	−0.120	−0.089	−0.021	−0.080	−0.040
	p = 0.138	p = 0.110	p = 0.667	p = 0.424	p = 0.471
	(0.081)	(0.055)	(0.050)	(0.100)	(0.056)
D_{TG}		0.038	0.058	0.041	
		p = 0.472	p = 0.330	p = 0.311	
		(0.053)	(0.059)	(0.040)	
g				1.619	
				p = 0.185	
				(1.221)	
D_u			−0.256**	−0.093	−0.203***
			p = 0.016	p = 0.499	p = 0.001
			(0.105)	(0.138)	(0.061)
Constant	0.104	0.113	0.112	0.072	0.065
	p = 0.236	p = 0.246	p = 0.247	p = 0.369	p = 0.170
	(0.088)	(0.097)	(0.097)	(0.080)	(0.047)
D_{PITR} maximising $D_{PITrate}$	-0.046	-0.060	-0.079	-0.062	–
BP test (p-value)	0.123	0.166	0.186	0.361	0.766
DW test (p-value)	0.000	0.000	0.000	0.000	0.000
Observations	108	108	108	108	108
R^2	0.117	0.129	0.172	0.184	0.130
Adjusted R^2	0.092	0.086	0.123	0.136	0.105
Residual Std. Error	0.237	0.237	0.233	0.231	0.235
F Statistic	4.597***	3.013**	3.490***	3.797***	5.199***

Note: Performed on regressors in decimal format. *p<0.1; **p<0.05; ***p<0.01

Table 5.21: OLS Regression Output for the Growth Rates in the PIT from 2001 Q1 to 2023 Q4

	Dependent variable:				
	D_{PITR}				
	(1)	(2)	(3)	(4)	(5)
$D_{PITrate}$	−0.444 p = 0.635 (0.936)	−0.635 p = 0.690 (1.587)	−0.875 p = 0.610 (1.712)	−0.766 p = 0.599 (1.454)	0.687*** p = 0.000 (0.145)
$D^2_{PITrate}$	−4.359 p = 0.244 (3.740)	−4.534 p = 0.420 (5.611)	−5.396 p = 0.367 (5.975)	−5.891 p = 0.308 (5.777)	
$D_{CITrate}$		0.859 p = 0.445 (1.123)	0.742 p = 0.461 (1.004)		
D_{TG}		0.030 p = 0.540 (0.050)	0.056 p = 0.354 (0.061)	0.046 p = 0.333 (0.048)	
g				1.880 p = 0.124 (1.220)	
D_u			−0.268*** p = 0.010 (0.104)	−0.094 p = 0.495 (0.138)	−0.222*** p = 0.001 (0.066)
Constant	0.106* p = 0.093 (0.063)	0.120 p = 0.260 (0.106)	0.114 p = 0.270 (0.103)	0.068 p = 0.367 (0.075)	0.064 p = 0.150 (0.045)
D_{PITR} maximising $D_{PITrate}$	-0.051	-0.070	-0.081	-0.065	–
BP test (p-value)	0.122	0.155	0.194	0.385	0.896
DW test (p-value)	0.000	0.000	0.000	0.000	0.000
Observations	92	92	92	92	92
R^2	0.108	0.120	0.164	0.187	0.123
Adjusted R^2	0.088	0.080	0.115	0.139	0.103
Residual Std. Error	0.254	0.255	0.251	0.247	0.252
F Statistic	5.385***	2.976**	3.370***	3.945***	6.233***

Note: Performed on regressors in decimal format. *p<0.1; **p<0.05; ***p<0.01

Table 5.22: OLS Regression Output for the Growth Rates in the CIT from 1997 Q1 to 2023 Q4

	Dependent variable:				
	D_{CITR}				
	(1)	(2)	(3)	(4)	(5)
$D_{CITrate}$	4.813	4.818	3.753	7.844**	0.886
	p = 0.145	p = 0.163	p = 0.321	p = 0.046	p = 0.298
	(3.270)	(3.425)	(3.761)	(3.872)	(0.846)
$D^2_{CITrate}$	44.013	44.493	33.713	66.337*	
	p = 0.161	p = 0.180	p = 0.358	p = 0.069	
	(31.117)	(32.914)	(36.472)	(35.959)	
$D_{PITrate}$		0.038	−0.009		
		p = 0.916	p = 0.982		
		(0.360)	(0.367)		
Partial	−0.117	−0.120	−0.075	−0.140	−0.055
	p = 0.238	p = 0.251	p = 0.546	p = 0.225	p = 0.598
	(0.098)	(0.104)	(0.123)	(0.114)	(0.104)
Windfall	0.554***	0.552***	0.578***	0.593***	0.626***
	p = 0.002	p = 0.003	p = 0.003	p = 0.001	p = 0.001
	(0.175)	(0.178)	(0.183)	(0.173)	(0.173)
D_{TG}		−0.003	0.003	0.019	
		p = 0.974	p = 0.970	p = 0.812	
		(0.083)	(0.084)	(0.081)	
g				4.052***	3.191**
				p = 0.007	p = 0.023
				(1.450)	(1.382)
D_u			−0.127	0.329	0.109
			p = 0.490	p = 0.169	p = 0.600
			(0.183)	(0.237)	(0.206)
Constant	0.081*	0.081*	0.072	0.013	0.003
	p = 0.053	p = 0.055	p = 0.104	p = 0.783	p = 0.955
	(0.041)	(0.042)	(0.044)	(0.047)	(0.047)
D_{CITR} maximising $D_{CITrate}$	-0.055	-0.054	-0.056	-0.059	–
BP test (p-value)	0.943	0.991	0.993	0.995	0.975
DW test (p-value)	0.545	0.476	0.453	0.669	0.563
Observations	108	108	108	108	108
R^2	0.127	0.127	0.131	0.194	0.166
Adjusted R^2	0.093	0.075	0.070	0.138	0.126
Residual Std. Error	0.339	0.343	0.344	0.331	0.333
F Statistic	3.736***	2.444**	2.153**	3.437***	4.072***

Note: Performed on regressors in decimal format. *p<0.1; **p<0.05; ***p<0.01

Table 5.23: OLS Regression Output for the Growth Rates in the CIT from 2001 Q1 to 2023 Q4

	Dependent variable:				
	D_{CITR}				
	(1)	(2)	(3)	(4)	(5)
$D_{CITrate}$	5.228	4.796	3.591	8.440**	1.797*
	p = 0.137	p = 0.196	p = 0.370	p = 0.044	p = 0.093
	(3.481)	(3.676)	(3.984)	(4.111)	(1.058)
$D^2_{CITrate}$	41.591	35.966	24.211	64.189*	
	p = 0.221	p = 0.331	p = 0.544	p = 0.099	
	(33.709)	(36.787)	(39.726)	(38.398)	
$D_{PITrate}$		−0.092	−0.139		
		p = 0.819	p = 0.734		
		(0.401)	(0.406)		
Windfall	0.541***	0.544***	0.577***	0.579***	0.612***
	p = 0.005	p = 0.005	p = 0.004	p = 0.003	p = 0.002
	(0.184)	(0.187)	(0.192)	(0.182)	(0.181)
D_{TG}		−0.031	−0.021	−0.009	
		p = 0.735	p = 0.822	p = 0.924	
		(0.091)	(0.093)	(0.089)	
g				4.428***	3.548**
				p = 0.006	p = 0.020
				(1.569)	(1.489)
D_u			−0.158	0.359	0.136
			p = 0.430	p = 0.172	p = 0.551
			(0.199)	(0.260)	(0.227)
Constant	0.094**	0.096**	0.084*	0.023	0.014
	p = 0.037	p = 0.037	p = 0.082	p = 0.644	p = 0.778
	(0.044)	(0.045)	(0.048)	(0.050)	(0.050)
D_{CITR} maximising $D_{CITrate}$	-0.063	-0.054	-0.074	-0.066	–
BP test (p-value)	0.907	0.985	0.989	0.987	0.956
DW test (p-value)	0.633	0.560	0.547	0.765	0.695
Observations	92	92	92	92	92
R^2	0.126	0.128	0.134	0.207	0.179
Adjusted R^2	0.096	0.077	0.073	0.151	0.142
Residual Std. Error	0.357	0.360	0.361	0.346	0.348
F Statistic	4.232***	2.521**	2.197*	3.706***	4.751***

Note: Performed on regressors in decimal format. *p<0.1; **p<0.05; ***p<0.01

The explanatory power measured by the R^2 is much smaller for the PIT Growth Rate models, which makes them more plausible to us. There is definitely no Laffer-type effect again. Statistically significant is only the Change in the Average Effective Personal Income Tax Rate in the last model, together with the change in the Unemployment Rate. There is no evidence for cross-effects of the growth of CITrate. We will use models (1) and (5) for later analysis.

A similar story goes for the Corporate Income Tax. In this case, the Growth Rate of CITrate is statistically significant only in the model (4). The Annual GDP Growth Rate is statistically significant here, whereas the change in Unemployment Rate is not. There is again no evidence for the Laffer-type effect. Cross-effects of the growth of PITrate are statistically insignificant. Models (1) and (4) will be utilised for forthcoming analysis.

5.5 Instrumental Variable Regression (IVR)

The OLS estimation is based on the assumptions mentioned in Table 5.11, one of the strongest ones is the "Zero Conditional Mean" or also called the "Exogeneity" assumption. When this assumption fails, the model fails, too. Thus, we need a method to solve the issue with possible endogeneity. Using so-called "Instrumental variables" and Two Stage Least Squares (2SLS) is one way to take care of it.

With this approach, we use additional variables (instruments) that help us ensure consistency and unbiasedness. Those variables are not in the base equation and must be correlated with the endogenous regressor. On the contrary, they must not be correlated with the error term.

Assume a model with an endogenous regressor y_2 and an instrument z_1 for this regressor. The base equation,

$$y_{1,t} = \beta_0 + \beta_1 y_{2,t} + \beta_2 x_{2,t} + \cdots + \beta_k x_{k,t} + \epsilon_t, \tag{5.47}$$

is known as the *structural equation*, to obtain an unbiased estimate $\hat{y}_{1,t}$, we form a so-called *reduced form equation*, which depicts the relationship between endogenous regressor, the instrument and other exogenous regressors,

$$y_{2,t} = \pi_0 + \pi_1 z_{1,t} + \pi_2 x_{2,t} + \cdots + \pi_k x_{k,t} + u_t. \tag{5.48}$$

Then we estimate the *reduced form equation* using OLS to get the fitted values for $\hat{y}_{2,t}$. In the second stage of the process (2SLS), we use $\hat{y}_{2,t}$ instead of $y_{2,t}$ to estimate the *structural equation*,

$$\hat{y}_{2,t} = \hat{\pi}_0 + \hat{\pi}_1 z_{1,t} + \hat{\pi}_2 x_{2,t} + \cdots + \hat{\pi}_k x_{k,t}, \tag{5.49}$$

$$y_{1,t} = \beta_0 + \beta_1 \hat{y}_{2,t} + \beta_2 x_{2,t} + \cdots + \beta_k x_{k,t} + \epsilon_t. \tag{5.50}$$

Using the predicted value, $\hat{y}_{2,t}$ addresses the endogeneity because it is constructed only from exogenous variables and instruments, which are assumed uncorrelated with ϵ_t. This method is only applicable when we have at least 1 instrument for each endogenous regressor and if $\pi_1 \neq 0$, otherwise we suffer from perfect multicollinearity as $\hat{y}_{2,t}$ would be a perfect linear combination of the exogenous regressors.

If we had more endogenous regressors, more *reduced form equations* would be utilised, in each of them, the instruments would be used instead of the endogenous variables. Fitted values would then be inserted into the *structural equation*.

Utilising the IVR and 2SLS method has its pros and cons. In case of endogeneity, we ensure consistency with IVR but lose efficiency. Variance and standard errors are larger for IVR estimators than for OLS estimators. Finally, if $y_{2,t}$ is exogenous, the IVR estimator is less efficient than the OLS estimator. For elaboration, see Wooldridge (2013) or Stock and Watson (2020). Nevertheless, we need a test to ensure the usage of IVR is necessary.

The endogeneity test of a regressor can be tested by the Durbin-Wu-Hausman Endogeneity Test (Wu-Hausman). The null hypothesis says that the regressor is exogenous, whereas the alternative hypothesis says it is endogenous. We also need a test to find whether the instrument is sufficiently correlated with the endogenous regressor, so-called Weak Instruments Test (WI) - the null hypothesis implies a weak correlation, whereas the alternative implies a strong correlation. Finally, the Sargan Test (Sargan) shows whether the instruments are correlated with the error term.[29] For more detailed description, see Hausman (1978), Sargan (1958), Stock and Yogo (2004), and Wooldridge (2013).

5.5.1 Stationarity Assuming IVR Models

We have concerns about the endogeneity of the developed models in Section 5.4.1. A decrease in tax revenues could lead to an increase in the tax rate as the politicians would try to regain their lost income. Furthermore, with fewer funds to receive and possibly larger deficits, the government could cut programmes creating jobs in the market or even leading to GDP growth. Therefore, even the control variables might be biased because of endogeneity.

Our instrumental variables for these issues will be the first and the second lags of the possibly endogenous explanatory variables (Trate, Trate^2, g, u). We assume the TG and our dummies to be exogenous. For the estimation, we use models described in Section 5.4.1 and the approach introduced in the previous section. The possibly endogenous explanatory variables are replaced by their fitted values from the regression with the instruments.

The relevant tests and goodness-of-fit measures are reported together with the regression results. The R^2 and the Adjusted R^2 lose their interpretation, as they can reach negative values. Standard errors are in parentheses. We continue using the robust SE where needed with the same decision-making process that was described in Section 5.4. For the outputs, see tables 5.24, 5.25 and 5.26.

The results of the Durbin-Wu-Hausman Endogeneity Test showed the evidence for the endogeneity only for the Trate. The endogeneity is not statistically significant for the PIT and the CIT.[30] Thus, the OLS models are better in estimating the relationships considering the Personal Income Tax and Corporate Income Tax.

For the Trate, our selected instruments proved to be useful, as shown by the respective Weak Instruments Tests. As our model is overidentified, we can compute the Sargan Test, which for our case proves that the instruments are uncorrelated with the error term and therefore are likely valid, except for the first regression.

The Laffer effect is statistically significant in most cases, the Unemployment Rate in all of them. The TG is not statistically significant, again, as well as the Annual GDP Growth Rate. The revenue-maximising tax rate fluctuates from 28.4% to 28.8%.

[29] This test is a special version of the J-test utilised in the GMM analysis.

[30] Except for 1 estimation for CIT.

Table 5.24: Instrumental Variable Regression Output for the Total Tax System

	Dependent variable: ln(TTR) (1)	(2)	(3)	(4)
Trate	205.611***	388.117**	95.089***	158.071
	p = 0.000	p = 0.050	p = 0.004	p = 0.140
	(50.762)	(197.605)	(32.826)	(106.963)
Trate^2	−361.754***	−680.780**	−165.039***	−275.354
	p = 0.000	p = 0.048	p = 0.005	p = 0.139
	(89.048)	(343.597)	(57.578)	(185.990)
Partial	−0.238***		−0.231***	
	p = 0.003		p = 0.003	
	(0.079)		(0.075)	
Windfall	0.214***	0.297*	0.079***	0.103
	p = 0.002	p = 0.077	p = 0.009	p = 0.169
	(0.068)	(0.167)	(0.030)	(0.075)
TG			−0.116	−0.122
			p = 0.315	p = 0.301
			(0.116)	(0.118)
g			0.373	−0.027
			p = 0.329	p = 0.959
			(0.382)	(0.522)
u			−4.300***	−3.855***
			p = 0.000	p = 0.001
			(0.722)	(1.132)
Constant	−16.462**	−42.444	−0.744	−9.701
	p = 0.023	p = 0.134	p = 0.874	p = 0.527
	(7.209)	(28.290)	(4.675)	(15.329)
Rev-max Tax Rate	0.284	0.285	0.288	0.287
BP test (p-value)	0.182	0.495	0.000	0.006
DW test (p-value)	0.000	0.000	0.000	0.000
WI (Trate) (p-value)	0.000	0.000	0.000	0.000
WI (Trate^2) (p-value)	0.000	0.000	0.000	0.000
WI (g) (p-value)	–	–	0.000	0.000
WI (u) (p-value)	–	–	0.000	0.000
Wu-Hausman (p-value)	0.000	0.000	0.011	0.016
Sargan (p-value)	0.022	0.662	0.294	0.987
Observations	106	88	106	88
R^2	0.478	−2.199	0.851	0.388
Adjusted R^2	0.457	−2.313	0.840	0.343
Residual Std. Error	0.157	0.238	0.085	0.106

Note: Performed on regressors in decimal format. *p<0.1; **p<0.05; ***p<0.01 Regressions (1) and (3) performed on data from 1997 Q1 to 2023 Q4; Regressions (2) and (4) performed on data from 2001 Q1 to 2023 Q4.

Table 5.25: Instrumental Variable Regression Output for the PIT

	Dependent variable:			
	ln(PITR)			
	(1)	(2)	(3)	(4)
PITrate	32.943***	32.767***	22.730***	22.542***
	p = 0.000	p = 0.000	p = 0.000	p = 0.000
	(8.420)	(8.022)	(4.592)	(4.668)
PITrate^2	−115.268***	−113.431***	−64.100***	−62.734***
	p = 0.001	p = 0.001	p = 0.000	p = 0.001
	(31.202)	(29.507)	(16.179)	(16.866)
CITrate			−1.395**	−1.457*
			p = 0.033	p = 0.073
			(0.654)	(0.811)
Partial	−0.848***		−0.750***	
	p = 0.000		p = 0.000	
	(0.036)		(0.046)	
TG			−0.018	−0.029
			p = 0.853	p = 0.771
			(0.094)	(0.101)
g			−0.224	−0.163
			p = 0.678	p = 0.773
			(0.538)	(0.563)
u			−3.964***	−3.865***
			p = 0.000	p = 0.000
			(0.686)	(0.766)
Constant	8.012***	8.011***	8.980***	8.995***
	p = 0.000	p = 0.000	p = 0.000	p = 0.000
	(0.530)	(0.507)	(0.280)	(0.306)
Rev-max Tax Rate	0.143	0.144	0.177	0.180
BP test (p-value)	0.031	0.076	0.005	0.007
DW test (p-value)	0.000	0.001	0.514	0.526
WI (PITrate)	0.000	0.000	0.000	0.000
WI (PITrate^2)	0.000	0.000	0.000	0.000
WI (g) (p-value)	–	–	0.000	0.000
WI (u) (p-value)	–	–	0.000	0.000
Wu-Hausman (p-value)	0.117	0.569	0.079	0.099
Sargan (p-value)	0.444	0.493	0.143	0.369
Observations	106	88	106	88
R^2	0.823	0.405	0.886	0.614
Adjusted R^2	0.817	0.391	0.877	0.585
Residual Std. Error	0.129	0.135	0.106	0.111

Note: Performed on regressors in decimal format. *p<0.1; **p<0.05; ***p<0.01 Regressions (1) and (3) performed on data from 1997 Q1 to 2023 Q4; Regressions (2) and (4) performed on data from 2001 Q1 to 2023 Q4.

Table 5.26: Instrumental Variable Regression Output for the CIT

	Dependent variable:			
	ln(CITR)			
	(1)	(2)	(3)	(4)
CITrate	15.098***	34.400***	30.004***	43.171***
	p = 0.010	p = 0.001	p = 0.001	p = 0.000
	(5.692)	(9.259)	(7.829)	(9.388)
CITrate^2	−29.563**	−68.757***	−54.168***	−81.953***
	p = 0.011	p = 0.001	p = 0.001	p = 0.000
	(11.386)	(18.934)	(14.456)	(18.876)
PITrate			−1.518	−0.900
			p = 0.344	p = 0.451
			(1.594)	(1.193)
Partial	−0.249**		−0.206*	
	p = 0.044		p = 0.079	
	(0.122)		(0.116)	
Windfall	0.549***	0.563***	0.402***	0.438***
	p = 0.000	p = 0.000	p = 0.001	p = 0.001
	(0.115)	(0.092)	(0.115)	(0.118)
TG			−0.120	−0.064
			p = 0.560	p = 0.813
			(0.206)	(0.270)
g			0.250	0.273
			p = 0.792	p = 0.749
			(0.943)	(0.851)
u			−6.493***	−5.907***
			p = 0.000	p = 0.000
			(1.341)	(1.349)
Constant	8.327***	6.061***	6.867***	5.247***
	p = 0.000	p = 0.000	p = 0.000	p = 0.000
	(0.684)	(1.086)	(0.878)	(1.076)
Rev-max Tax Rate	0.255	0.250	0.277	0.263
BP test (p-value)	0.817	0.926	0.072	0.004
DW test (p-value)	0.080	0.005	0.686	0.320
WI (CITrate) (p-value)	0.000	0.000	0.000	0.000
WI (CITrate^2) (p-value)	0.000	0.000	0.000	0.000
WI (g) (p-value)	–	–	0.000	0.000
WI (u) (p-value)	–	–	0.000	0.000
Wu-Hausman (p-value)	0.244	0.258	0.134	0.021
Sargan (p-value)	0.402	0.456	0.059	0.090
Observations	106	88	106	88
R^2	0.412	0.325	0.526	0.516
Adjusted R^2	0.389	0.301	0.487	0.473
Residual Std. Error	0.223	0.195	0.204	0.169

Note: Performed on regressors in decimal format. *p<0.1; **p<0.05; ***p<0.01 Regressions (1) and (3) performed on data from 1997 Q1 to 2023 Q4; Regressions (2) and (4) performed on data from 2001 Q1 to 2023 Q4.

5.5.2 IVR Models Solving for Stationarity

As it was for the level models, we are concerned about the endogeneity in our model. A smaller Growth Rate of Tax Revenue could lead the government to increase the Tax Rate and therefore accelerate its Growth Rate. Similarly, a fall in the Growth Rate of Tax Revenue could make the government stop some of its social programmes, which might have led to a lower Annual GDP Growth Rate or smaller Change in the Unemployment Rate. Therefore, we build an IVR model containing the lagged values of the Growth Rates of Tax Rates, Annual GDP Growth Rate and the Growth Rate of the Unemployment Rate as instruments.

Table 5.27: IVR Output for the Growth Rates in the Total Tax System

	Dependent variable:			
	D_{TTR}			
	(1)	(2)	(3)	(4)
D_{Trate}	1.226***	1.201***	1.101***	1.092***
	p = 0.000	p = 0.000	p = 0.000	p = 0.000
	(0.306)	(0.305)	(0.118)	(0.176)
D^2_{Trate}	−0.764	−0.569		
	p = 0.745	p = 0.821		
	(2.349)	(2.510)		
Partial	−0.010		0.008	
	p = 0.433		p = 0.398	
	(0.013)		(0.009)	
Windfall	−0.035***	−0.034***	−0.021***	−0.021**
	p = 0.000	p = 0.002	p = 0.005	p = 0.015
	(0.008)	(0.011)	(0.007)	(0.009)
D_u			−0.065***	−0.061**
			p = 0.000	p = 0.015
			(0.015)	(0.025)
Constant	0.029***	0.027***	0.023***	0.023***
	p = 0.002	p = 0.008	p = 0.001	p = 0.000
	(0.009)	(0.010)	(0.006)	(0.005)
D_{TTR}-max D_{Trate}	0.802	1.055	–	–
BP test (p-value)	0.007	0.007	0.017	0.031
DW test (p-value)	0.000	0.000	0.000	0.000
WI (D_{Trate}) (p-value)	0.018	0.019	0.119	0.105
WI (D^2_{Trate}) (p-value)	0.090	0.126	–	–
WI D_u (p-value)	–	–	0.000	0.000
Wu-Hausman (p-value)	0.843	0.924	0.000	0.000
Sargan (p-value)	0.501	0.528	0.079	0.177
Observations	106	88	106	88
R^2	0.857	0.857	0.907	0.900
Adjusted R^2	0.851	0.852	0.903	0.897
Residual Std. Error	0.031	0.032	0.025	0.027

Note: Performed on regressors in decimal format. *p<0.1; **p<0.05; ***p<0.01 Regressions (1) and (3) performed on data from 1997 Q1 to 2023 Q4; Regressions (2) and (4) performed on data from 2001 Q1 to 2023 Q4.

Table 5.28: Instrumental Variable Regression Output for the Growth Rates in the PIT

	Dependent variable:			
	D_{PITR}			
	(1)	(2)	(3)	(4)
$D_{PITrate}$	−0.137 p = 0.923 (1.399)	1.305** p = 0.015 (0.521)	0.923*** p = 0.000 (0.190)	0.786*** p = 0.000 (0.122)
$D^2_{PITrate}$	−3.923 p = 0.495 (5.735)	2.114 p = 0.274 (1.917)		
Partial	−0.132 p = 0.108 (0.082)		−0.066 p = 0.201 (0.051)	
D_u			−0.150* p = 0.060 (0.080)	−0.115** p = 0.030 (0.052)
Constant	0.106 p = 0.237 (0.090)	0.006 p = 0.771 (0.019)	0.071 p = 0.133 (0.047)	0.020* p = 0.063 (0.011)
D_{PITR}-max $D_{PITrate}$	-0.017	-0.309	–	–
BP test (p-value)	0.131	0.645	0.794	0.551
DW test (p-value)	0.000	0.055	0.000	0.297
WI ($D_{PITrate}$) (p-value)	0.000	0.000	0.000	0.000
WI ($D^2_{PITrate}$) (p-value)	0.000	0.000	–	–
WI D_u (p-value)	–	–	0.319	0.000
Wu-Hausman (p-value)	0.659	0.257	0.319	0.000
Sargan (p-value)	0.756	0.971	0.580	0.180
Observations	106	88	106	88
R^2	0.112	0.324	0.120	0.399
Adjusted R^2	0.086	0.308	0.094	0.385
Residual Std. Error	0.239	0.102	0.238	0.097

Note: Performed on regressors in decimal format. *p<0.1; **p<0.05; ***p<0.01 Regressions (1) and (3) performed on data from 1997 Q1 to 2023 Q4; Regressions (2) and (4) performed on data from 2001 Q1 to 2023 Q4.

Table 5.29: Instrumental Variable Regression Output for the Growth Rates in the CIT

	Dependent variable:			
	D_{CITR}			
	(1)	(2)	(3)	(4)
$D_{CITrate}$	5.529 p = 0.174 (4.034)	5.310* p = 0.069 (2.872)	11.082** p = 0.038 (5.255)	9.040* p = 0.062 (4.773)
$D_{CITrate^2}$	50.114 p = 0.221 (40.664)	45.255 p = 0.130 (29.562)	98.027* p = 0.054 (50.258)	74.464* p = 0.091 (43.599)
Partial	−0.101 p = 0.385 (0.116)		−0.180 p = 0.205 (0.141)	
Windfall	0.550*** p = 0.003 (0.177)	0.569*** p = 0.000 (0.124)	0.573*** p = 0.002 (0.176)	0.601*** p = 0.001 (0.175)
D_{TG}			0.029 p = 0.729 (0.084)	0.039 p = 0.638 (0.083)
g			4.302** p = 0.033 (1.988)	3.776* p = 0.054 (1.931)
D_u			0.450 p = 0.136 (0.298)	0.270 p = 0.287 (0.252)
Constant	0.085* p = 0.058 (0.044)	0.066** p = 0.042 (0.032)	0.021 p = 0.689 (0.051)	0.013 p = 0.808 (0.051)
D_{CITR}-max $D_{CITrate}$	-0.055	-0.059	-0.057	-0.061
BP test (p-value)	0.945	0.189	0.995	0.987
DW test (p-value)	0.533	0.225	0.653	0.637
WI $D_{CITrate}$ (p-value)	0.000	0.000	0.000	0.000
WI $D_{CITrate^2}$ (p-value)	0.000	0.000	0.000	0.000
WI (g) (p-value)	–	–	0.000	0.000
WI D_u (p-value)	–	–	0.000	0.000
Wu-Hausman (p-value)	0.951	0.827	0.820	0.824
Sargan (p-value)	0.713	0.395	0.886	0.791
Observations	106	88	106	106
R^2	0.123	0.248	0.183	0.178
Adjusted R^2	0.088	0.221	0.125	0.128
Residual Std. Error	0.342	0.239	0.335	0.334

Note: Performed on regressors in decimal format. *p<0.1; **p<0.05; ***p<0.01 Regressions (1) and (3) performed on data from 1997 Q1 to 2023 Q4; Regressions (2) and (4) performed on data from 2001 Q1 to 2023 Q4.

To start with, only the more complex models for the overall tax system and one model for PIT give statistically significant results of endogeneity. At the same time, for the total tax system, instruments for the Growth Rate of the Average Overall Tax Rate do not seem statistically significant. A statistically significant Laffer effect definitely does not exist for the growth rates in these models. Based on various results of our models, which are mostly in favour of exogeneity, we take the OLS models as superior to the IVR ones.

The only difference lies in the models for the total tax system. We process to estimate the GMM models for all the taxes in order to make our findings more robust.

5.6 Generalised Method of Moments (GMM)

The last method utilised for our analysis is the Generalised Method of Moments. The generally used version of the method was introduced by Hansen (1982). In case of an overidentification,[31] the GMM is more efficient than the IVR as it better handles the additional given information. Furthermore, it does not require the data to be normally distributed (Hansen, 1982; Hayashi, 2000).

The method works as follows. Let $\mathbf{w}_t$ be an $(h \times 1)$ vector of variables observed at time t, let $\boldsymbol{\theta}$ stand for $(a \times 1)$ vector of unknown coefficients. and let $\mathbf{h}(\boldsymbol{\theta}, \mathbf{w}_t)$ be a $(r \times 1)$ vector-valued function. Let θ_0 denote the true population value of θ. This value is characterised by the property that

$$\mathbb{E}(\mathbf{h}(\theta, \mathbf{w_t})) = 0, \tag{5.51}$$

according to which the GMM estimates the model.

Furthermore, let $\mathbf{y_T} = (w'_T, w'_{T-1}, \dots, w'_1)$ stand for a $(Th \times 1)$ vector including all observations in a sample of size T. Then the function $\mathbf{g}(\boldsymbol{\theta}, \mathbf{y}_T)$ provides the sample average of $\mathbf{h}(\theta, \mathbf{w_t})$ as follows,

$$\mathbf{g}(\boldsymbol{\theta}, \mathbf{y}_T) = \frac{1}{T}\sum_{t=1}^{T} \mathbf{h}(\boldsymbol{\theta}, \mathbf{w}_t). \tag{5.52}$$

The core idea behind GMM is to choose $\boldsymbol{\theta}$ such that the sample moment $\mathbf{g}(\theta, \mathbf{y_T})$ is as close as possible to the population moment of zero. Mathematically, the GMM estimator $\hat{\theta}_\mathbf{T}$ is the value of $\boldsymbol{\theta}$ which follows the optimisation problem

$$Q(\boldsymbol{\theta}, \mathbf{y}_T) = \arg\min_{\boldsymbol{\theta}} [\mathbf{g}(\boldsymbol{\theta}, \mathbf{y}_T)]' \mathbf{W}_T [\mathbf{g}(\boldsymbol{\theta}, \mathbf{y}_T)], \tag{5.53}$$

where $Q(\boldsymbol{\theta}, \mathbf{y}_T)$ stands for the vector-valued function of the optimal $\boldsymbol{\theta}$ and $\mathbf{W}_t$ denotes a sequence of positive definite weighting matrices.

The optimal value for the weighting matrix $\mathbf{W}_t$ is given by the inverse of the asymptotic variance matrix, $\mathbf{S}^{-1}$, when

$$\mathbf{S} = \lim_{T \to \infty} T \times E([\mathbf{g}(\boldsymbol{\theta}, \mathbf{y}_T)][\mathbf{g}(\boldsymbol{\theta}, \mathbf{y}_T)]'). \tag{5.54}$$

The equation (5.53) can therefore be rewritten as

$$Q(\boldsymbol{\theta}, \mathbf{y}_T) = \arg\min_{\boldsymbol{\theta}} [\mathbf{g}(\boldsymbol{\theta}, \mathbf{y}_T)]' \mathbf{S}^{-1} [\mathbf{g}(\boldsymbol{\theta}, \mathbf{y}_T)]. \tag{5.55}$$

[31] Having more instruments than endogenous regressors.

If those conditions are satisfied, the GMM estimator is consistent, asymptotically normal and with a correct choice of the weighting matrix $\mathbf{W}_T$ also asymptotically efficient (Hamilton, 1994).

An important statistic for the GMM models is the J-statistic. It serves as a goodness-of-fit measure when we dispose of more moment conditions than parameters to estimate. Thus, it helps to determine whether our model's overidentifying restrictions are valid. A J-statistic above the critical values indicates the model's misspecification (Stock & Watson, 2020).

For the derivations and more elaborate discussion about the principle of GMM, see Hamilton (1994) and Hansen (1982).

5.6.1 Stationarity Assuming GMM Models

We estimated overidentified models in Section 5.5.1. The GMM is better at handling the additional information received from the overidentification. Furthermore, it does not require normally distributed data. It also automatically uses the HAC to obtain relevant p-values.

The nature of our models is unchanged from the model utilised for the Instrumental Variable Regression. Formally, our moment conditions state,

$$\mathbb{E}(h(\boldsymbol{\theta}, z_{i,t})) = 0, \quad h(\theta, z_{i,t}) = z_{i,t}(y_t - x'_{i,t}\theta), \tag{5.56}$$

where $\mathbf{z_{i,t}}$ is the vector containing the values of the i^{th} instrument in time. We use the same instruments as we did in Section 5.5.1.

We report the J-Test p-values, which denote whether the instruments are exogenous and whether the model is correctly specified.[32] For the regression results, see tables 5.30, 5.31 and 5.32.

[32]This is the null hypothesis.

Table 5.30: Generalised Method of Moments Output for the Total Tax System

	Dependent variable:			
	ln(TTR)			
	(1)	(2)	(3)	(4)
Trate	199.768***	369.426**	77.389***	192.734**
	p = 0.000	p = 0.012	p = 0.000	p = 0.015
	(49.426)	(146.487)	(19.079)	(78.690)
Trate^2	−352.576***	−648.767**	−135.073***	−336.049**
	p = 0.000	p = 0.012	p = 0.000	p = 0.015
	(86.422)	(255.470)	(32.613)	(137.920)
Partial	−0.313***		−0.279***	
	p = 0.000		p = 0.000	
	(0.068)		(0.050)	
Windfall	0.216***	0.292*	0.067**	0.121
	p = 0.005	p = 0.056	p = 0.017	p = 0.137
	(0.076)	(0.152)	(0.028)	(0.081)
TG			−0.138	−0.132*
			p = 0.126	p = 0.076
			(0.090)	(0.074)
g			0.698**	−0.072
			p = 0.050	p = 0.850
			(0.356)	(0.380)
u			−4.436***	−3.532***
			p = 0.000	p = 0.002
			(0.581)	(1.092)
Constant	−15.531**	−39.724*	1.867	−14.637
	p = 0.028	p = 0.058	p = 0.504	p = 0.193
	(7.041)	(20.910)	(2.788)	(11.226)
Rev-max Tax Rate	0.283	0.285	0.286	0.287
J-Test (p-value)	0.031	0.739	0.161	0.966
Observations	106	88	106	88

Note: Performed on regressors in decimal format. *p<0.1; **p<0.05; ***p<0.01 Regressions (1) and (3) performed on data from 1997 Q1 to 2023 Q4; Regressions (2) and (4) performed on data from 2001 Q1 to 2023 Q4.

Table 5.31: Generalised Method of Moments Output for the PIT

	Dependent variable:			
	ln(PITR)			
	(1)	(2)	(3)	(4)
PITrate	32.292***	32.462***	23.076***	22.801***
	p = 0.001	p = 0.001	p = 0.000	p = 0.000
	(8.395)	(8.484)	(2.913)	(2.801)
PITrate^2	−112.089***	−111.585***	−62.379***	−61.865***
	p = 0.001	p = 0.001	p = 0.000	p = 0.000
	(30.842)	(30.980)	(10.588)	(10.036)
CITrate			−1.721***	−1.618***
			p = 0.000	p = 0.001
			(0.349)	(0.457)
Partial	−0.852***		−0.739***	
	p = 0.000		p = 0.000	
	(0.034)		(0.031)	
TG			0.003	−0.012
			p = 0.977	p = 0.918
			(0.114)	(0.119)
g			−0.374	−0.255
			p = 0.274	p = 0.449
			(0.341)	(0.336)
u			−4.787***	−4.578***
			p = 0.000	p = 0.000
			(0.449)	(0.487)
Constant	8.037***	8.015***	9.028***	9.024***
	p = 0.000	p = 0.000	p = 0.000	p = 0.000
	(0.533)	(0.541)	(0.180)	(0.173)
Rev-max Tax Rate	0.144	0.145	0.185	0.184
J-Test (p-value)	0.344	0.391	0.340	0.585
Observations	106	88	106	88

Note: Performed on regressors in decimal format. *p<0.1; **p<0.05; ***p<0.01 Regressions (1) and (3) performed on data from 1997 Q1 to 2023 Q4; Regressions (2) and (4) performed on data from 2001 Q1 to 2023 Q4.

Table 5.32: Generalised Method of Moments Output for the CIT

	Dependent variable:			
	ln(CITR)			
	(1)	(2)	(3)	(4)
CITrate	18.424***	25.950***	30.873***	32.827***
	p = 0.001	p = 0.001	p = 0.000	p = 0.000
	(5.489)	(7.419)	(5.231)	(7.334)
CITrate^2	−36.190***	−51.517***	−56.824***	−61.656***
	p = 0.002	p = 0.001	p = 0.000	p = 0.000
	(11.015)	(15.143)	(9.655)	(14.194)
PITrate			−0.309	−0.559
			p = 0.797	p = 0.629
			(1.202)	(1.155)
Partial	−0.172		−0.138	
	p = 0.168		p = 0.160	
	(0.124)		(0.098)	
Windfall	0.553***	0.556***	0.398***	0.441***
	p = 0.000	p = 0.000	p = 0.000	p = 0.000
	(0.091)	(0.106)	(0.081)	(0.097)
TG			−0.337*	−0.210
			p = 0.100	p = 0.229
			(0.204)	(0.175)
g			0.664	1.172**
			p = 0.186	p = 0.030
			(0.501)	(0.539)
u			−7.031***	−5.934***
			p = 0.000	p = 0.000
			(0.949)	(0.864)
Constant	7.929***	7.051***	6.768***	6.497***
	p = 0.000	p = 0.000	p = 0.000	p = 0.000
	(0.658)	(0.872)	(0.568)	(0.829)
Rev-max Tax Rate	0.255	0.252	0.272	0.266
J-Test (p-value)	0.231	0.373	0.103	0.111
Observations	106	88	106	88

Note: Performed on regressors in decimal format. *p<0.1; **p<0.05; ***p<0.01 Regressions (1) and (3) performed on data from 1997 Q1 to 2023 Q4; Regressions (2) and (4) performed on data from 2001 Q1 to 2023 Q4.

The estimation of the overall tax system model proved the Laffer effect to be statistically significant, with the revenue-maximising tax rate being around 28.5%. The Unemployment Rate is statistically significant again, which is true for the Annual GDP Growth Rate, too. The J-test indicates correctly selected instruments for all regressions except the first one.

Considering the PIT, this method also proved the existence of a statistically significant Laffer effect. The revenue-maximising rate varies with the regressors used. The cross-effect of CITrate is still statistically significant. For the CIT, the results indicate the existence of a statistically significant Laffer effect with the revenue-maximising rate ranging from 25.2% to 27.1%. Models are correctly specified as shown by the J-Test, there is very little evidence for the existence of cross-effects as PITrate is statistically insignificant at the 5% level. The Unemployment Rate is statistically significant as expected, and for the last regression, the Annual GDP Growth Rate is as well.

5.6.2 GMM Models Solving for Stationarity

The last set of models is dedicated to the Growth rates. We use the same variables and instruments as in Section 5.5.2. Our moment conditions are unchanged from the previous section, with the obvious difference in utilised instruments.

Table 5.33: GMM Output for the Growth Rates in the Total Tax System

	Dependent variable:			
	D_{TTR}			
	(1)	(2)	(3)	(4)
D_{Trate}	0.860***	0.923***	0.839***	0.873***
	p = 0.001	p = 0.000	p = 0.000	p = 0.000
	(0.248)	(0.189)	(0.122)	(0.122)
D^2_{Trate}	1.980	1.237		
	p = 0.256	p = 0.365		
	(1.742)	(1.362)		
Partial	−0.018		−0.011	
	p = 0.142		p = 0.340	
	(0.012)		(0.012)	
Windfall	−0.028***	−0.030***	−0.026***	−0.030***
	p = 0.009	p = 0.002	p = 0.010	p = 0.002
	(0.011)	(0.010)	(0.010)	(0.009)
D_u			−0.062***	−0.049**
			p = 0.001	p = 0.013
			(0.018)	(0.020)
Constant	0.023**	0.026***	0.036***	0.037***
	p = 0.030	p = 0.009	p = 0.000	p = 0.000
	(0.010)	(0.010)	(0.006)	(0.006)
D_{TTR}-max D_{Trate}	-0.217	-0.373	–	–
J-Test (p-value)	0.560	0.732	0.037	0.111
Observations	106	88	106	88

Note: Performed on regressors in decimal format. *p<0.1; **p<0.05; ***p<0.01 Regressions (1) and (3) performed on data from 1997 Q1 to 2023 Q4; Regressions (2) and (4) performed on data from 2001 Q1 to 2023 Q4.

The regressions results are in tables 5.33, 5.34 and 5.35. For the overall tax system and the Personal Income Tax, there is again no statistically significant evidence of the Laffer effect. Surprisingly, the analysis of the Growth Rates in the Corporate Income Tax shows a significant opposite effect, the existence of a change in the CITrate that would lead to the smallest change in the Corporate Income Tax Revenue.

This effect was already almost confirmed in the IVR analysis. The changes in the Unemployment Rate are statistically only for the overall tax system and the PIT. For CIT, the Annual GDP Growth Rate is statistically significant. J-test signals invalidity of the overidentifying restrictions for the extended models for PIT and one of the extended models for the overall tax system.

For more elaborate comments on the results obtained from all the models, see the next section, Results and Discussion.

Table 5.34: Generalised Method of Moments Output for the Growth Rates in the Personal Income Tax

	Dependent variable:			
	D_{PITR}			
	(1)	(2)	(3)	(4)
$D_{PITrate}$	1.067***	1.055***	0.773***	0.817***
	p = 0.000	p = 0.000	p = 0.006	p = 0.003
	(0.195)	(0.192)	(0.281)	(0.273)
$D^2_{PITrate}$	−1.342	−0.792		
	p = 0.695	p = 0.844		
	(3.420)	(4.011)		
Partial	−0.016		0.002	
	p = 0.511		p = 0.869	
	(0.024)		(0.013)	
D_u			−0.091***	−0.092***
			p = 0.000	p = 0.000
			(0.015)	(0.014)
Constant	0.033**	0.032*	0.029***	0.029***
	p = 0.038	p = 0.056	p = 0.000	p = 0.000
	(0.016)	(0.017)	(0.006)	(0.006)
D_{PITR}-max $D_{PITrate}$	0.397	0.666	–	–
J-Test (p-value)	0.123	0.136	0.014	0.028
Observations	106	88	106	88

Note: Performed on regressors in decimal format. *p<0.1; **p<0.05; ***p<0.01 Regressions (1) and (3) performed on data from 1997 Q1 to 2023 Q4; Regressions (2) and (4) performed on data from 2001 Q1 to 2023 Q4.

Table 5.35: Generalised Method of Moments Output for the Growth Rates in the Corporate Income Tax

	Dependent variable:			
	D_{CITR}			
	(1)	(2)	(3)	(4)
$D_{CITrate}$	0.717***	0.729***	1.018***	1.019***
	p = 0.000	p = 0.000	p = 0.000	p = 0.000
	(0.132)	(0.149)	(0.009)	(0.011)
$D_{CITrate^2}$	−4.024	−3.169*	0.332**	0.330**
	p = 0.231	p = 0.097	p = 0.021	p = 0.022
	(3.355)	(1.905)	(0.143)	(0.143)
Partial	−0.033*		−0.000	
	p = 0.058		p = 0.904	
	(0.017)		(0.001)	
Windfall	−0.024	−0.024	−0.002	−0.002
	p = 0.184	p = 0.164	p = 0.141	p = 0.151
	(0.018)	(0.017)	(0.001)	(0.001)
D_{TG}			0.000	0.000
			p = 0.822	p = 0.740
			(0.000)	(0.000)
g			0.960***	0.951***
			p = 0.000	p = 0.000
			(0.017)	(0.015)
D_u			−0.000	−0.001
			p = 0.946	p = 0.638
			(0.002)	(0.002)
Constant	0.048***	0.044***	−0.000	0.000
	p = 0.000	p = 0.000	p = 0.981	p = 0.857
	(0.012)	(0.006)	(0.001)	(0.001)
D_{CITR}-max $D_{CITrate}$	0.089	0.115	-1.533	-1.546
J-Test (p-value)	0.843	0.794	0.808	0.392
Observations	106	88	106	88

Note: Performed on regressors in decimal format. *p<0.1; **p<0.05; ***p<0.01 Regressions (1) and (3) performed on data from 1997 Q1 to 2023 Q4; Regressions (2) and (4) performed on data from 2001 Q1 to 2023 Q4.

6 Results and Discussion

In this section, we connect the findings made throughout our work and present some overall results. To begin with, in Section 5.3.1 we have shown the issue with stationarity. This led us to creating 2 separate branches of our models, one of which took stationarity as granted and the other utilising the growth rates instead of the levels of our variables. That being said, there is no statistically correct way to estimate the models for level variables. Therefore, there is no way to find a plausible, unbiased revenue-maximising tax rate for the presented taxes in the Czech Republic.

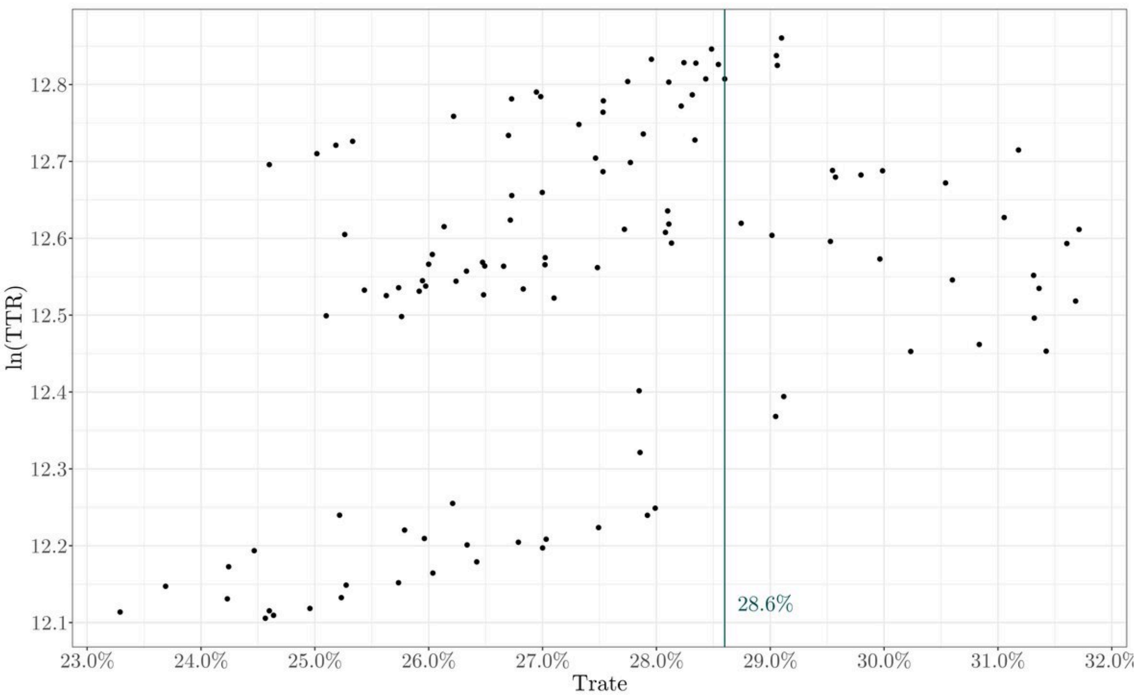
6.1 Stationarity Assuming Models

The stationarity assumption, just partly supported by our data, allowed us to model the relationships for level variables. For most of the models, a statistically significant Laffer effect was found. For both the overall tax system and individual taxes, it was shown that the Trust in the Government does not have a statistically significant effect on the tax revenue.

6.1.1 Overall Tax System

All the methods we utilised showed the existence of a Laffer Curve and a revenue-maximising tax rate for the overall tax system. We also proved endogeneity issues with the model, which were solved by utilising the Instrumental Variable Regression and then Generalised Method of Moments. We consider the second to be the best as it better uses the additional information, and the J-statistics are valid.

Figure 6.1: Values of ln(TTR) Conditional on Trate



Note: Author's own elaboration based on data provided by the institutions mentioned in Section 5.1 and on variables calculated by the procedures described in Section 5.2 and 5.3.1.

The revenue-maximising tax rate is therefore 28.6%, it is visualised in Figure 6.1.[33] For the summary of estimated revenue-maximising rates, see Table 6.1.

Table 6.1: Total Tax Revenue Maximising Average Overall Tax Rate

Estimation	OLS		IVR		GMM		Actual Trate
	Long	Short	Long	Short	Long	Short	2024 Q4
(1)	0.286	0.286	0.284	0.285	0.283	0.285	
(2)	0.286	0.286	–	–	–	–	
(3)	0.285	0.286	–	–	–	–	0.262
(4)	0.303	0.302	–	–	–	–	
(5)	0.299	0.302	0.288	0.287	0.286	0.287	

Note: Author's own elaboration based on estimated regression results and data from the data set.

Based on our analysis, the Czech Republic experienced periods with the Average Overall Tax Rate being higher than the revenue-maximising tax rate, with the last one dating back to 2020 Q1 and Q4. The last observed Trate in our data set was 26.2% in 2023 Q4.

There is also statistically significant evidence for the positive relationship between the Annual GDP Growth Rate and the Total Tax Revenue and the negative effect of an increase in the Unemployment Rate on the Total Tax Revenue. All effects are considered *ceteris paribus*. These links have been continuously proven throughout the analysis.

6.1.2 Personal Income Tax

For the Personal Income Tax, the endogeneity was not proved. When considering the OLS analysis, the Laffer effect was found statistically significant with the maximising rate approximately 17.5%. The summary of estimated revenue-maximising tax rates is observable in Table 6.2.

Table 6.2: Personal Income Tax Revenue Maximising Average Effective Personal Income Tax Rate

Estimation	OLS		IVR		GMM		Actual Trate
	Long	Short	Long	Short	Long	Short	2024 Q4
(1)	0.144	0.144	0.143	0.144	0.144	0.145	
(2)	0.154	0.175	–	–	–	–	
(3)	0.156	0.175	–	–	–	–	0.088
(4)	0.169	0.175	–	–	–	–	
(5)	0.156	0.157	0.177	0.180	0.185	0.184	

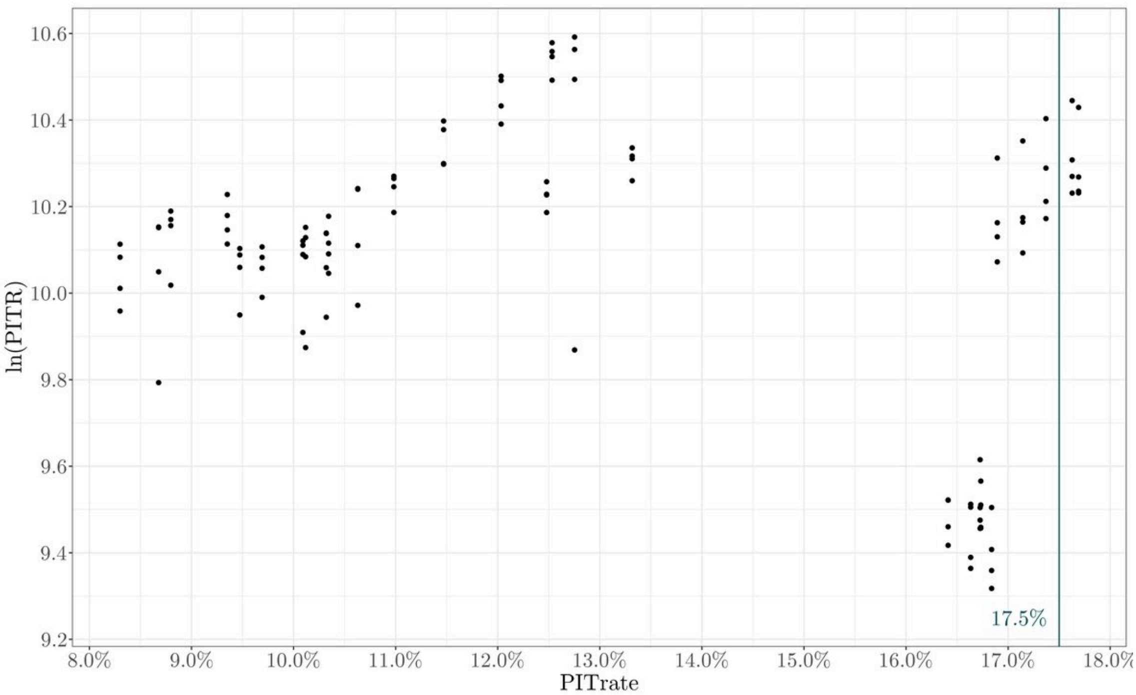
Note: Author's own elaboration based on estimated regression results and data from the data set.

The latest Average Effective Personal Income Tax Rate from our data set reached only 8.80% in 2023 Q4, which is far below the revenue-maximising tax rate. It

[33]We did not add a fitted values line as it is also affected by the control variables.

is important to note that the revenue-maximising tax rate is beyond almost all the Average Effective Personal Income Tax Rate that have ever been experienced in the Czech Republic, as observable in Figure 6.2. To construct a plausible Laffer Curve, we typically need datapoints from both sides.

Figure 6.2: Values of ln(PITR) Conditional on PITrate



Note: Author's own elaboration based on data provided by the institutions mentioned in Section 5.1 and on variables calculated by the procedures described in Section 5.2 and 5.3.1.

We also wanted to test whether there exists a cross-effect of CITrate. We failed to prove such a relationship in more complex models estimated for the longer period. On the contrary, all of those estimated for the shorter period signalled a statistically significant negative effect of CITrate. Such a relationship could have an economic explanation. As firms must pay a larger portion of their income, they have fewer funds to spend on employees' wages. Therefore, the tax base shrinks and consequently the tax does as well.

There is a negative link to the Unemployment Rate - as more people are unemployed, fewer people pay taxes, which leads to smaller Personal Income Tax Revenue. The Annual GDP Growth Rate is not statistically significant in the case of PIT. The effects of the windfall tax were not analysed here as it primarily affected only the CIT.

6.1.3 Corporate Income Tax

Considering the Corporate Income Tax, endogeneity was tested as statistically significant only for the most complex model estimated with the shorter TS. All the models proved the existence of a statistically significant Laffer Curve with the revenue-maximising tax rate 26.6% as estimated by the GMM for the endogenous model. For all the estimated revenue-maximising tax rates, see Table 6.3.

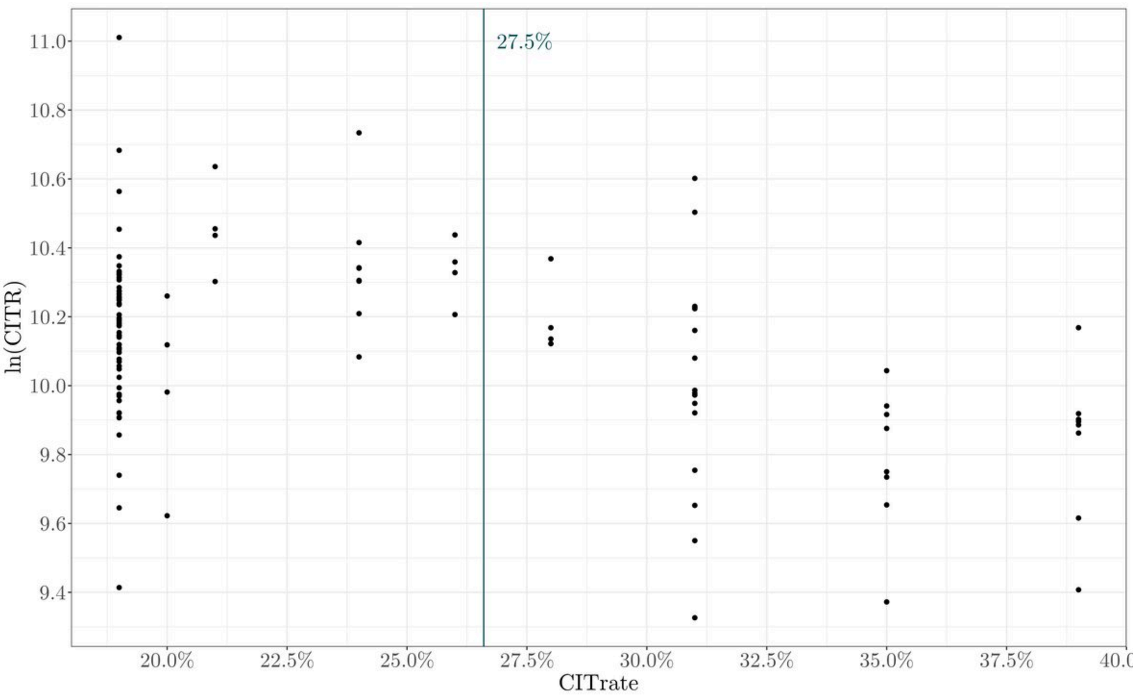
Table 6.3: Corporate Income Tax Revenue Maximising Corporate Income Tax Rate

Estimation	OLS		IVR		GMM		Actual Trate
	Long	Short	Long	Short	Long	Short	2024 Q4
(1)	0.258	0.251	0.255	0.250	0.255	0.252	
(2)	0.258	0.249	–	–	–	–	
(3)	0.256	0.248	–	–	–	–	0.21
(4)	0.283	0.267	–	–	–	–	
(5)	0.275	0.263	0.277	0.263	0.272	0.266	

Note: Author's own elaboration based on estimated regression results and data from the data set and (Czech Republic, 2023).

In this case, we had enough datapoints from both sides of the Laffer Curve as visualised in Figure 6.3. Even the higher estimated revenue-maximising tax rates, around 28%, would still have some other datapoints on both sides. The last Corporate Income Tax Rate from our data set reaches 19%. Nevertheless, effective from 2024, it has increased to 21%.

Figure 6.3: Values of ln(CITR) Conditional on CITrate



Note: Author's own elaboration based on data provided by the institutions mentioned in Section 5.1 and on variables calculated by the procedures described in Section 5.2 and 5.3.1.

No statistically significant cross-effects of PITrate were discovered in this case. The Unemployment Rate negatively affects the revenues. The GMM analysis was the only one that found a statistically significant contribution of the Annual GDP Growth Rate to the CITR, which makes sense as firms experience larger profits during economic booms, from which they pay their taxes. What is more, a growing economy encourages entrepreneurship, leading to the formation of new taxpayers.

6.2 Models Solving for Stationarity

In the second part of our analysis, we estimated the models with the growth rates. This crucially changed the interpretation but also made the models unbiased as they use stationary variables. We eliminated the risk of spurious regressions.

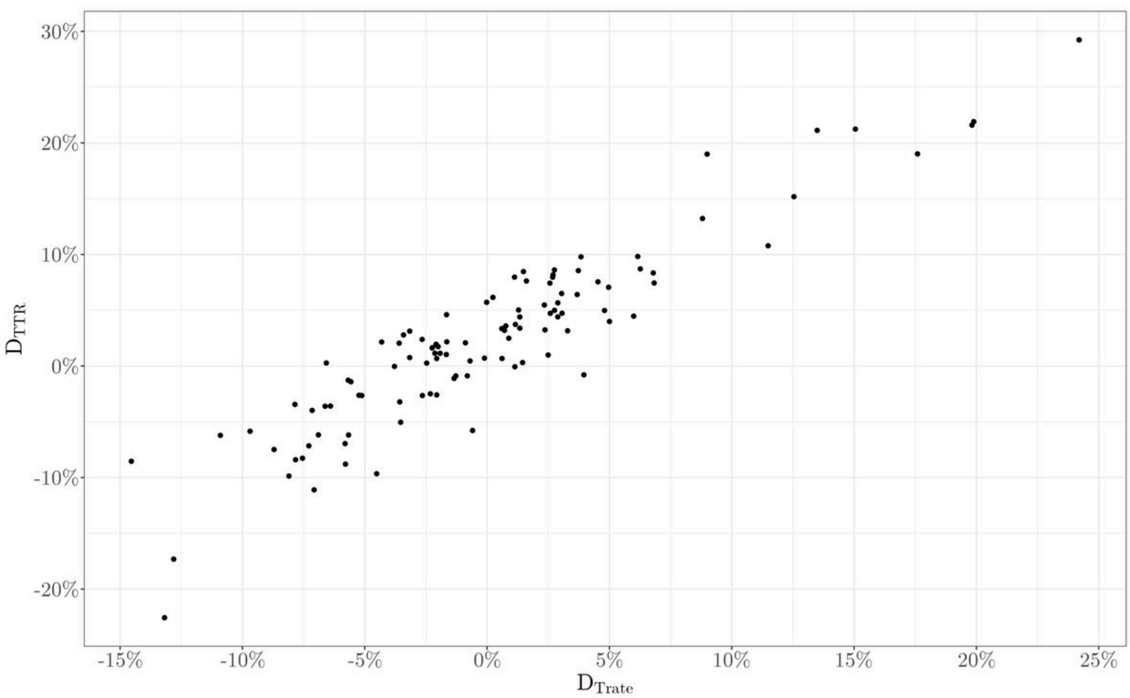
Nevertheless, these models show less consistent results. Our original goal in this analysis was to find whether there is a growth rate in tax rates that would accelerate the growth of tax revenues the most. There is no statistically significant evidence for such an effect in all the estimated models.

Generally, the simpler models performed better as they showed similar goodness-of-fit with more regressors being statistically significant. Based on these results, the relationship between the Growth Rate of a Tax Revenue and the Growth Rate of a Tax Rate is strictly linear, as shown in figures 6.4, 6.5 and 6.6.

Tax revenues are therefore linearly affected by their relevant tax rates. The Total Tax Revenue and the PITR are also statistically significantly determined by the growth rate of the Unemployment Rate, whereas the effect on CITR is statistically insignificant. On the contrary, Annual GDP Growth Rate is a statistically significant explanatory variable for the Corporate Income Tax Revenue, while it is not for Personal Income Tax Revenue.

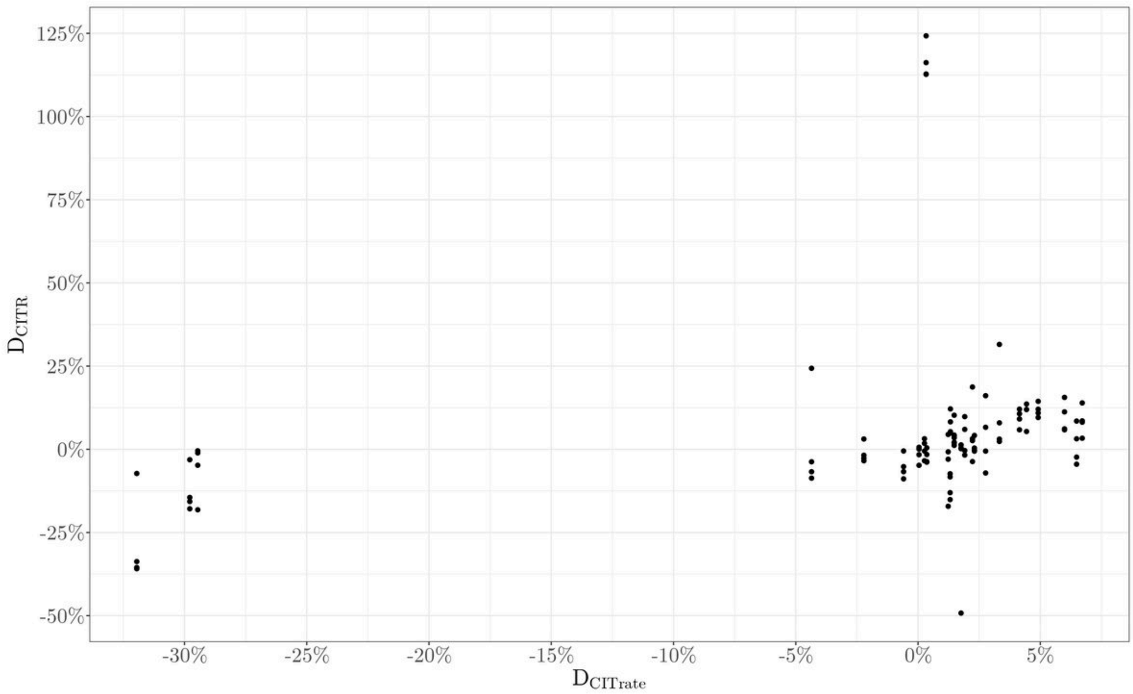
OLS regressions for the overall tax system raised questions because of their incredibly high R^2 and adjusted R^2, especially the model (4). This could have happened because of the way that $Trate$ was originally computed in Data Processing, which included the GDP. Thus, we have to take these results with a grain of salt.

Figure 6.4: Values of D_{TTR} Conditional on D_{Trate}



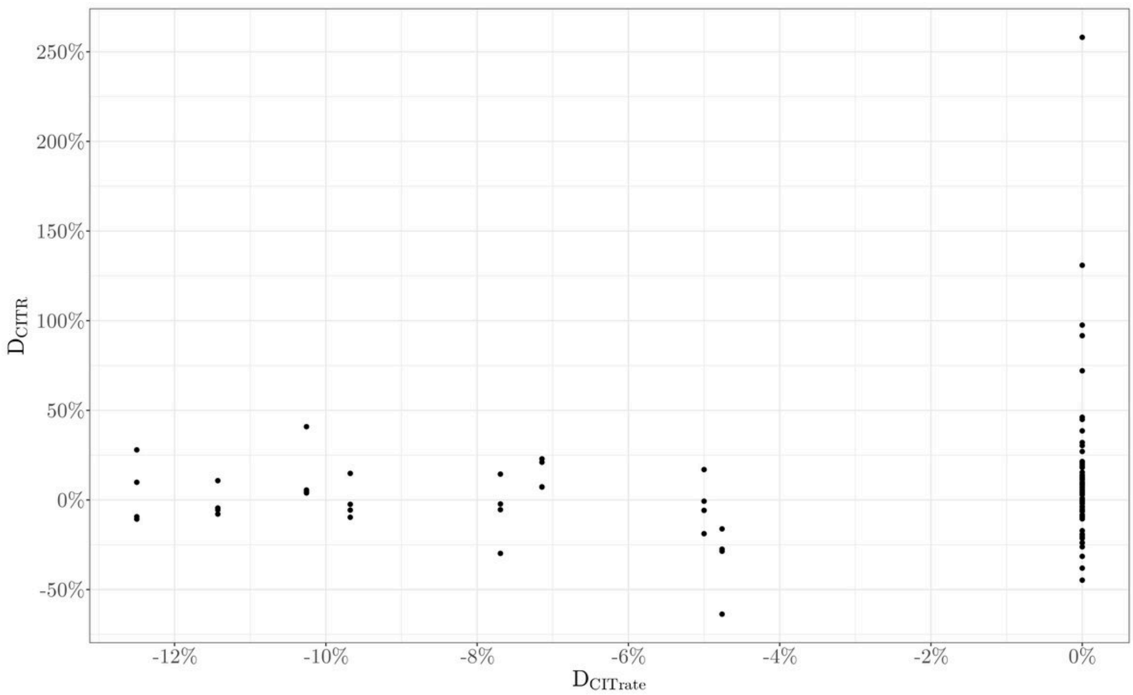
Note: Author's own elaboration based on data provided by the institutions mentioned in Section 5.1 and on variables calculated by the procedures described in Section 5.2 and 5.3.1.

Figure 6.5: Values of D_{PITR} Conditional on $D_{PITrate}$



Note: Author's own elaboration based on data provided by the institutions mentioned in Section 5.1 and on variables calculated by the procedures described in Section 5.2 and 5.3.1.

Figure 6.6: Values of D_{CITR} Conditional on $D_{CITrate}$



Note: Author's own elaboration based on data provided by the institutions mentioned in Section 5.1 and on variables calculated by the procedures described in Section 5.2 and 5.3.1.

7 Conclusion

The original crucial goal of this thesis was to obtain the tax revenue-maximising tax rate for the Czech tax system. This rate was to be found for the overall tax system, for the Personal Income Tax and the CIT.

Before analysing the system and individual taxes, we explained some basic terms and showed various economic consequences of taxation. Furthermore, we delved into the theory of optimal taxation and showed the contributions of the greats such as Diamond and Mirrlees (1971) and Ramsey (1927).

We followed up on this topic by introducing another viewpoint on the tax systems. We showed the theory reborn by Laffer (2004) and Wanniski (1978), who talked about the existence of a tax rate that would maximise the tax revenue. We also showed the current empirical evidence for this theory, both inside and outside the Czech Republic, and presented some extensions of it. The fourth section then focused on a historical overview of the Czech tax system and gave special attention to the Income Taxes, later analysed from the perspective of the Laffer Curve.

We began the part dedicated to the Estimating Laffer Curves for the Czech Republic by describing our data and computing multiple key variables for our model. We decided to utilise the TTR to GDP ratio as a representative Average Overall Tax Rate. For the Corporate Income Tax, we simply employed the Corporate Income Tax Rate determined by the relevant law.

The most complicated calculation was performed to obtain the value of Average Effective Personal Income Tax Rate. We computed it as a weighted average of the Personal Income Tax rates affecting the mean wages in various sectors in the Czech economy. This process could have been improved if better microdata were available. Dušek et al. (2013) created a model for this type of data, which would generate more representative values of the Average Effective Personal Income Tax Rate. Nevertheless, since the period of our analysis is longer than the period in which the data required by the methodology are published, we had to choose a different approach.

After processing the data, we checked for the issues with it. We found a structural break in some of our time series and therefore introduced binary variables to control for it. Furthermore, we thoroughly tested the stationarity of our variables in time. The results indicated non-stationarity for several of our time series. This led us to divide the analysis into two separate branches.

The first one analysed our variables at levels under the assumption of stationarity. As it was shown, there is no other way the revenue-maximising tax rates could be estimated. It is important to remember that in case of non-stationarity of the time series, we risk encountering a spurious regression, which would invalidate our results. Nevertheless, without such an assumption, estimating the revenue-maximising tax rates for the Czech Republic is impossible.

This analysis found the revenue-maximising tax rate for the overall tax system to be, under the assumption as mentioned earlier, 28.6%. Furthermore, the tax revenues are affected positively by rises in the Annual GDP Growth Rate and negatively by rises in the Unemployment Rate.

The Personal Income Tax has its revenue-maximising Average Effective Personal Income Tax Rate at 17.5%, and the revenues are further negatively affected by the Unemployment Rate. There is no statistically significant effect of Annual GDP Growth Rate. We also discovered some evidence for the existence of the cross-effect of the Corporate Income Tax Rate. The analysis of this tax should be interpreted with caution, as there are just a few data points in the *prohibitive range* of the Laffer Curve.

The revenue-maximising Corporate Income Tax Rate was estimated at 27.5%. The Personal Income Tax Revenue is also positively influenced by the rises in the Annual GDP Growth Rate and negatively affected by the Unemployment Rate. We found no statistically significant cross-effects of the Average Effective Personal Income Tax Rate.

The second branch of our study used the growth rates of our variables. Originally, we tried to find a growth rate in the tax rates that would accelerate the growth rate in the tax revenues the most. We did not find such evidence for any of the studied constructs.

We found a linear relationship between the tax rate and the tax revenue for the overall tax system, Personal Income Tax, and Corporate Income Tax. The growth rate of the Total Tax Revenue and the growth rate of the Personal Income Tax Revenue are further determined by the growth rate of the Unemployment Rate, whereas the growth rate of the Corporate Income Tax Revenue is not. On the contrary, only the growth rate of Corporate Income Tax Revenue is statistically significantly affected by the Annual GDP Growth Rate.

Consequently, in Section 6, we have shown that the tax rates in the Czech Republic are below the revenue-maximising tax rates. This means there is room to raise the tax revenues by increasing the tax rates. However, in the long term, such a behaviour might not be economically optimal as the revenue-maximising tax rates are not the optimal tax rates. This topic was briefly broached in Section 2.

This study prepares the ground for subsequent studies for the Czech Republic, which would focus on applying the maximising tax rates and their implications. The tax rates would have various effects on the Annual GDP Growth Rate, Unemployment Rate and other important economic indicators. The analysis of the growth rates might be useful for short-term predictions and policy adjustments.

We presented the latest and most robust results of the Laffer Curve estimation, thanks to the longer and more frequent data used and the multiple estimation methods that were applied. Understanding how the tax rates and other macroeconomic indicators affect the tax revenues is crucial for the right policy-making. This thesis contributes to the debate by offering the latest empirical insights.

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List of Acronyms

2SLS Two Stage Least Squares

ADF Augmented Dickey-Fuller Test

BP test Breusch-Pagan Test

CIT Corporate Income Tax

CITR Corporate Income Tax Revenue

CITrate Corporate Income Tax Rate

CPI Consumer Price Index

CZK Czech Koruna

DW test Durbin-Watson Test

ECM Error Correction Model

EU European Union

g Annual GDP Growth Rate

GDP Gross Domestic Product

GMM Generalised Method of Moments

HAC Newey-West's Heteroskedasticity and Autocorrelation Consistent Standard Errors

HC White's Heteroskedasticity Consistent Standard Errors

IVR Instrumental Variable Regression

KPSS Kwiatkowski–Phillips–Schmidt–Shin Test

OECD Organisation for Economic Co-operation and Development

OLS Ordinary Least Squares

PIT Personal Income Tax

PITR Personal Income Tax Revenue

PITrate Average Effective Personal Income Tax Rate

PP Phillips-Perron Test

Sargan Sargan Test

SSC Social Security Contributions

TG Trust in the Government

TR Tax Revenue

Trate Average Overall Tax Rate

TS Time Series

TTR Total Tax Revenue

u Unemployment Rate

US United States

VAT Value Added Tax

WI Weak Instruments Test

Wu-Hausman Durbin-Wu-Hausman Endogeneity Test

Appendix

A.1 Data Provided by the Czech Ministry of Finance

Table A.1: Monthly Total Tax Revenue (cumulative figures in bil. CZK)

Month	1995	1996	1997	1998	1999	2000	2001
January	38.22	44.20	43.07	49.75	46.63	48.73	55.00
February	62.20	71.32	70.11	80.81	81.80	82.53	90.11
March	96.13	107.83	106.76	123.54	126.69	136.05	140.07
April	133.83	148.47	153.90	167.20	173.98	180.01	187.21
May	164.23	182.59	193.80	203.55	213.39	221.22	230.63
June	211.13	229.35	241.51	258.60	273.90	285.90	292.03
July	247.14	276.28	287.60	315.80	322.37	333.94	353.26
August	278.22	308.74	323.65	353.31	363.34	376.18	400.54
September	312.47	347.39	367.73	395.94	413.63	420.08	454.65
October	355.68	394.03	418.12	443.84	462.02	480.32	506.19
November	393.88	431.43	457.26	486.57	509.98	528.12	550.88
December	439.97	482.82	508.95	537.41	567.28	586.21	626.22

Table A.2: Tax Revenue Excluding SSC (cumulative figures in bil. CZK)

Month	1995	1996	1997	1998	1999	2000	2001
January	24.09	27.24	25.67	29.41	27.47	28.49	32.14
February	34.37	40.56	37.26	43.11	41.48	45.01	48.19
March	54.71	62.86	58.41	67.67	69.28	73.80	76.01
April	79.60	88.21	88.06	94.26	98.19	99.78	102.26
May	96.27	107.27	108.27	112.97	119.79	122.09	125.45
June	127.03	135.02	135.72	147.01	159.59	165.53	163.63
July	147.79	164.84	164.00	177.77	189.71	193.06	203.22
August	163.56	181.19	180.10	195.81	212.27	216.22	228.97
September	184.62	202.86	204.10	220.17	238.13	241.39	263.12
October	211.68	232.82	235.63	250.46	267.53	276.89	292.86
November	230.96	253.86	256.30	274.52	293.75	304.13	315.56
December	255.38	283.05	287.43	305.70	329.26	337.32	356.61

Table A.3: Social Security Contributions (cumulative figures in bil. CZK)

Month	1995	1996	1997	1998	1999	2000	2001
January	12.37	14.18	16.21	17.69	18.30	19.13	21.09
February	23.09	26.63	30.40	33.49	34.06	35.46	38.96
March	33.82	38.88	44.11	48.71	49.65	51.94	57.08
April	45.96	51.89	59.06	64.61	66.33	69.07	76.06
May	56.82	64.94	74.02	80.75	83.13	86.66	95.23
June	69.29	80.80	92.53	99.16	102.07	106.16	117.09
July	82.97	97.56	108.78	116.48	119.77	124.99	137.09
August	94.86	112.14	124.64	133.47	137.25	143.41	157.65
September	106.87	126.26	140.01	149.74	154.27	161.44	176.90
October	119.67	140.31	155.72	166.03	171.33	179.89	196.94
November	133.59	155.60	172.33	183.38	189.39	199.09	217.73
December	154.32	174.32	191.01	203.91	210.89	222.18	242.32

Table A.4: Personal Income Tax Revenues (cumulative figures in bil. CZK)

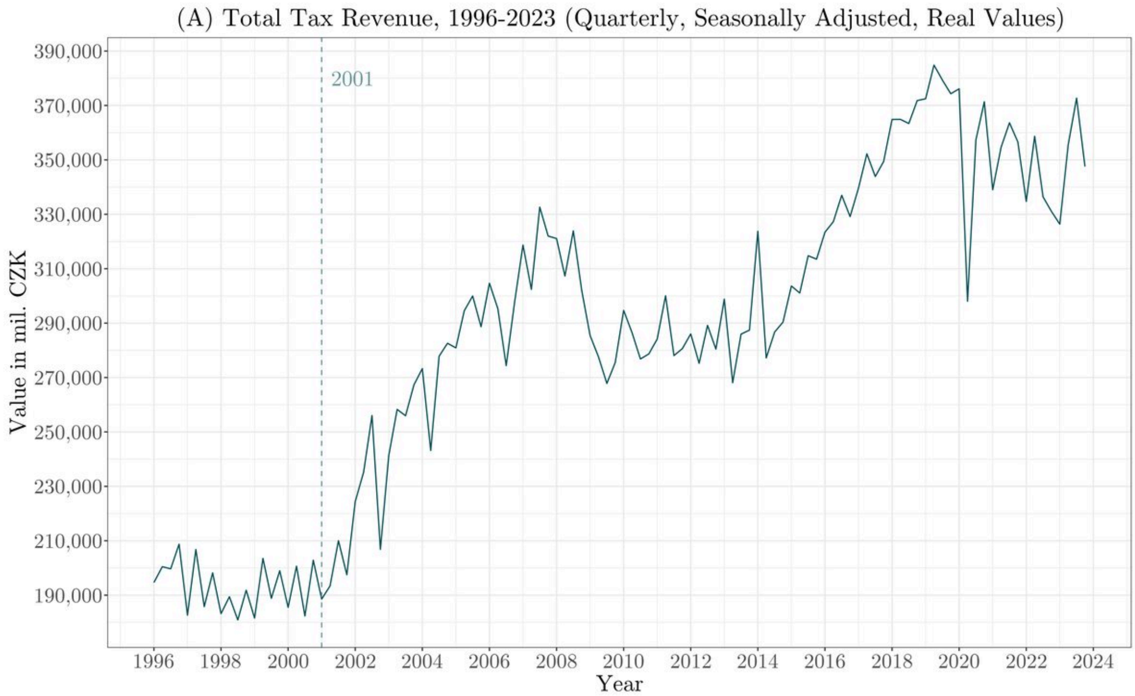
Month	1995	1996	1997	1998	1999	2000	2001
January	2.63	2.96	4.74	5.27	5.54	4.18	7.72
February	2.89	6.79	7.16	8.07	8.31	6.77	12.94
March	3.23	8.29	8.73	9.94	10.21	8.76	19.38
April	3.83	9.83	10.63	12.04	12.32	10.86	23.90
May	4.18	11.82	14.49	14.51	14.78	13.40	29.01
June	4.68	14.38	15.86	17.60	17.64	16.40	37.40
July	5.44	17.01	19.12	21.20	20.93	19.63	44.69
August	5.98	19.37	21.72	24.02	23.66	22.39	50.65
September	6.49	21.61	24.28	26.69	26.22	25.05	57.11
October	7.35	24.07	27.08	29.72	28.92	27.85	62.94
November	7.80	26.36	29.82	32.54	31.55	30.77	68.99
December	8.49	29.73	33.20	36.28	35.13	34.73	78.59

Table A.5: Corporate Income Tax Revenues (cumulative figures in bil. CZK)

Month	1995	1996	1997	1998	1999	2000	2001
January	2.60	2.37	1.68	1.63	1.67	1.72	1.92
February	7.07	4.82	3.71	3.96	4.24	4.35	4.22
March	12.59	9.47	7.49	8.82	9.84	9.39	8.81
April	17.95	14.27	12.04	12.35	13.21	11.74	13.11
May	21.89	16.62	14.00	14.22	15.73	14.85	15.61
June	36.74	24.88	23.33	27.64	31.68	30.84	26.67
July	40.01	30.93	25.95	32.80	33.34	32.79	37.32
August	42.78	31.87	26.00	33.49	34.88	34.26	42.89
September	47.95	34.96	30.21	38.26	39.04	38.14	54.22
October	53.67	39.50	33.91	42.10	42.20	43.56	56.15
November	56.96	41.61	35.78	44.92	45.37	46.25	56.96
December	64.19	48.57	41.64	51.10	51.31	52.28	68.88

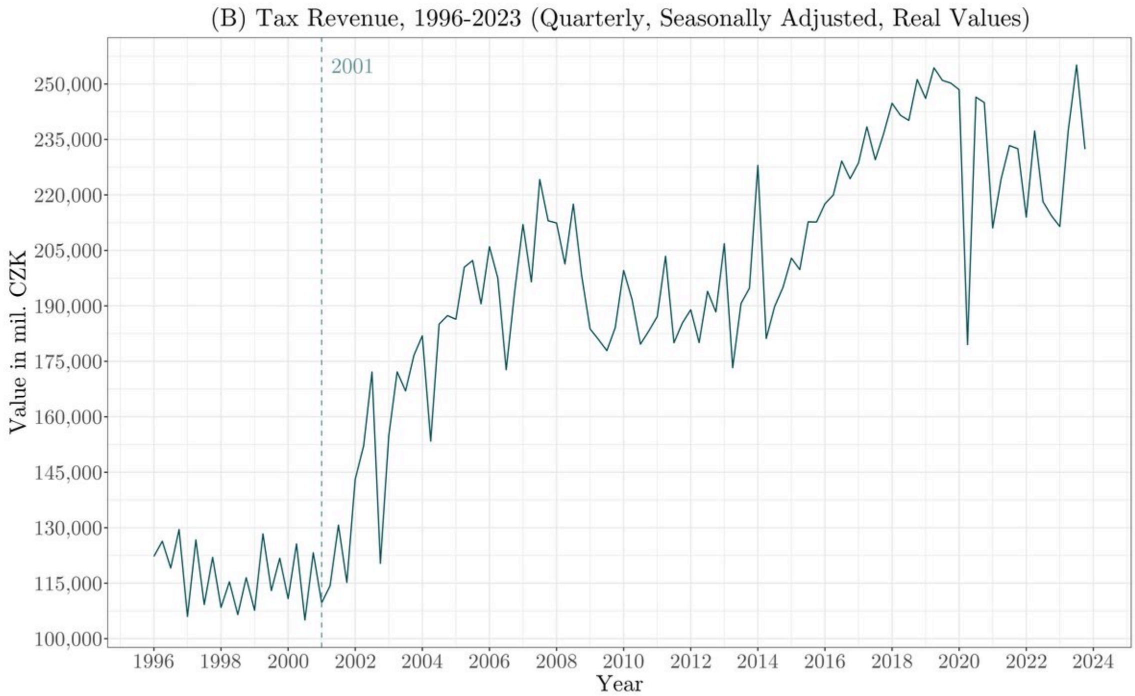
A.2 Development of our Key Variables

Figure A.1: Total Tax Revenue



Note: Author's own elaboration based on data provided by the institutions mentioned in Section 5.1 and on variables calculated by the procedures described in Section 5.2.

Figure A.2: Tax Revenue

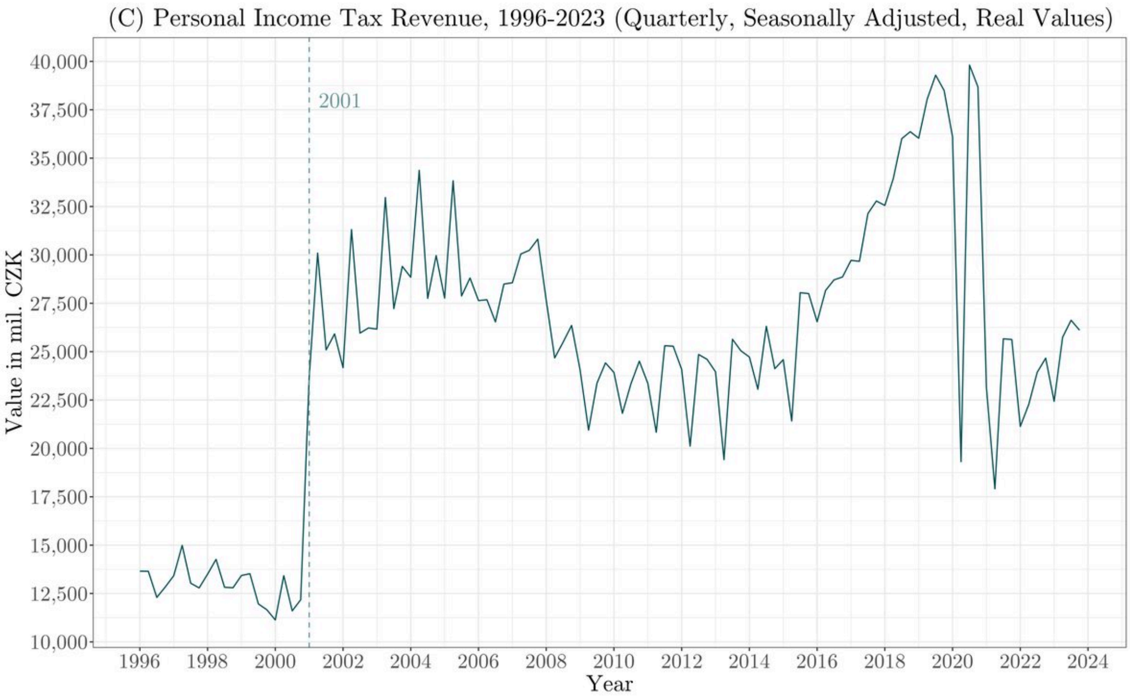


Note: Author's own elaboration based on data provided by the institutions mentioned in Section 5.1 and on variables calculated by the procedures described in Section 5.2.

Figure A.3: Social Security Contributions

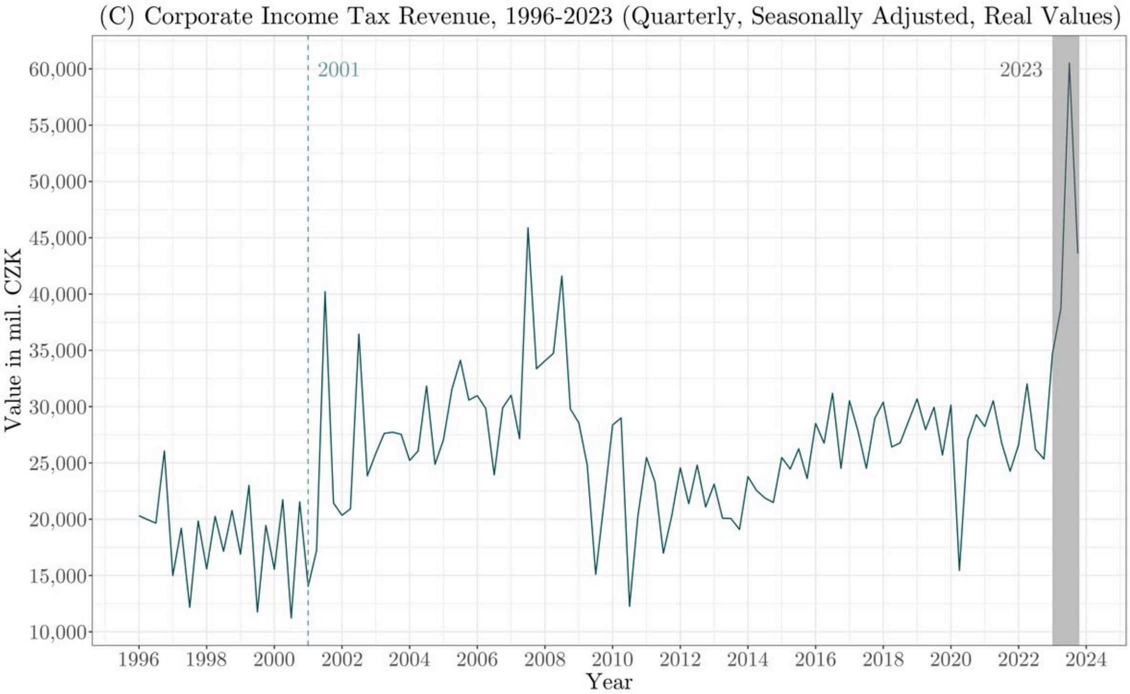
Note: Author's own elaboration based on data provided by the institutions mentioned in Section 5.1 and on variables calculated by the procedures described in Section 5.2.

Figure A.4: Personal Income Tax Revenue



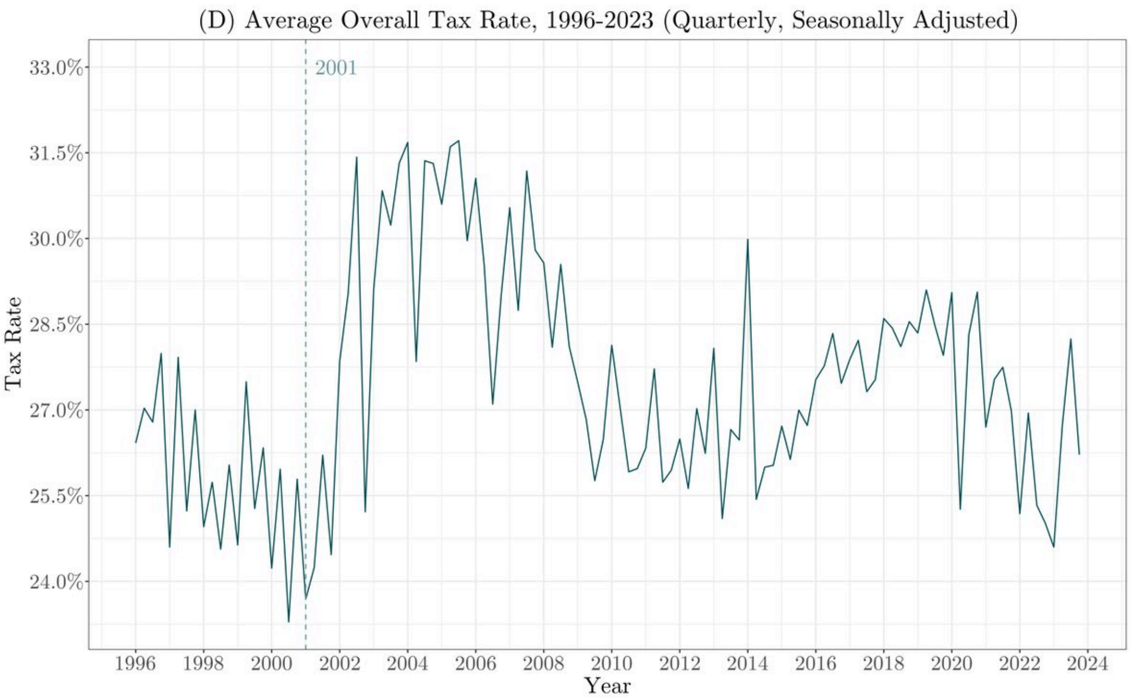
Note: Author's own elaboration based on data provided by the institutions mentioned in Section 5.1 and on variables calculated by the procedures described in Section 5.2.

Figure A.5: Corporate Income Tax Revenue



Note: Author's own elaboration based on data provided by the institutions mentioned in Section 5.1 and on variables calculated by the procedures described in Section 5.2.

Figure A.6: Average Overall Tax Rate

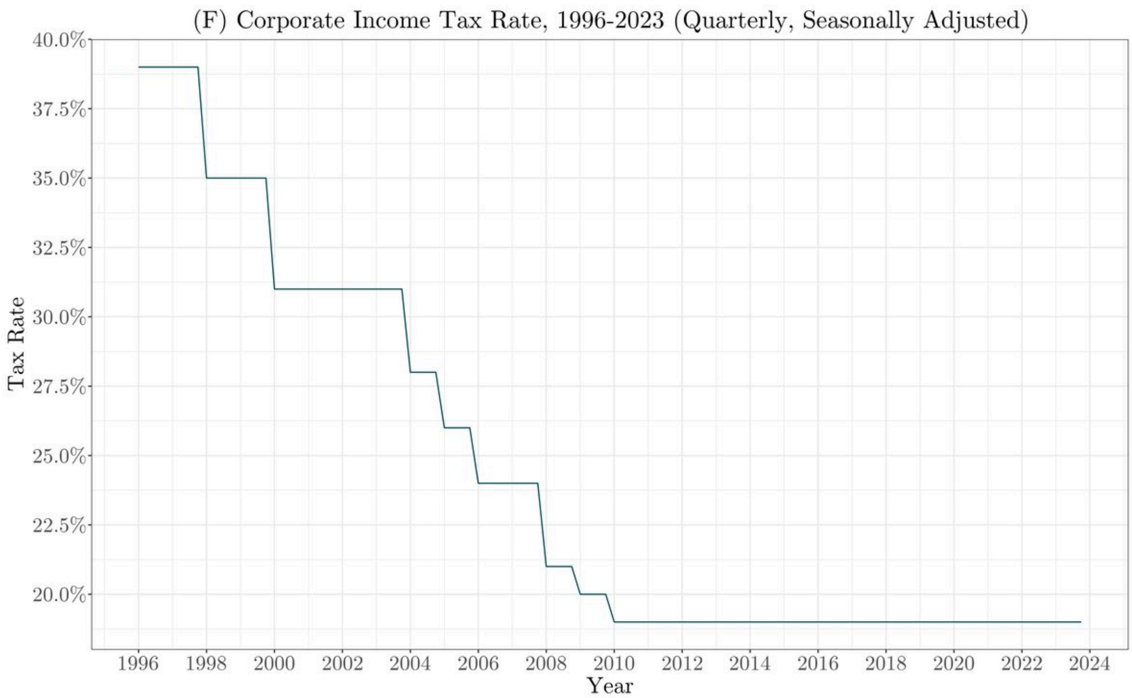


Note: Author's own elaboration based on data provided by the institutions mentioned in Section 5.1 and on variables calculated by the procedures described in Section 5.2.

Figure A.7: Average Effective Personal Income Tax Rate

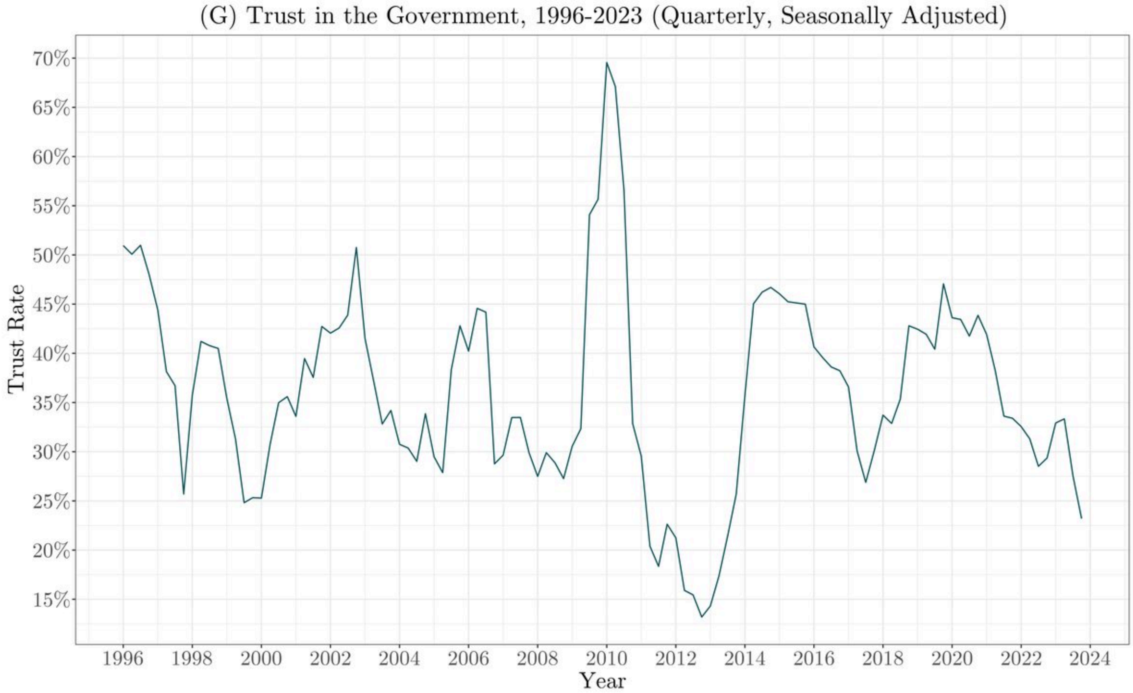
Note: Author's own elaboration based on data provided by the institutions mentioned in Section 5.1 and on variables calculated by the procedures described in Section 5.2.

Figure A.8: Corporate Income Tax Rate



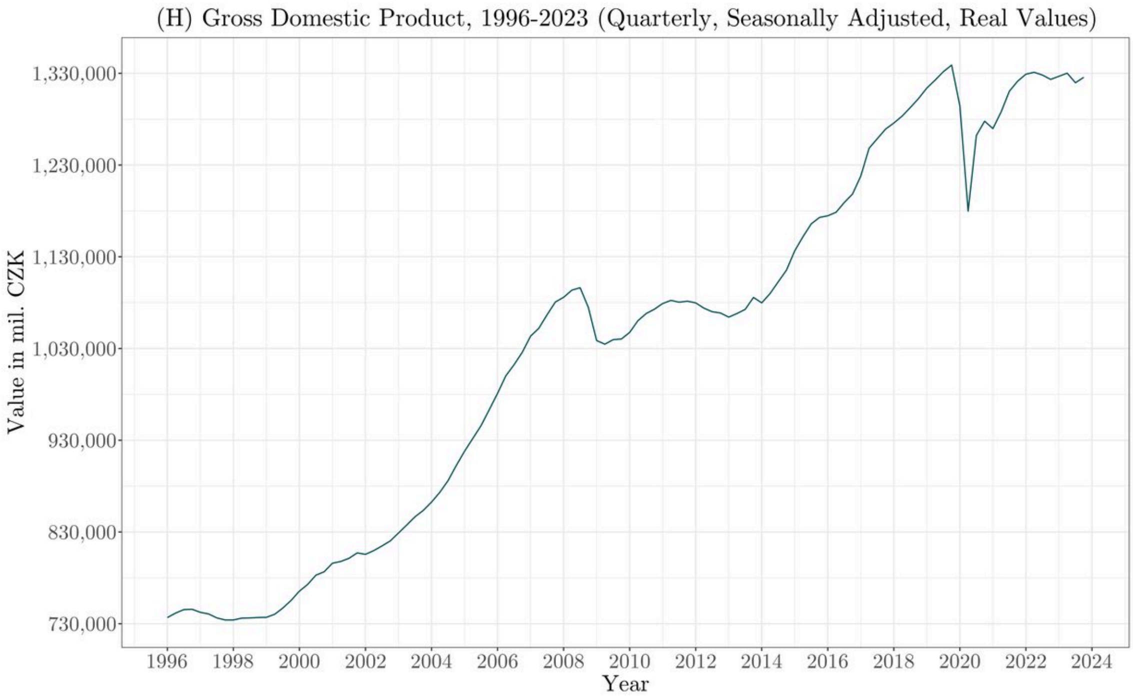
Note: Author's own elaboration based on data provided by the institutions mentioned in Section 5.1 and on variables calculated by the procedures described in Section 5.2.

Figure A.9: Trust in the Government



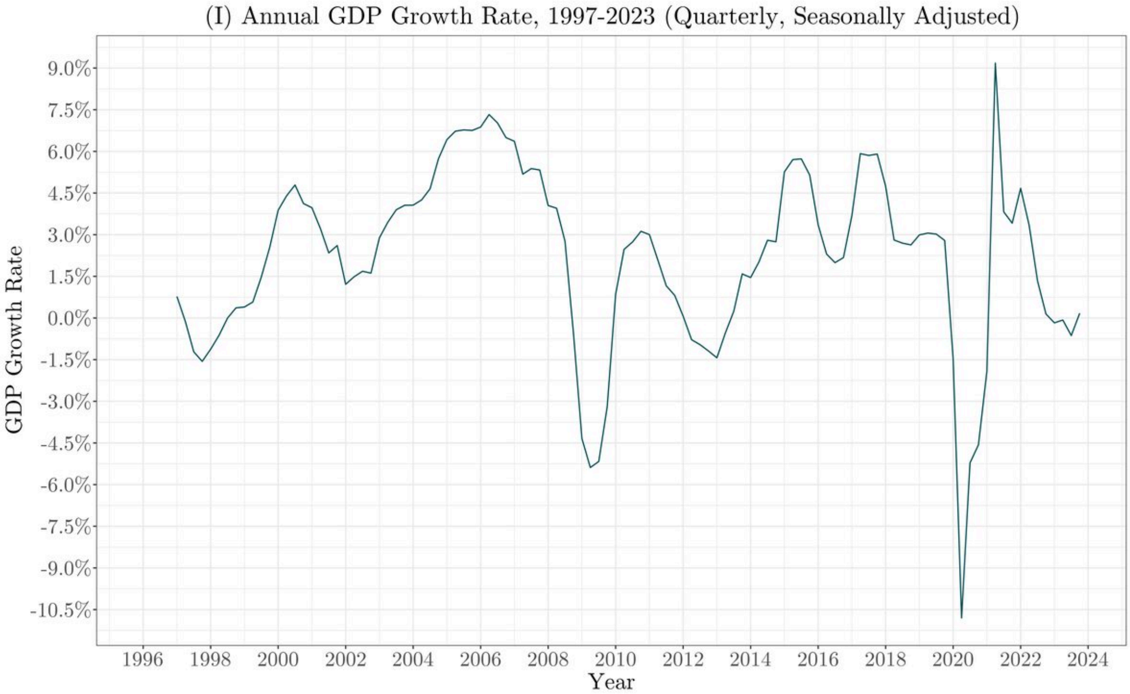
Note: Author's own elaboration based on data provided by the institutions mentioned in Section 5.1 and on variables calculated by the procedures described in Section 5.2.

Figure A.10: Gross Domestic Product



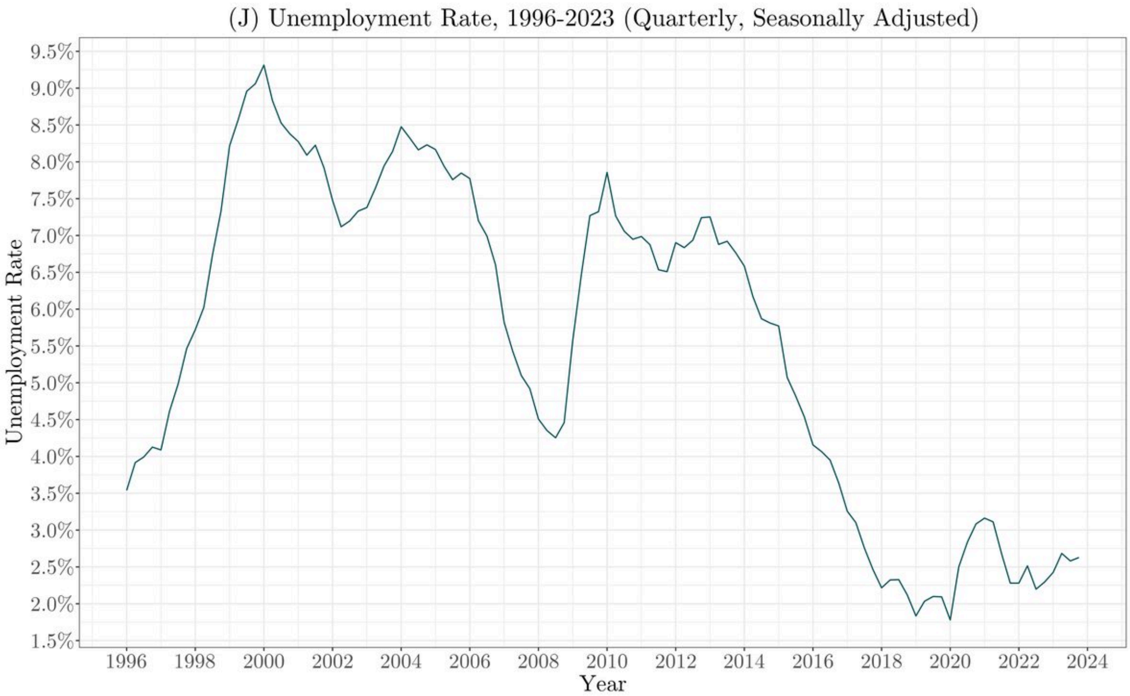
Note: Author's own elaboration based on data provided by the institutions mentioned in Section 5.1 and on variables calculated by the procedures described in Section 5.2.

Figure A.11: Annual GDP Growth Rate



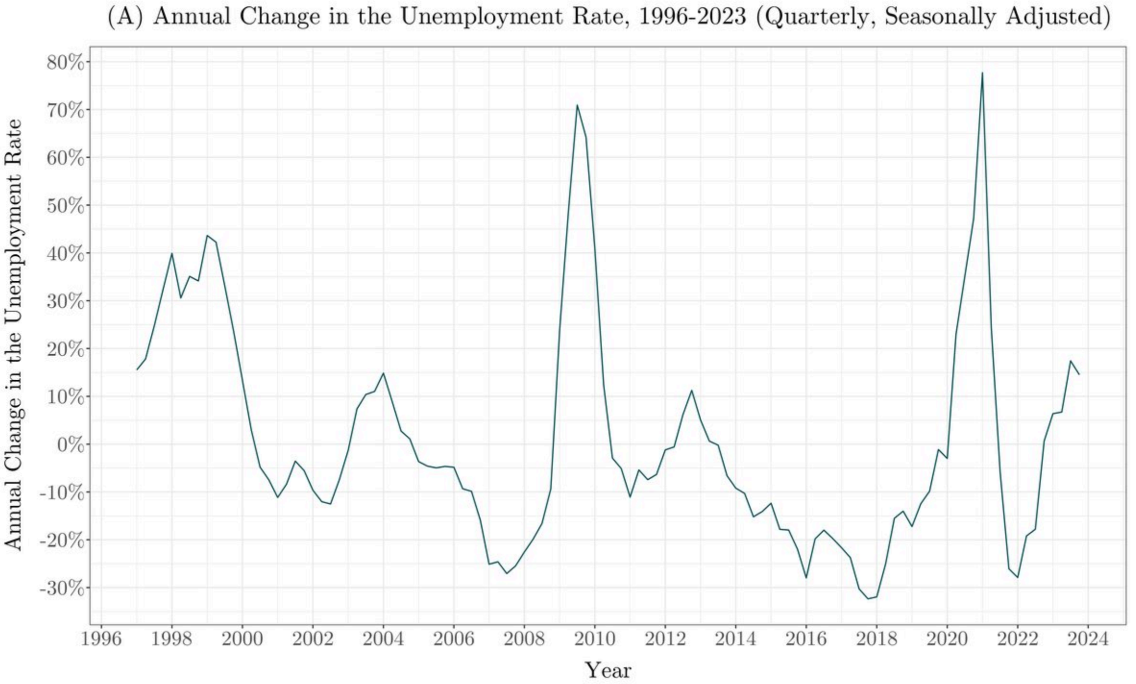
Note: Author's own elaboration based on data provided by the institutions mentioned in Section 5.1 and on variables calculated by the procedures described in Section 5.2.

Figure A.12: Unemployment Rate



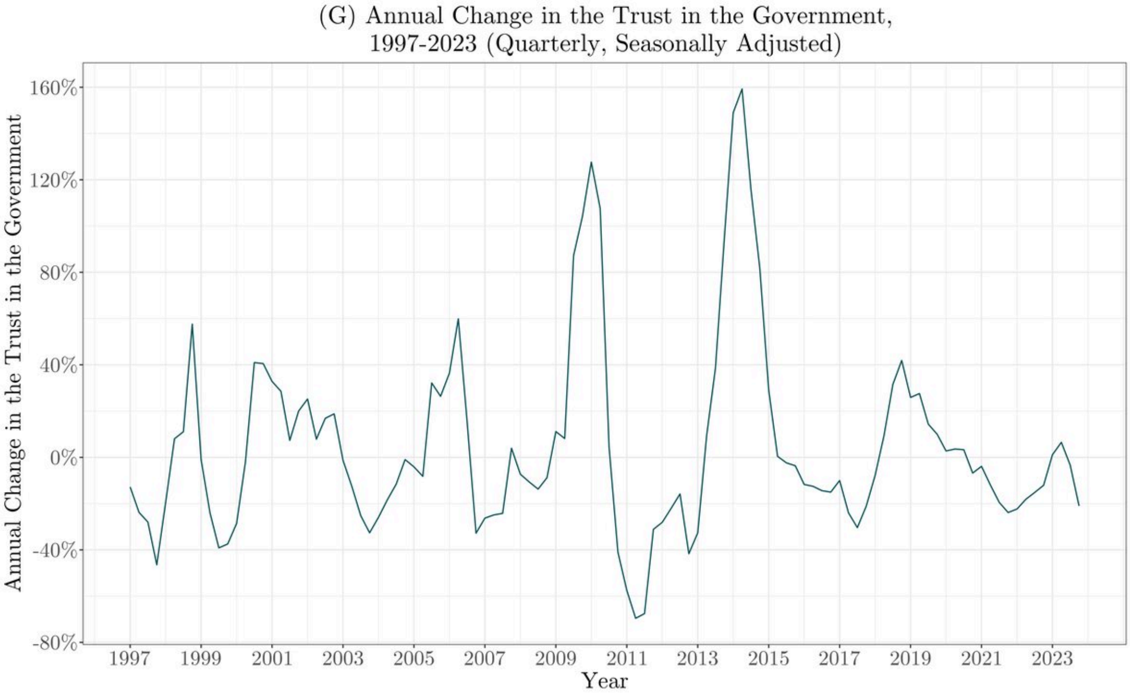
Note: Author's own elaboration based on data provided by the institutions mentioned in Section 5.1 and on variables calculated by the procedures described in Section 5.2.

Figure A.13: Annual Change in the Unemployment Rate



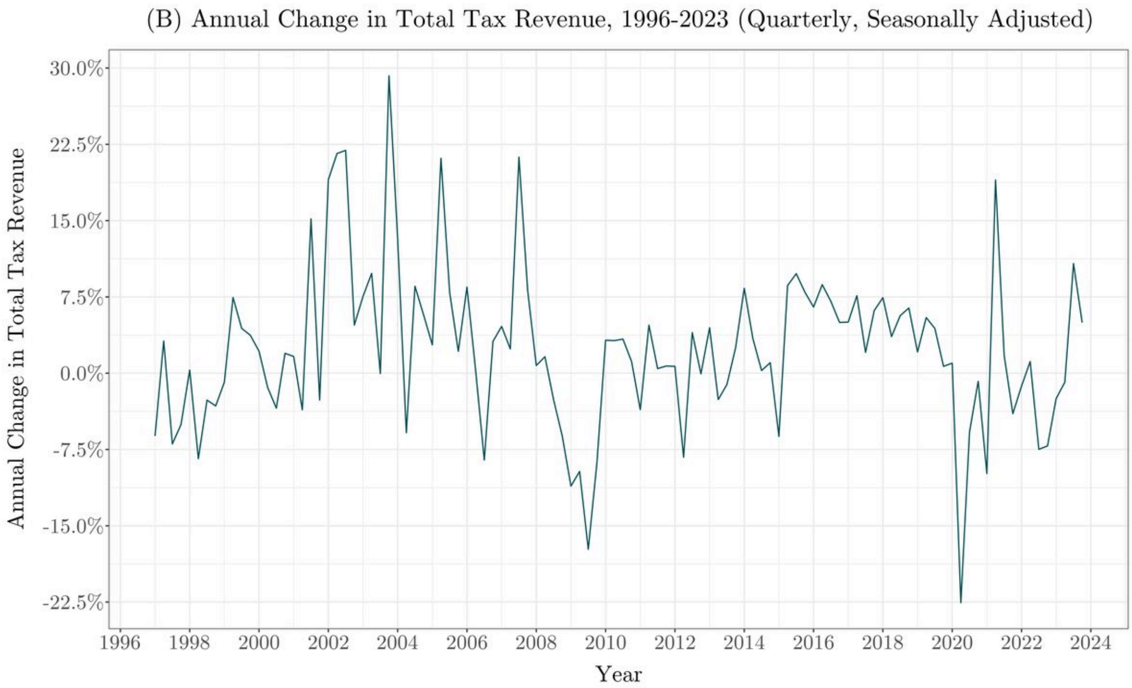
Note: Author's own elaboration based on data provided by the institutions mentioned in Section 5.1 and on variables calculated by the procedures described in Section 5.2 and 5.3.1.

Figure A.14: Annual Change in the Trust in the Government



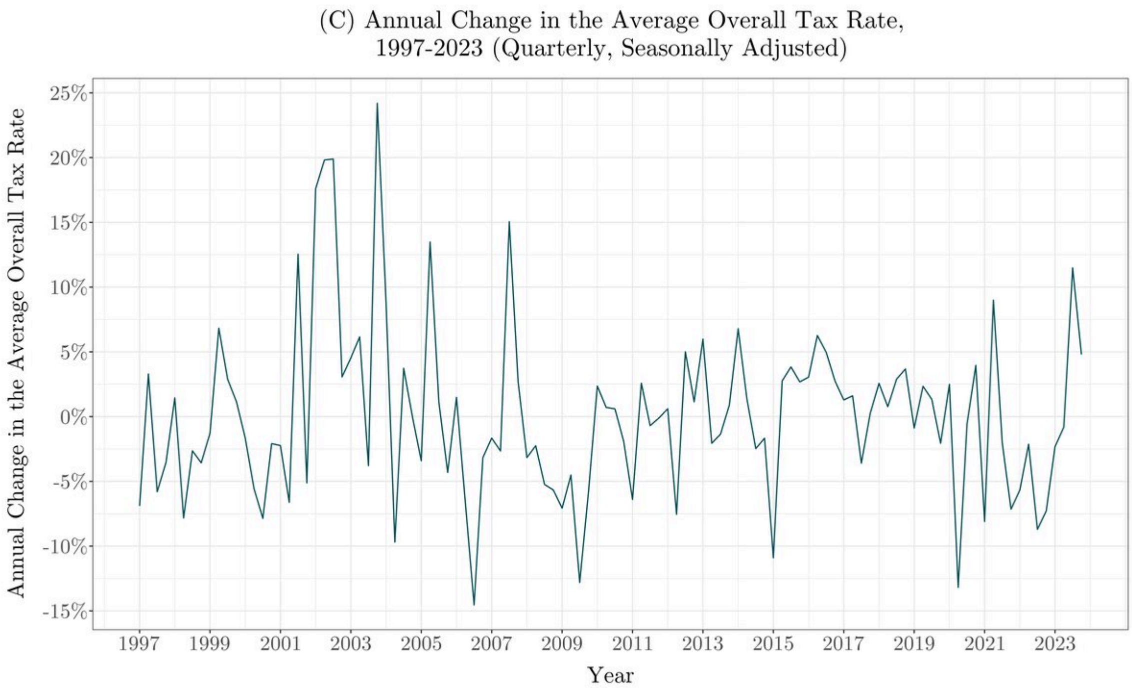
Note: Author's own elaboration based on data provided by the institutions mentioned in Section 5.1 and on variables calculated by the procedures described in Section 5.2 and 5.3.1.

Figure A.15: Annual Change in the Total Tax Revenue



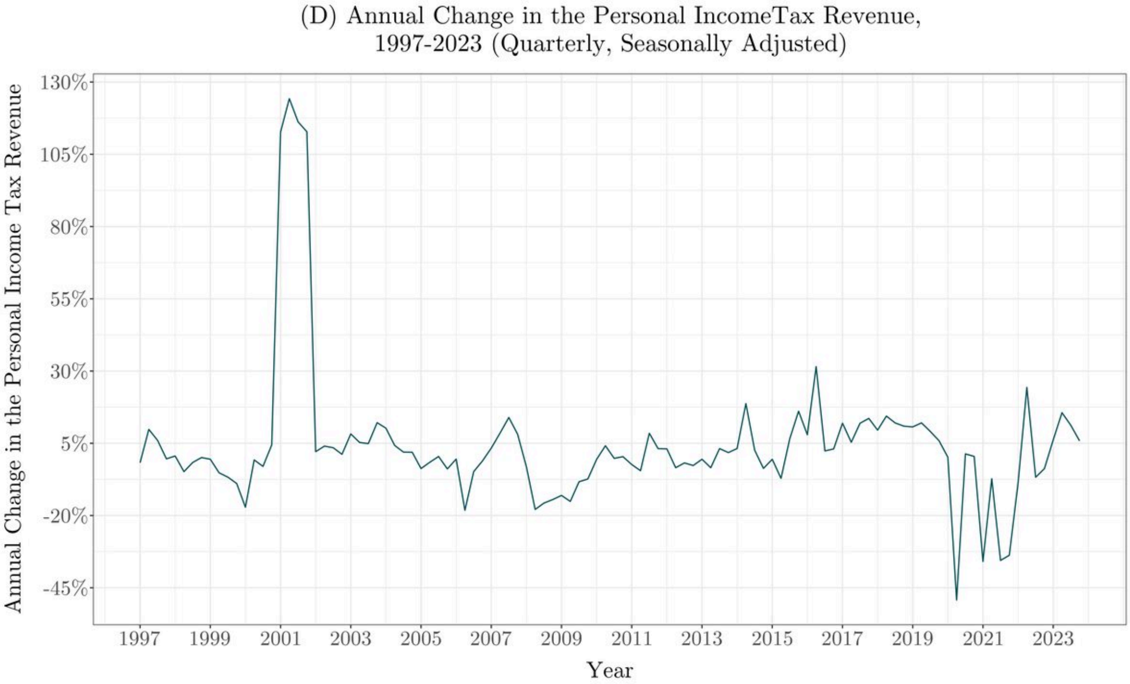
Note: Author's own elaboration based on data provided by the institutions mentioned in Section 5.1 and on variables calculated by the procedures described in Section 5.2 and 5.3.1.

Figure A.16: Annual Change in the Average Overall Tax Rate



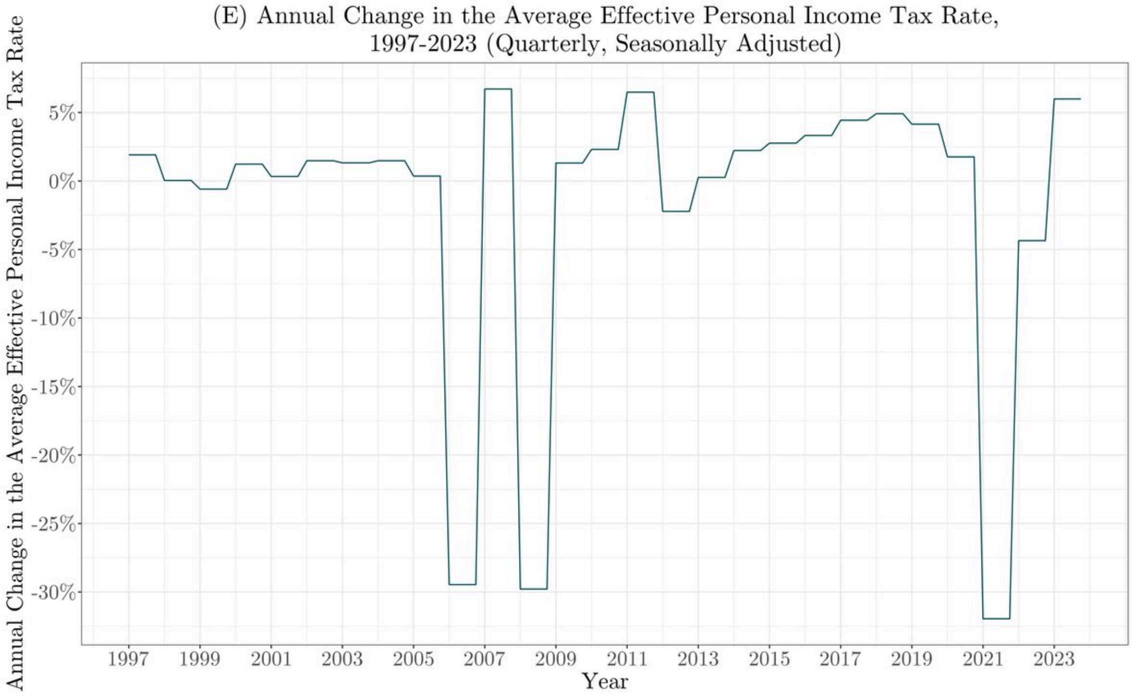
Note: Author's own elaboration based on data provided by the institutions mentioned in Section 5.1 and on variables calculated by the procedures described in Section 5.2 and 5.3.1.

Figure A.17: Annual Change in the Personal Income Tax Revenue



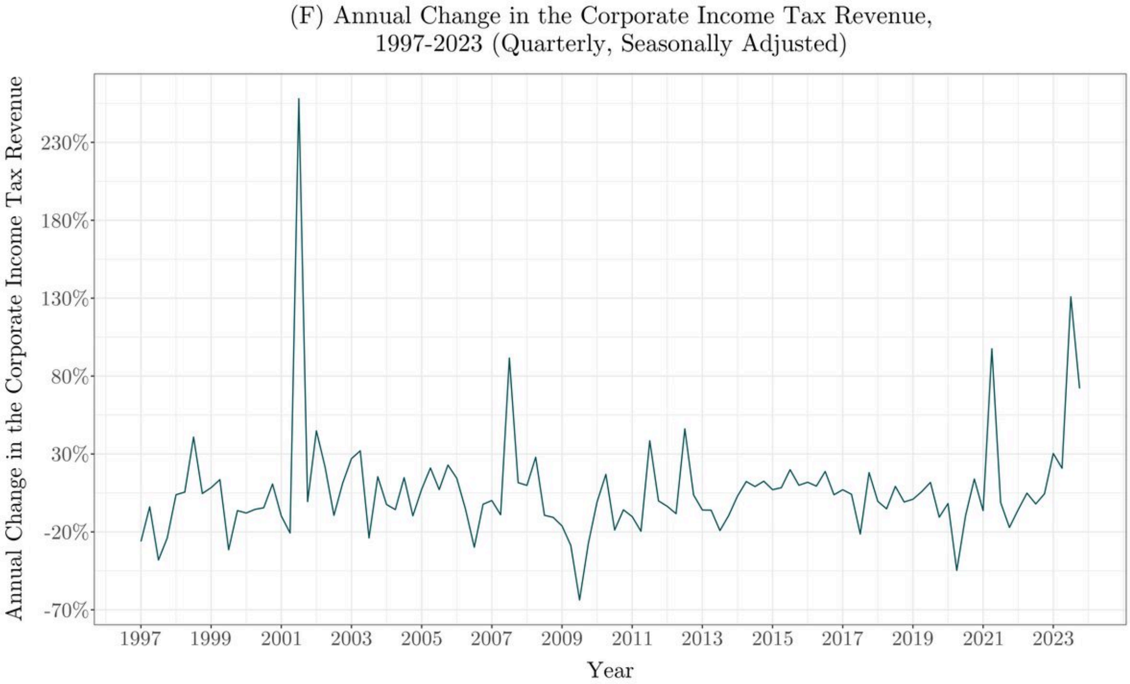
Note: Author's own elaboration based on data provided by the institutions mentioned in Section 5.1 and on variables calculated by the procedures described in Section 5.2 and 5.3.1.

Figure A.18: Annual Change in the Average Effective Personal Income Tax Rate



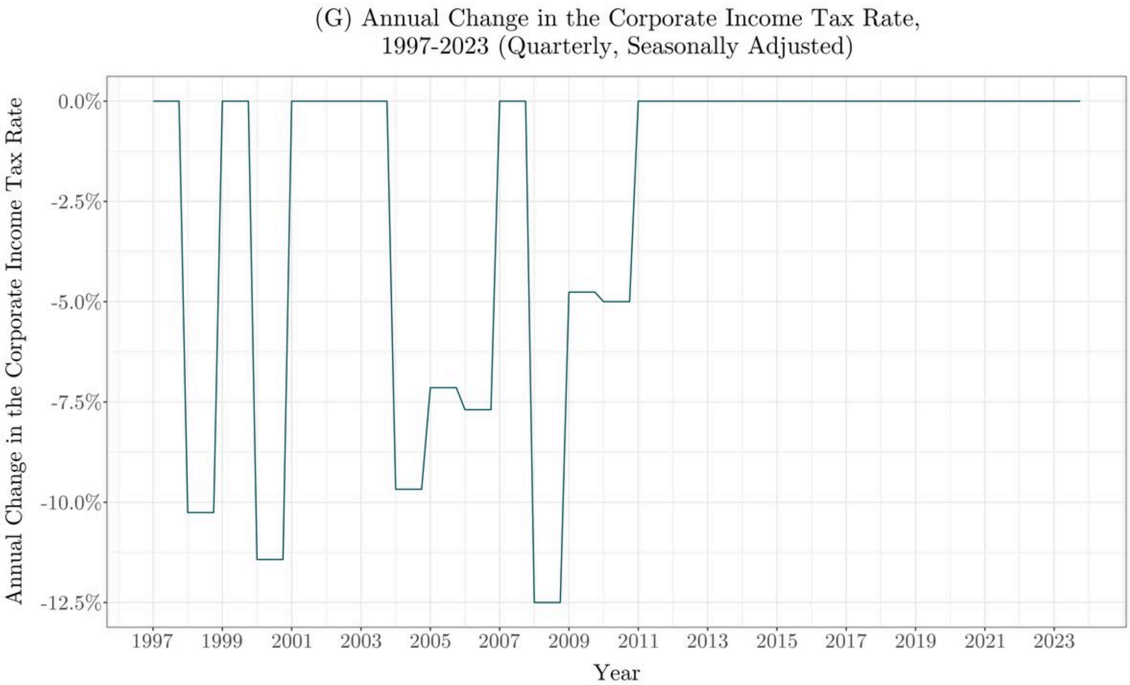
Note: Author's own elaboration based on data provided by the institutions mentioned in Section 5.1 and on variables calculated by the procedures described in Section 5.2 and 5.3.1.

Figure A.19: Annual Change in the Corporate Income Tax Revenue



Note: Author's own elaboration based on data provided by the institutions mentioned in Section 5.1 and on variables calculated by the procedures described in Section 5.2 and 5.3.1.

Figure A.20: Annual Change in the Corporate Income Tax Rate



Note: Author's own elaboration based on data provided by the institutions mentioned in Section 5.1 and on variables calculated by the procedures described in Section 5.2 and 5.3.1.

A.3 Variables Utilised for the Computation of the Average Effective Personal Income Tax Rate (PITrate)

Table A.6: Income Tax Rates in the Czech Republic (1996-2009)

Year	Personal credit	First bracket		Second bracket		Third bracket		Fourth bracket		Rest
		Rate	Upper Cap	Rate	Upper Cap	Rate	Upper Cap	Rate	Upper Cap	Rate
1996	–	0.15	7,000	0.20	12,000	0.25	17,000	0.32	47,000	0.40
1997	–	0.15	7,000	0.20	14,000	0.25	21,000	0.32	63,000	0.40
1998	–	0.15	7,620	0.20	15,250	0.25	22,850	0.32	68,550	0.40
1999	–	0.15	8,500	0.20	17,000	0.25	26,000	0.32	92,000	0.40
2000	–	0.15	8,500	0.20	17,000	0.25	26,000	0.32	–	–
2001	–	0.15	9,100	0.20	18,200	0.25	27,600	0.32	–	–
2002	–	0.15	9,100	0.20	18,200	0.25	27,600	0.32	–	–
2003	–	0.15	9,100	0.20	18,200	0.25	27,600	0.32	–	–
2004	–	0.15	9,100	0.20	18,200	0.25	27,600	0.32	–	–
2005	–	0.15	9,100	0.20	18,200	0.25	27,600	0.32	–	–
2006	600	0.12	10,100	0.19	18,200	0.25	27,600	0.32	–	–
2007	600	0.12	10,100	0.19	18,200	0.25	27,600	0.32	–	–
2008	2,070	0.15	–	–	–	–	–	–	–	–
2009	2,070	0.15	–	–	–	–	–	–	–	–
2010	2,070	0.15	–	–	–	–	–	–	–	–
2011	2,070	0.15	–	–	–	–	–	–	–	–
2012	2,070	0.15	–	–	–	–	–	–	–	–
2013	2,070	0.15	103,536	0.22	–	–	–	–	–	–
2014	2,070	0.15	103,768	0.22	–	–	–	–	–	–
2015	2,070	0.15	106,444	0.22	–	–	–	–	–	–
2016	2,070	0.15	108,024	0.22	–	–	–	–	–	–
2017	2,070	0.15	119,916	0.22	–	–	–	–	–	–
2018	2,070	0.15	126,064	0.22	–	–	–	–	–	–
2019	2,070	0.15	130,796	0.22	–	–	–	–	–	–
2020	2,070	0.15	139,340	0.22	–	–	–	–	–	–
2021	2,320	0.15	141,764	0.22	–	–	–	–	–	–
2022	2,570	0.15	155,644	0.22	–	–	–	–	–	–
2023	2,570	0.15	161,296	0.22	–	–	–	–	–	–

Note: The 0.22 (22%) rate from 2013-2023 was legalised as a 0.15 (15%) rate with a 0.07 (7%) for the part of the monthly income exceeding the First Bracket Upper Cap.